Weighted configurations and the nonexistence of arithmetic ball quotient compactifications

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Abstract

We prove that, for every $n \geq 9$, no ordered GIT quotient of n points on $\mathbb{P}^1$ with positive rational weights smaller than one and of total weight two is isomorphic to the Baily–Borel compactification of an arithmetic complex ball quotient. The weights may lie on stability walls, and the lattice may have torsion and ramify along divisors. No compatibility with a prescribed period map is assumed. The proof compares homomorphisms from the tangent bundle to line bundles of negative degree with normal bundles of arithmetic fixed subvarieties. We describe these homomorphisms through coincident marked points and prove vanishing statements for their kernels. The comparison, together with integrality of Chern classes and the invariant theory of finite reflection groups, gives the nonexistence theorem without a Fano assumption on the GIT quotient.

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1 Introduction

Let $n \geq 9$, and choose rational weights

$$0 < \mu_i < 1, \qquad \sum_{i=1}^{n} \mu_i = 2. \tag{1.1}$$

For $I \subset \{1, \dots, n\}$, write $\mu(I) = \sum_{i \in I} \mu_i$. After clearing denominators, let

$$P_n = (\mathbb{P}^1)^n, \qquad X_\mu = P_n /\!/_{\mathcal{O}(m\mu_1, \dots, m\mu_n)} \mathrm{SL}(2, \mathbb{C}).$$

The labels are retained, and the quotient is independent of the integer $m > 0$ used to clear denominators. A configuration is stable precisely when the weight at every point is smaller than one. We use $\mathbb{B}^d$ for the complex ball of dimension d, and the superscript BB for the Baily–Borel compactification.

Theorem 1.1. For every $n \geq 9$ and every weight vector satisfying (1.1), there is no arithmetic lattice $\Gamma \subset \mathrm{PU}(1, n-3)$ and no algebraic isomorphism

$$X_\mu \simeq (\mathbb{B}^{n-3}/\Gamma)^{\mathrm{BB}}.$$

We impose no compatibility with label permutations or with a specified period map. Failure of an integrality condition for a Deligne–Mostow period map alone would not prove the theorem [10]. The weights may lie on stability walls. The lattice may have torsion, need not be a normal or congruence subgroup of a prescribed group, and may ramify along divisors. No Fano hypothesis is imposed on X_μ.

The case of twelve equal weights answers Question 2.3 of Hulek–Kondö–Maeda [15]. The companion paper [23] proves the nonexistence theorem for seven equal weights by a comparison involving all fixed components. It cites the common tangent and arithmetic propositions proved here. The proof of Theorem 1.1 does not use that result.

The universal marked projective line determines homomorphisms from the tangent bundle to lines of negative degree and controls their kernels. The original arithmetic lattice gives fixed balls whose normal directions are invariant polynomials, with finite characters and central periods that descend together. We compare these structures on complete curves of the full normalizer. Three reductions shorten the proof. The marked degrees first give effective curve decompositions and the initial degree estimates without an external theorem on generators of the Mori cone. Next, a persistent irreducible factor has equal basic degrees and rank at most 2, and its fixed compactification is a quotient with one or two labels merged. Strong induction from $n = 9, 10$ excludes every such factor for $n \geq 11$. Finally, after the stabilizers are of type A, write $N = n - 2$ and

$$b_N = \prod_{p|N} p^{\lceil 2v_p(N)/3 \rceil}, \qquad C_N = \frac{2N^2}{b_N}. \tag{1.2}$$

Chern integrality and complete moving marked curves give $K_W + C_N L$ nef, including a possible chamber model isomorphic to projective space. For $N \geq 11$,

$$C_N < \frac{N^2 - 4N + 1}{2},$$

contradicting a uniform type A comparison on fixed curves.

Sections 2 and 3 give the common geometric and arithmetic comparisons. Sections 4 and 5 establish the initial cases, and Section 6 completes the induction. The appendices retain the local algebra, projective comparisons, and finite ranges used in the proof. Throughout, L is the automorphic rational line bundle of the original lattice or its specified pullback. The notation $H = -K_W$ does not assert ampleness; in a calculation on projective space its hyperplane class is identified separately.

2 Marked curves, arithmetic groups, and cusps

Put $d = n - 3$, $N = d + 1 = n - 2$, and $\mu(I) = \sum_{i \in I} \mu_i$. Throughout this section $n \geq 6$. Stability means that the weight at every point is less than 1. Stable configurations have at least three distinct points, so the action of $\mathrm{PGL}(2, \mathbb{C})$ is free.

2.1 The marked family and its quotients onto lines of negative degree

Lemma 2.1. There is a smooth chamber quotient $W = X_\nu$ and a small projective morphism $\pi : W \to X_\mu$ such that K_W is π-nef. The original stable quotient X^{s}_μ is an open subset of W. At a genuine wall vertex the local model is

$$\mathrm{Cone}\big(\mathbb{P}^{|I|-2} \times \mathbb{P}^{|I^c|-2}, \mathcal{O}(1,1)\big).$$

The resolution retains the smaller factor $\mathbb{P}^u$, where $u = \min(|I|, |I^c|) - 2$, and

$$\mathrm{codim}_W \mathrm{Exc}(\pi) \geq \lceil (n-2)/2 \rceil.$$

Proof. After fixing one mark at each of $0, \infty$, the transverse torus weights are $(1^{|I|-1}, -1^{|I^c|-1})$. Products of opposite coordinates generate its invariant ring, with the two by two minors as

relations. Keeping the smaller factor gives normal bundle $\mathcal{O}(-1)^{\oplus(v+1)}$, $v = \max(|I|, |I^c|) - 2$, and degree against $-K_W$ $u - v \le 0$. All these choices are made simultaneously by

$$\nu = (1+\epsilon)\mu - \epsilon(2/n, \dots, 2/n) + \epsilon^2\eta, \qquad \sum_i \eta_i = 0,$$

with η generic and $\epsilon > 0$ sufficiently small. A wall subset of size less than $n/2$ becomes long; η separates the equal sizes. The exceptional codimension is $d - u$. In particular, it is at least four for $n \ge 9$. □

The principal $\mathrm{PGL}(2, \mathbb{C})$-bundle of stable configurations defines $\mathcal{P} \to W$, with its labeled sections σ_i. For a nonconstant map $f : C \to W$ from a smooth projective connected curve, write its pullback as $\mathbb{P}_C(V)$, using the convention of one-dimensional quotients. A section supplies a divisor of relative degree one, so such a bundle of rank 2 V exists. Write $V \twoheadrightarrow L_i$ for the marked quotients and put $M_i = L_i^2(\det V)^{-1}$, $m_i = \deg M_i$, and $a = -K_W \cdot f_*C$. The infinitesimal action and the brackets of distinct sections give

$$\begin{gathered} 0 \longrightarrow \mathrm{End}_0(V) \longrightarrow \bigoplus_i M_i \longrightarrow f^*T_W \longrightarrow 0, \\ a = \sum_i m_i, \qquad m_i + m_j = 2\deg D_{ij} \ge 0 \quad (\sigma_i \ne \sigma_j). \end{gathered} \tag{2.1}$$

Here D_{ij} is the effective zero divisor of the bracket. If a set I of sections coincides, its smooth coincidence locus satisfies

$$N_{Z_I^{\mathrm{s}}/W} = M_I \otimes_{\mathbb{C}} (\mathbb{C}^I/\mathbb{C}\mathbf{1}). \tag{2.2}$$

Proposition 2.2. Suppose Q is a line bundle of negative degree and $\mathrm{Hom}(f^*T_W, Q) \ne 0$. Every homomorphism $f^*T_W \to Q$ factors through the stalkwise quotient

$$D = M^{\oplus(\ell-1)}, \qquad \psi : f^*T_W \twoheadrightarrow D, \qquad \psi\, df = 0.$$

Its $\ell \ge 2$ labels form a set I of identical sections of weight $\nu(I) < 1$, and $M = M_I$ has degree $-e < 0$. For $j \notin I$, write D_j for the bracket divisor with this section and $\delta = \sum_{j\notin I} \deg D_j$. Then

$$M_j = M^{-1}(2D_j), \quad a = (n-2\ell)e + 2\delta, \quad \mathrm{Hom}(f^*T_W, Q) = H^0(C, QM^{-1})^{\oplus(\ell-1)}. \tag{2.3}$$

In particular, $\deg Q = -b < 0$ implies $e \ge b$. If this Hom space has dimension one, then $\ell = 2$ and $a \ge (n-4)b$.

Proof. By (2.1), two negative marked lines cannot belong to distinct sections. Thus all negative lines arise from one section, and its brackets give the first two formulas. All outside lines have positive degree and map trivially to Q. Evaluation of $\mathrm{End}_0(V)$ onto the common tangent line is surjective at every stalk, and its image in the I components is exactly the diagonal. Quotienting those components by the diagonal proves the assertion and the Hom formula. If there is only one such component it is removed entirely, so $\ell \ge 2$. The derivatives of identical sections coincide, proving $\psi\, df = 0$. □

Proposition 2.3. Put $F = f^*T_{Z_I^{\mathrm{s}}}$, let $T_f \subset F$ be the saturated image of df, and put $\mathcal{K} = F/T_f$. Then

$$f^*T_W = F \oplus D, \qquad N_f^{\mathrm{sat}} = \mathcal{K} \oplus D, \qquad \bigoplus_{j\notin I} M_j \twoheadrightarrow F \twoheadrightarrow \mathcal{K}.$$

For every line bundle Q and every permitted q,

$$\deg Q < qe \implies \mathrm{Hom}(\textstyle\bigwedge^q F, Q) = \mathrm{Hom}(\bigwedge^q \mathcal{K}, Q) = 0.$$

For $C = \mathbb{P}^1$, all line summands of F have degree at least e. A line direct summand of f^*T_W, or of N_f^{sat}, of degree less than e is isomorphic to M. If f is immersed, N_f^{sat} is the ordinary normal bundle.

Proof. Split the I components of (2.1) into their diagonal and the subbundle with sum zero. The infinitesimal action has zero projection to the latter, giving the displayed splitting. Stalkwise evaluation onto the diagonal removes that diagonal and gives the outside surjection onto F. Its exterior powers have summands of degrees at least qe, proving the vanishing. On $\mathbb{P}^1$, this also says that $F(-e)$ is globally generated. For a line summand of degree less than e, its retraction vanishes on F, or on $\mathcal{K}$, and factors through D. The resulting idempotent of D, after tensoring by M^{-1}, is a constant matrix; its image of rank 1 is trivial. □

Proposition 2.4. Let $\mathcal{Q}$ be a vector bundle of rank q on C, with negative determinant degree, admitting a surjection from f^*T_W or N_f^{sat}. Then the coincidence set and bundles in Proposition 2.3 exist. If the restriction to F, respectively $\mathcal{K}$, has generic rank k, then

$$\deg\det\mathcal{Q} \geq (2k-q)e \geq -qe. \tag{2.4}$$

If $\deg\det\mathcal{Q} < -(q-2)e$, the surjection factors through D and $\ell \geq q+1$. If $\deg\det\mathcal{Q} = -qe$, then $\mathcal{Q} = M^{\oplus q}$. If f takes values in the stable locus of the original equal weight quotient with $n = 7, 8$ and $q = 2$, we have $\ell = 3$, $\mathcal{Q} = M^{\oplus 2}$, and $e = a/\varepsilon_n$, where $\varepsilon_7 = 1$, $\varepsilon_8 = 2$. On that stable locus for $n = 6$, no such quotient of rank 2 exists.

Proof. Negative determinant degree requires a repeated negative section: stalkwise evaluation removes a single occurrence. Nonzero minors give determinant degree at least ke for the saturated image of F, or $\mathcal{K}$; the quotient generated by D contributes at least $-(q-k)e$. This proves (2.4) and the strict case. Equality makes $M^{-1}\mathcal{Q}$ globally generated of determinant degree zero, hence trivial. For $q = 2$, negativity gives $k = 0$ and $\ell \geq 3$. Original equal weight stability forces $\ell = 3$ and $D_j = 0$ for $n = 7, 8$, so $D \twoheadrightarrow \mathcal{Q}$ is an isomorphism and (2.3) gives e; $n = 6$ forbids $\ell \geq 3$. □

2.2 Effective curve classes without a Mori theorem for moduli spaces

For the equal weight quotients X_n, with $\mu_i = 2/n$, put $s_n = \lfloor (n-1)/2 \rfloor$ and $\varepsilon_n = n - 2s_n$.

Proposition 2.5. The smooth stable quotient X_n^{s} is simply connected, and the descent H of $\mathcal{O}_{P_n}(2, \ldots, 2)$ is ample and equals $-K$. For odd n, we have $X_n = X_n^{\mathrm{s}}$ and $\mathrm{Pic}(X_n) = \{b \in \mathbb{Z}^n : \sum_i b_i \in 2\mathbb{Z}\}$, $H = (2, \ldots, 2)$. For even $n = 2h$, the descent J of $\mathcal{O}(1, \ldots, 1)$ is very ample, $\mathrm{Pic}(X_n) = \mathbb{Z}J$, and $H = 2J = -K_{X_n}$. The smooth locus equals the stable locus; each of the $\frac{1}{2}\binom{n}{h}$ singularities is the cone over $\mathbb{P}^{h-2} \times \mathbb{P}^{h-2}$ with polarization $\mathcal{O}(1,1)$. Its link is simply connected, and its two small resolutions are crepant. For curves $f : C \to X_n^{\mathrm{s}}$, Proposition 2.2 gives $\ell \leq s_n$, $a \geq \varepsilon_n e$, and vanishing whenever $\deg Q < -a/\varepsilon_n$.

Proof. General position and the homotopy sequence of the principal bundle give simple connectedness. The character $(-1)^{\sum b_i}$ gives odd descent. Even descent requires $\sum_{i\in I} b_i = \sum_{i\notin I} b_i$ at every closed orbit, forcing constant multidegree; perfect matching brackets give very ampleness [13, Theorem 2.3]. The tangent sequence gives $H = -K$. Lemma 2.1, the primitive circle class $(1,1)$, and equal factor dimensions give the cones, links, and crepancy. Stability and (2.3) give the curve bounds. □

Proposition 2.6. Let $n = 2h+1$, and let E be the line bundle of multidegree b on X_n. Set $q = \frac{1}{2}\sum_i b_i$ and $d_I = q - \sum_{i\in I} b_i$ for $|I| = h$. Then $Z_I = \mathbb{P}^{h-1}$, $M_I = \mathcal{O}(-1)$, $H|_{Z_I} = \mathcal{O}(1)$, $E|_{Z_I} = \mathcal{O}(d_I)$. The conditions $d_I \geq 0$ for every I, global generation of E, and nefness of E are equivalent. If ℓ_I is a line in Z_I, then $\overline{\mathrm{NE}}(X_n) = \sum_I \mathbb{R}_{\geq 0}\ell_I$. Every ample integral E has $E - H$ globally generated.

Proof. The affine quotient at infinity gives $\mathbb{P}((\mathbb{C}^{h+1}/\mathbb{C}\mathbf{1})^\vee)$, represented by $\mathcal{O}\oplus\mathcal{O}(1)$, proving the restrictions. Partitioning the other labels gives $b_j = d_I + d_{I'} \geq 0$. List b_i copies of label i consecutively in a circle of length $2q$, and pair opposite positions. The inequalities prevent coincident pairs, so bracket products generate E; $q = 0$ is immediate. Pass to real classes by clearing denominators and limits. Since $H\ell_I = 1$, cone duality gives the Mori cone; ampleness gives $d_I \geq 1$. □

Proposition 2.7. For equal weights, if $|I| = \ell$ and $n > 2(\ell+1)$, there is a complete embedded $\mathbb{P}^1 \subset Z_I^{\mathrm{s}}$ on which M_I is trivial. Thus no finite surjection onto Z_I^{s} can make M_I^{-1} ample.

Proof. Moving one outside label with the others distinct and fixed gives an embedded stable $\mathbb{P}^1$, since $\ell + 1 < n/2$. Three fixed labels trivialize the common tangent line, whose pullback to every complete curve above it still has degree zero. □

Proposition 2.8. The cone $\overline{\mathrm{NE}}(W)$ is generated by coordinate curves γ_i, moving a single mark with all pairs containing that mark allowable, and by lines β_J in the positive-dimensional closed loci

$$Z_J = \mathbb{P}^{n-|J|-2}, \qquad \nu(J) < 1, \qquad \nu(J) + \nu_j > 1 \quad (j \notin J).$$

For these curves,

$$-K_W \cdot \gamma_i = 2, \qquad -K_W \cdot \beta_J = n - 2|J|, \qquad N_{Z_J/W} = \mathcal{O}(-1)^{\oplus(|J|-1)}.$$

More precisely, if all $m_i \geq 0$, then

$$[f_*C] = \tfrac{1}{2}\sum_i m_i[\gamma_i].$$

If I is the set of negative sections, possibly of size one, there is a complete rational curve β_I with marked degrees -1 on I and 1 elsewhere, and

$$[f_*C] = e[\beta_I] + \sum_{j\notin I} \deg D_j[\gamma_j], \qquad [\beta_I] = [\beta_J] + \sum_{j\in J\setminus I}[\gamma_j] \tag{2.5}$$

for every maximal short set $J \supset I$. Every coordinate curve with positive coefficient in these formulas exists.

Proof. The M_i generate $\mathrm{Pic}(W)_{\mathbb{Q}}$. Pull a rational section of a line bundle B to the principal bundle and close its invariant divisor in $(\mathbb{P}^1)^n$. Its bundle is $\mathcal{O}(b_1,\dots,b_n)$; its linearization agrees with the standard $\mathrm{SL}(2,\mathbb{C})$-linearization because $\mathrm{SL}(2,\mathbb{C})$ has no characters. Squaring and descending gives $B^2 = \bigotimes_i M_i^{b_i}$. Thus marked degrees determine numerical curve classes.

If $m_i > 0$ and all degrees are nonnegative, each distinct section has a bracket of positive degree with i, so the corresponding pair actually collides and must be allowable. Identical sections also form an allowable pair by stability. Moving i, keeping the other marks general, therefore gives γ_i, with degrees $2\delta_{ij}$. This proves the first formula.

In the negative case all outside pairs are allowable, by their positive bracket degrees or stability. On $\mathbb{P}^1$, take the quotient of $\mathcal{O}\oplus\mathcal{O}(1)$ onto $\mathcal{O}$ for I, and the quotients $(s_j, 1)$ onto $\mathcal{O}(1)$ elsewhere, with s_j general linear forms. The common section is disjoint from the others; every other collision is an allowable pair. This constructs β_I. If $\deg D_j > 0$, the original collision of j with I is stable, and all pairs containing j are allowable; hence γ_j exists. Comparing marked degrees proves the first equality in (2.5).

For J maximal short, put its common section at infinity. Every proper subset of the outside labels is short, since $\nu(J) + \nu_j > 1$. The affine coordinates, modulo translations and scalars, give $Z_J = \mathbb{P}^{n-|J|-2}$ and $M_J = \mathcal{O}(-1)$, whence the normal bundle and degree against $-K_W$. In (2.5) the outside labels of J are at least three: their total weight exceeds one, but every pair

outside I is allowable. Thus this Z_J has positive dimension and β_J exists. All pairs containing $j \in J \setminus I$ are allowable, either inside J or outside I. Comparison of marked degrees proves the second equality. Every curve class is now in the cone on the indicated finite set, which proves the assertion after closure. In particular no line in a zero-dimensional coincidence locus has been included. □

Corollary 2.9. If $W \neq \mathbb{P}^d$, every ample line bundle A_X on X_μ satisfies

$$K_W + (n-4)\pi^* A_X \text{ is nef.}$$

Proof. Only negative extremal rays require checking. They are noncontracted because K_W is π-nef, so $\pi^* A_X$ has positive integral degree on their indicated generators. A maximal singleton would give $Z_J = W = \mathbb{P}^d$; otherwise their degrees against $-K_W$ are at most $n-4$. □

Lemma 2.10. Let $A = \pi^* A_X$ for an ample line bundle on X_μ, and suppose the general point of $f(C)$ is in X_μ^{s}. Put $b = \deg f^* A$. Then $a \leq (n-2)b$; if all marked degrees are nonnegative, $a \leq 2b$. A nonzero map $f^* T_W \to Q$ with $\deg Q = q \leq 0$ implies $a \leq (n-4)b$. More generally, if such a map satisfies

$$a - (n-4)q > 2b,$$

then a repeated negative section exists, $q < e$, and $e + \delta \leq b$. The splitting of Proposition 2.3 applies, and all maps to lines of degree less than e vanish on F.

Proof. The perturbation in Lemma 2.1 does not make an original pair of weight at least one allowable. All required γ_i in the preceding proof therefore lie in the original stable locus. Also $\mu(I) < 1$, and the constructed β_I has only allowable outside pair collisions and never meets its common section. Thus A has positive integral degree on each curve appearing in the first decomposition of (2.5). The decompositions give $a \leq 2b$ or $e + \delta \leq b$, respectively, and hence $a \leq (n-2)b$. For $q \leq 0$, all outside lines of degree at least e map trivially to Q, and diagonal evaluation requires $\ell \geq 2$. This gives $a = (n-2\ell)e + 2\delta \leq (n-4)b$. For the last assertion, nonnegative marked degrees give no map when $q < 0$, and $a - (n-4)q \leq 2b$ when $q \geq 0$. If a negative degree exists but $q \geq e$, then $a \leq (n-2)e + 2\delta \leq (n-4)q + 2b$, again a contradiction. Thus $q < e$, the outside columns vanish, and diagonal evaluation gives $\ell \geq 2$. Proposition 2.3 concludes. □

2.3 The original arithmetic group and its automorphic bundle

Suppose $X_\mu \simeq (\mathbb{B}^d/\Gamma)^{\mathrm{BB}}$ with Γ arithmetic. We keep this group throughout. The ball parametrizes negative lines, and $\mathcal{L}$ denotes the tautological automorphic bundle, with rational class L. The orbifold canonical class is NL. A normalized reflection acts identically on its mirror.

Proposition 2.11. The original Γ, all finite point stabilizers, and all full cusp stabilizers are generated by ball reflections. There is a Hermitian space (V, h) of dimension N over a CM extension K/F, of signature $(1, d)$ at the distinguished embedding and definite at all other real embeddings of F. The normalized reflections generate a group $\widehat{\Gamma} \subset \mathrm{U}(h, K)$ preserving one full $\mathcal{O}_K$-lattice and mapping onto Γ, with finite scalar kernel $C_0 \subset \mu(K)$. For $c = |C_0|$, $\mathcal{L}^c$ is an actual ample line bundle on X_μ. In the nonuniform case $F = \mathbb{Q}$ and $c \mid w_K = |\mu(K)| \in \{2, 4, 6\}$.

Proof. The genuine wall link is simply connected: it is the circle bundle of primitive Chern class $(1, 1)$ over a product of two positive-dimensional projective spaces. Its second Betti number is one, but a finite linear quotient link is a rational homology sphere. Thus a genuine wall vertex cannot be an interior ball quotient point. The rational smooth projective variety W is simply connected, by weak factorization [1]. Removing the cusp inverse images, of codimension at

least two, preserves its fundamental group. Brown–Higgins [9, Propositions 1.8 and 2.8(iii)] now makes Γ generated by stabilizers. Interior smoothness and Chevalley–Shephard–Todd [11, Theorem 1.3] show that they are generated by reflections. The local cusp argument in Lemma 2.16, which does not use arithmeticity, gives the same assertion for the full cusp groups.

Choose a normal torsion-free subgroup of Γ of index b_0. Reflection orders divide b_0, so determinants of normalized lifts lie in μ_{b_0}, and their scalar kernel lies in μ_{Nb_0}. Their generated group is consequently a discrete finite central extension of the original arithmetic lattice. Apply Stover's first-type theorem [28, Corollary 4.4] to this group generated by reflections; its determinant-one subgroup need not itself be generated by reflections. This gives the stated Hermitian arithmetic datum.

For completeness, the entire projective group is rational over F, not just a finite-index subgroup. Put $G = \mathrm{PGU}(h)$, $\mathfrak{g} = \mathrm{Lie}(G)$, and choose an arithmetic $\Delta \subset G(F)$ commensurable with Γ. For $\gamma \in \Gamma$, suitable finite-index subgroups $\Delta_1, \Delta_2 \subset \Delta$ satisfy $\gamma\Delta_1\gamma^{-1} = \Delta_2$. Borel density [6] and absolute irreducibility of the adjoint representation give $\operatorname{span}_F \mathrm{Ad}(\Delta_i) = \mathrm{End}_F(\mathfrak{g})$. Skolem–Noether therefore writes $\mathrm{Ad}(\gamma) = rB$, with $B \in \mathrm{GL}_F(\mathfrak{g})$ and r real at the distinguished place. Preservation of the bracket gives $B[X, Y] = r[BX, BY]$; a nonzero component gives $r \in F$. Thus $\gamma \in G(F)$. Hilbert 90 for the center $\mathrm{Res}_{K/F}\mathbb{G}_m$ of $\mathrm{GU}(h)$ supplies a K-rational similitude representative. For a reflection its mirror eigenvalue has multiplicity d; because $2d > N$, that eigenvalue lies in K. Dividing by it gives a normalized unitary reflection over K. The normalized lifts preserve a full lattice: take a finite-index normal integral subgroup and sum the finitely many translates of its invariant lattice. This uses the same lattice for the full group.

Interior stabilizers act on the tautological fiber through C_0, so $\mathcal{L}^c$ descends to the coarse interior. At a cusp let ℓ generate its rational isotropic line. Every normalized cusp reflection fixes ℓ. With h linear in its second variable,

$$s([v]) = v/h(\ell, v)$$

is a nonzero tautological frame whose c-th power is invariant under the full cusp group. On a small cusp neighborhood U, glue $\mathcal{O}_U$ to the interior bundle by $1 \mapsto s^c$ on the punctured neighborhood. This first constructs an actual holomorphic line bundle M on X_μ. A sufficiently divisible ample Baily–Borel bundle [4], killing its finite determinant character, agrees with M^r on the interior for some $r > 0$. The isomorphism and its inverse extend across the finitely many cusps by normality, since $d \geq 2$. They remain inverse; GAGA and the ampleness of that Baily–Borel bundle show that M is algebraic and ample. Finally, a nonzero isotropic K-vector remains isotropic at every embedding. In the nonuniform case this excludes all other real places of F, so $F = \mathbb{Q}$. □

2.4 Fixed balls and persistent factors

Lemma 2.12. Let T be a finite reflection group in Γ, let $V_0 \subset V$ be its common fixed space, and put $E = V_0^{\perp}$, $s = \dim E$, $m = d - s \geq 1$. Assume $\mathbb{B}(V_0) \neq \varnothing$ and T is its full generic stabilizer. The effective image Π of its full normalizer is arithmetic and gives a finite map

$$j : (\mathbb{B}(V_0)/\Pi)^{\mathrm{BB}} \longrightarrow X_\mu$$

birational onto its image. The normalizer has finite image on E, after retaining C_0. If every full interior stabilizer along this ball is $T \times J_x$, with J_x acting trivially on E, the interior map is immersed and its ordinary normal bundle has components indexed by the basic polynomial invariant degrees q of T. A component of multiplicity u_q has finite projective monodromy and first Chern class $-qu_q j^*L$. The same assertion applies over the original stable open subset of W.

Proof. The spaces V_0, E are defined over K. Its matrices on E preserve the same $\mathcal{O}_K$-lattice and are bounded at every Archimedean embedding. Minkowski discreteness therefore makes that image finite. The algebraic stabilizer of their orthogonal decomposition has a compact factor on E; its arithmetic subgroup projects to an arithmetic lattice on the ball by Borel–Harish-Chandra [7]. The effective full normalizer is now a discrete overgroup of this lattice and hence has finite index over it. After a finite-index passage for this construction, a rational flag of hyperplanes and the local extension in [20, Section 5] give the cusp extension; the flag lies in $E^{\perp}$, and all its normals are positive. When $F \neq \mathbb{Q}$ there are no cusps and properness is immediate. The extended map is proper because its source is projective; its interior fibers are discrete by discreteness and the local finite point groups, hence compact and finite. There are finitely many cusp preimages, so the extended map is finite. A generic identification carries the full generic stabilizer T to itself and therefore belongs to the full normalizer, proving birationality. Under the product hypothesis the local coarse map is $y \mapsto (0, y)$ in invariant coordinates for T. The linear indecomposable invariants of degree q transform by finite matrices times the inverse q-th automorphic factor. Decomposable invariants have zero normal differential. This gives the stated ordinary normal bundles, including their finite characters. □

Lemma 2.13. On a weighted quotient of $m \geq 5$ labels, a marked relative tangent line M_* with an antiample rational Cartier extension forces

$$Z = \mathbb{P}^{m-3}, \qquad M_* = \mathcal{O}(-1).$$

Proof. First take chamber weights. If $\mu_* + \mu_j < 1$, keep all marks except j at distinct fixed points and move j over $\mathbb{P}^1$. At each unstable pair collision j, k, blow up that collision and contract the original fiber. The complementary marks then coincide with weight $2 - \mu_j - \mu_k < 1$, so the completed family is stable. The centers avoid $*$; each contraction increases its self-intersection by one. Thus M_* has nonnegative degree on a complete nonconstant curve, a contradiction. All pairs with $*$ are therefore long. Putting $*$ at infinity and dividing affine coordinates by translations and scalars gives $\mathbb{P}^{m-3}$, with $M_* = \mathcal{O}(-1)$. At a genuine wall vertex the local class group is $\mathrm{Pic}(\mathbb{P}^{a-2} \times \mathbb{P}^{b-2})/\mathbb{Z}[\mathcal{O}(1,1)] \simeq \mathbb{Z}$. The marked character is its non-torsion generator, with sign determined by the side containing $*$. It cannot have a rational Cartier extension. At a wall with a zero-dimensional factor the chamber retaining that factor is locally isomorphic to the wall quotient. The simultaneous perturbation away from equal weights makes these choices consistently, and the chamber argument applies. □

Proposition 2.14. Under the product hypothesis of Lemma 2.12, suppose T is irreducible on E and $s \leq d - 2$. Its image is a closed coincidence locus $Z_I = \mathbb{P}^{d-s}$, with $|I| = s + 1$. More precisely, the fixed compactification is isomorphic to the GIT quotient of the original merged weights

$$\mu^I = (\mu(I), (\mu_j)_{j \notin I}), \qquad 0 < \mu(I) < 1, \qquad \sum \mu_i^I = 2,$$

on $n - s$ labels, including its stability walls. Its automorphic rational bundle is the restriction of the original $\mathcal{L}$. All its basic polynomial invariant degrees equal one number δ, and the marked tangent line has antiample rational Cartier extension with class $-\delta L|_{Z_I}$. In particular $s \leq 2$. For a compact lattice an irreducible persistent factor of rank $d - 1 \geq 3$ is also impossible. For equal weights, a factor satisfying the product hypothesis is also impossible when $n > 2(s+2)$.

Proof. In the fixed compactification choose a general complete intersection curve avoiding the finite boundary and wall vertices, and not contained in any proper inverse image of a coincidence locus. After a finite cover its normal bundle is a sum of negative lines with degrees $-q_\alpha t$, $t > 0$. Proposition 2.2 factors its stalkwise normal quotient through $M_I^{\oplus(|I|-1)}$, so $|I|-1 \geq s$. The curve selection implies that the entire image lies in Z_I; dimension gives the reverse inequality. The selected curve lies in the original stable open set, so $\mu(I) < 1$. Thus $|I| - 1 = s$, and exactness

of $\mathrm{SL}(2,\mathbb{C})$-invariants identifies the closed diagonal image with the normal GIT quotient of the original merged weights μ^I. Finite birationality then identifies the fixed compactification with this quotient. Negative lines in V_0 are the same lines in V, so their actual automorphic rational bundles agree. The ordinary normal bundle is $M_I^{\oplus s}$. Its semistability on the curve forces all the q_α to coincide. The reflection classification [11, Table 1] has no irreducible example of equal basic degrees in rank at least 3.

On the stable interior, the determinant transition differs from the automorphic factor of degree $-s\delta$ by a finite character. For an integer $r > 0$ killing that character and all scalar denominators, there is an actual isomorphism

$$M_I^{sr} \simeq (\mathcal{L}|_{Z_I})^{-s\delta r}.$$

Since $\dim Z_I \geq 2$, this constructs the rational Cartier extension across the finite omitted set; its inverse is ample. Lemma 2.13 gives the projective space assertion. In the compact case of rank $d-1$, use the whole fixed curve, which has no boundary, for the same quotient onto a line of negative degree and semistability argument. The resulting contradiction of equal basic degrees occurs before any use of Lemma 2.13; that lemma requires at least five merged labels. For equal weights, Proposition 2.7 excludes the antiample extension when $n > 2(s+2)$. □

Lemma 2.15. In the nonuniform case, a finite reflection group of rank $s \leq d-2$ embeds, with its same reflection representation and orders, in the horizontal group of an actual full cusp stabilizer of the original Γ.

Proof. Its fixed Hermitian space over the imaginary quadratic field has dimension at least three. The underlying indefinite rational quadratic form has at least six variables, so Meyer [27, Chapter IV, Section 3.2] supplies a fixed rational isotropic vector. The Heisenberg lattice is torsion-free, hence the finite group injects in the horizontal quotient, with unchanged nonzero roots and eigenvalues. This argument does not apply in rank $d-1$. □

2.5 The full central period and Segre cusps

These local arguments apply to any nonuniform ball lattice. For a full cusp group P, let N_P be its intersection with the Heisenberg group, $F_P = P/N_P$, and

$$P \cap Z = N_P \cap Z = c_*\mathbb{Z}, \qquad q_* = \exp(2\pi i w/c_*).$$

The affine action on $A_P = \mathbb{C}^r/\Lambda_P$ is faithful and has no nonidentity pure translations; no common fixed point is assumed. The actual translation multipliers $\exp(2\pi i a_\lambda(z)/c_*)$ define a negative line bundle $\mathcal{V}_*$, with dual ample bundle M_* and its induced F_P-linearization. The invariant metric is $|q_*|^2\exp(4\pi\|z\|^2/c_*)$, of negative curvature. The cusp contracts the zero section of $\mathcal{V}_*/F_P$, with ring

$$R_P = \bigoplus_{k\geq 0} H^0(A_P, M_*^k)^{F_P}. \tag{2.6}$$

It is enough to use a high Veronese to construct the contraction; generation in degree one is not assumed. This is the horoball germ [22, Sections 2.1–2.3] and, in the arithmetic case, the Baily–Borel germ [20, Section 4.3]. For a neat subgroup with central period $c_0 = e_0 c_*$, its bundle satisfies $q_* = q_0^{e_0}$ and $f_0^* M_* = M_0^{e_0}$. Both the constant multipliers and the remaining finite action are retained in these identities.

Lemma 2.16. If the punctured cusp germ is smooth and simply connected, P is generated by ball reflections and F_P is a reflection group. A smooth cusp has irreducible horizontal representation. A Segre cusp $\mathrm{Cone}(\mathbb{P}^a \times \mathbb{P}^b, \mathcal{O}(1,1))$, $a, b \geq 1$, has exactly two irreducible horizontal factors, of ranks a, b. An irreducible factor, with its actual polarized lift, is strongly crystallographic and its ring of invariant sections is polynomial.

Proof. Apply Brown–Higgins to a contractible deep horoball and then Chevalley–Shephard–Todd at its smooth points. The first Betti number of the link excludes fixed horizontal vectors. Write the orthogonal irreducible factors as F_j of ranks r_j. Their invariant cohomology is $\mathbb{C}[\omega_j]_j/(\omega_j^{r_j+1})_j$, $\deg \omega_j = 2$, by [14, Theorem 1.2]. The Gysin sequence for the negative bundle gives $b_{2i} = h_i - h_{i-1}$, where $\sum h_i t^i = \prod_j (1+t+\cdots+t^{r_j})$, in the first half of the link dimensions. Comparison with the spherical link or with the link of the Segre cone gives, respectively, one factor or two factors of ranks a, b. Rains [25, Proposition 2.6] applies to the actual bundle lift, making the action strongly crystallographic; [25, Theorem 2.7] gives the polynomial ring. Evaluation and averaging on free horizontal orbits give invariant sections in every sufficiently large degree. Thus their generator degrees have greatest common divisor one in this actual grading. If a fiber kernel μ_g is first removed, the correct new coordinate is $q_{\mathrm{new}} = q_{\mathrm{old}}^g$, with period $c_{\mathrm{new}} = c_{\mathrm{old}}/g$, and generator degrees divided by g. □

Lemma 2.17. Let $(F_P)_x$ be the full stabilizer of $x \in A_P$, and let χ_x be its character on $(\mathcal{V}_*)_x$. Its kernel T lifts to the full finite stabilizer of a nonzero point over x in a sufficiently deep horoball. If T has no horizontal fixed vector, its fixed ball is a disk ending at this cusp. For the disk modulo its full normalizer, the coarse end coordinate is

$$Q = q_*^{\kappa_x}, \qquad \kappa_x = |\operatorname{im} \chi_x|.$$

Proof. Choose a lift of a nonzero fiber point. Unique correction by the freely acting Heisenberg lattice lifts exactly $\ker \chi_x$. Its fixed Hermitian plane gives the disk when the horizontal fixed space is zero. The full central translations retain the period c_*, and the residual stabilizer acts by χ_x, giving the coordinate Q. □

Lemma 2.18. Let $\mathbb{C}^*$ act on $\mathbb{C}^{p+q}$, $p, q \geq 2$, with positive weights u_i and negative weights $-v_j$. Assume that the greatest common divisor after deleting any one weight is one. If the affine quotient is smooth off the vertex and has embedding dimension pq there, all weights equal one.

Proof. At the orbit $X_i Y_j \neq 0$ the transverse stabilizer is $\mu_{\gcd(u_i, v_j)}$. The deletion condition excludes ineffective elements and pseudoreflections, so smoothness forces $\gcd(u_i, v_j) = 1$. The pq pair monomials $X_i^{v_j} Y_j^{u_i}$ are therefore indecomposable. If $u_i > 1$, take $j \neq k$ and $1 \leq e < u_i$ with $v_j + e v_k \equiv 0 \pmod{u_i}$. The invariant $X_i^{(v_j + e v_k)/u_i} Y_j Y_k^e$ has an indecomposable factor divisible by none of the pair monomials, contradicting the embedding dimension. Exchange signs to conclude. □

Theorem 2.19. For a Segre cusp as in Lemma 2.16, there are actual products $A_P = A_1 \times A_2$, $F_P = F_1 \times F_2$, of dimensions a, b. The subgroup P_j generated by all reflections whose roots lie in the j-th factor has the same least positive central period as P. For this period, its ring of invariant sections is polynomial on $r_j + 1$ generators of degree one, where $r_1 = a, r_2 = b$. The contraction is that of $\mathcal{O}_{\mathbb{P}^a \times \mathbb{P}^b}(-1,-1)$. For $r_j \geq 3$, F_j is S_{r_j+1} or $G(m_j, 1, r_j)$, with $m_j \in \{2, 3, 4, 6\}$. The full F_P occurs as a finite stabilizer arbitrarily deep in the cusp.

Proof. A reflection's displacement and central multiplier belong to its root factor, and orthogonal reflections commute. Since P is generated by reflections, $P = P_1 P_2$, $N_P = N_1 N_2$, and $\Lambda_P = \Lambda_1 \oplus \Lambda_2$. This gives the actual product of abelian varieties. Write the factor central periods as $c_j = d_j c_*$, so $\gcd(d_1, d_2) = 1$. Rains gives $R_1 = \mathbb{C}[x_0, \ldots, x_a]$ and $R_2 = \mathbb{C}[y_0, \ldots, y_b]$, with degrees α_i, β_j of greatest common divisor one. Including constants and characters in the multipliers gives

$$q_* = q_1^{d_1} q_2^{d_2}, \qquad M_* = M_1^{d_1} \boxtimes M_2^{d_2}, \qquad R_P = \bigoplus_{k \geq 0} (R_1)_{d_1 k} \otimes (R_2)_{d_2 k}.$$

Thus R_P is the torus invariant ring of weights $d_2\alpha_i, -d_1\beta_j$.

The residual $\mu_{d_j} = P_jZ_*/P_j$ acts freely off the vertex of the j-th factor cone. Otherwise combine a fixed factor point with a free horizontal orbit in the other factor. Its finite stabilizer lies in P_jZ_*, but smoothness of the punctured germ makes it generated by reflections in P_j, a contradiction. On the coordinate axis $x_i \neq 0$, the residual group μ_{d_1} has stabilizer of order $\gcd(d_1, \alpha_i)$, since its action is $x_i \mapsto \zeta^{\alpha_i}x_i$. Freeness therefore gives $\gcd(d_1, \alpha_i) = 1$ for every i, and similarly $\gcd(d_2, \beta_j) = 1$ for every j. After deleting the positive weight $d_2\alpha_i$, the remaining weights have greatest common divisor

$$\gcd\left(d_2 \gcd_{k\neq i} \alpha_k, d_1 \gcd_j \beta_j\right) = \gcd\left(d_2 \gcd_{k\neq i} \alpha_k, d_1\right) = 1.$$

Deleting a negative weight gives the same conclusion with the factors exchanged. This argument includes $a = 1$ or $b = 1$, because each factor still has a generator after one weight is deleted. The analytic germ comparison gives smoothness away from the vertex and embedding dimension $(a + 1)(b + 1)$. Lemma 2.18 forces $d_1 = d_2 = \alpha_i = \beta_j = 1$. Consequently $A_j/F_j = \mathbb{P}^{r_j}$, all stabilizer fiber characters are trivial, and the negative bundles descend to $\mathcal{O}(-1)$ and $\mathcal{O}(-1, -1)$. The faithful affine actions have no pure translations. The classification of smooth quotients [3, Theorem 1.1(3)] gives a common fixed point, since the generators have fixed points; [2, Theorem 1.1] gives the indicated irreducible groups. Their product has trivial fiber character. Lemma 2.17, followed by central imaginary translation, realizes the full group arbitrarily deep. □

3 One comparison for all fixed curves

The common comparisons in this section apply for $n \geq 6$, subject to the hypotheses of each statement. Put $d = n-3$, $N = d+1$, $H = -K_W$, $a = H \cdot f_*C$, and $t = L \cdot f_*C > 0$. Here W is the smooth chamber model of Lemma 2.1, and $f : C \to W$ comes from the fixed ball quotient of Lemma 2.12 in the original lattice. We do not assume $a > 0$, immersion, or separate descent of an automorphic factor and a finite character. All cusp section degrees below use the least positive central period of the full cusp group.

3.1 The determinant and the selected inverse direction

Proposition 3.1. Let X be the Baily–Borel compactification of an arithmetic ball quotient of dimension d, and let K be the full generic stabilizer of a fixed ball of dimension $m = d-k \geq 1$. Assume that K is generated by reflections and has rank k. Let $f : Y \to X$ be the finite morphism from the compactified quotient by its full effective normalizer in the original lattice. Assume that K_X is $\mathbb{Q}$-Cartier near $f(Y)$, and that the general point of every interior divisor of Y maps to the smooth locus of X. Put $H = -K_X$, and write D_G for the sum of the basic invariant degrees of a reflection group G. There is an effective $\mathbb{Q}$-divisor on Y with class

$$f^*H + (D_K + m + 1)f^*L.$$

Its interior support consists of the divisors with a larger full stabilizer. For $m > 1$, let Z_K be the sum of all reduced images of the specified representation type, and let $n_{K,J}$ count axes in the normal representation of J with full pointwise group of type K. Then

$$(H + (D_K + m + 1)L)Z_K = \sum_{\operatorname{rank} J = k+1} (D_J - D_K - 1)\frac{|K|n_{K,J}}{|J|}Z_J.$$

There is no cusp divisor term when $m > 1$.

For $m = 1$, write $Y = C$, let $S \subset C$ be the cusp set, and assume that X is smooth near $f(C)$, with polynomial rings of invariant sections at its ends. Put $a = \deg f^*H$, $t = \deg f^*L$, and $e_y = |N_{J_y}(K_y)|/|K_y|$, where J_y is the full interior stabilizer and K_y the pointwise stabilizer of the corresponding local axis. At an end c, let $b_{c,1}, \dots, b_{c,d}$ be the generator degrees measured by the full least positive central period, and put $A_c = \sum_i b_{c,i}$. Let κ_c be the order of the full fiber character, so that the coarse curve coordinate and ambient central coordinate satisfy $Q = q^{\kappa_c}$. Then

$$a + (D_K + 2)t = \sum_{y \in C \setminus S,\ J_y \neq K_y} \frac{D_{J_y} - D_K - 1}{e_y} + \sum_{c \in S} \frac{A_c}{\kappa_c}. \tag{3.1}$$

No immersion or separate descent of an invariant line is required. If H is nef, every component with $m > 1$ contains an interior stratum of rank $k + 1$.

Proof. Lemma 2.12 gives Y and the finite coefficient representations on invariant normal directions. Indecomposable invariants of degree e transform by $\mathcal{L}^{-e}$ with their finite representation; terms in the square of the invariant ideal disappear from the normal differential. Repeated degrees are kept in one vector bundle. The tangent determinant along the fixed ball has automorphic factor $\mathcal{L}^{-(m+1)}$ and a finite character. Thus the source determinant of the quotient differential is $\mathcal{L}^{-(D_K+m+1)}$ with a finite character. Take a sufficiently divisible power N to remove this character, make NK_X Cartier, and make $\mathcal{L}^N$ descend to the actual Baily–Borel line bundle. The determinant then gives a nonzero section of the line bundle

$$f^*\mathcal{O}_X(-NK_X) \otimes (f^*\mathcal{L}^N)^{\otimes(D_K+m+1)}.$$

The finite coefficient representations are removed on a normal neat cover; their automorphic factors are retained.

At an interior divisor with full group J, discard the common fixed directions and write u for the remaining axis coordinate, η_i for the basic invariants of K, and F_j for those of J. Homogeneity and generic invertibility give

$$\det \frac{\partial(F_1, \dots, F_{k+1})}{\partial(u, \eta_1, \dots, \eta_k)}(u, 0) = c_0 u^{D_J - D_K - 1}, \qquad c_0 \neq 0.$$

The residual axis group has order $e = |N_J(K)|/|K|$, so the divisor coefficient after division by N is $(D_J - D_K - 1)/e$. Changes of invariant coordinates multiply the determinant by a unit, and the tangent column is included. The section is regular in codimension one away from cusps and extends across the remaining interior subset by normality of Y. When $m > 1$, the cusp set also has codimension at least two, so the same extension applies. In a toroidal cover, a boundary divisor maps to a point and has zero pushforward as a divisor on Y.

For $m = 1$, extend in the logarithmic radial frame at each end. The frame on the full punctured cusp in Proposition 2.11 introduces no power of its central coordinate. At an end the contraction and its constant differential are

$$U_j = q^{b_{c,j}} \bar{\theta}_j(\boldsymbol{\eta}), \qquad J_c = [(b_{c,j}\bar{\theta}_j(0))_j, D_{\boldsymbol{\eta}}\bar{\theta}(0)]. \tag{3.2}$$

The differential is $\operatorname{diag}(q^{b_{c,j}})J_c$, and J_c is invertible by smoothness of the punctured quotient with full stabilizer K. If $Q = (q')^r$ on the neat cover, its determinant order is rA_c/κ_c, giving A_c/κ_c after descent. This proves effectivity and the curve formula. For $m > 1$, summing the local axis orbits gives $\sum 1/e = |K|n_{K,J}/|J|$; pushing the divisors forward proves the cycle formula. Finally, $(f^*H + (D_K + m + 1)f^*L)(f^*L)^{m-1} > 0$ if H is nef, so the effective divisor is nonzero and contains the asserted interior stratum. □

Lemma 3.2. For $r \geq 5$, the linear normalizer of $G(m, p, r)$, $m \geq 2$, $p \mid m$, lies in $\mathbb{C}^*G(m, 1, r)$ and preserves its individual symmetric and product invariant lines.

Proof. The bounds $m^k k!(r-k)!/p$, $k>0$, and $r!$, $k=0$, for point stabilizers are logarithmically convex. The strict inequality $m^{r-1} > pr$ identifies the coordinate axes by maximal stabilizers. A normalizer permutes these axes; phased transpositions then force all diagonal ratios into μ_m. □

Lemma 3.3. If $u \mapsto \zeta_e u$ acts on a line by ζ_e^a, $0 \le a < e$, its coarse extension has degree equal to its orbifold degree minus a/e.

Proof. The invariant frame is u^a times a function of u^e. □

For $r = d-1 \ge 3$, $m \ge 2$, and $p \mid m$, put $T = G(m,p,r)$ and

$$f_j = e_j(x_1^m, \dots, x_r^m)\ (1 \le j < r), \quad P_0 = (x_1 \cdots x_r)^{m/p}, \quad D = \frac{mr(r-1)}{2} + \frac{rm}{p}.$$

Set $k = rm$ if $p = 1$, and $k = (r-1)m$ otherwise.

Proposition 3.4. Suppose every larger interior group on a fixed disk is $G(m,p,r+1)$, with the prescribed coordinate axis, or $T \times \mu_e$, with μ_e trivial on the normal space. Suppose every end is a smooth cusp whose actual horizontal stabilizer H_c has $\ker \chi_c = T$, $|\operatorname{im} \chi_c| = \kappa_c$. Suppose the full normalizer preserves the degree k indecomposable direction and its effective curve stabilizers fix that direction. Let R count monomial enlargements, s count ends, and $\sigma = \sum(1 - e^{-1})$ count product enlargements. Put

$$B_c = \frac{\sum_i b_{c,i}}{\kappa_c}, \quad \nu_c = \left\lfloor \frac{\max_i b_{c,i}}{\kappa_c} \right\rfloor, \quad \beta = r + \frac{1}{p} - \frac{1}{m}.$$

There is an actual line E_k of degree $-kt$, and

$$2g - 2 + s + (1 - m^{-1})R + \sigma = 2t, \tag{3.3}$$

$$a = \sum_c B_c + \beta R + \sigma - (D+2)t, \tag{3.4}$$

$$0 \ne f^* T_W \longrightarrow E_k(D_R + \sum_c \nu_c [c]), \qquad h = -kt + R + \sum_c \nu_c. \tag{3.5}$$

Proof. On an invariant line of degree q, a transition is $j_\gamma^{-q}\chi_q(\gamma)$. Scalar lifts cancel between the two factors, so their product is effective. A neat cover of degree v, of any genus, computes degree $-qvt$. This proves the assertion for E_k; Riemann–Hurwitz for the disk gives (3.3). Put $\delta = m/p$. The local map at an enlargement is

$$g_1 = u^m + f_1, \quad g_j = f_j + u^m f_{j-1}\ (2 \le j < r), \quad g_r = P_0^p + u^m f_{r-1}, \quad Z = u^\delta P_0. \tag{3.6}$$

Its axis Jacobian is $mu^{rm+\delta-1}$, so Proposition 3.1 gives β. The residual action sends P_0 to $\zeta_m^{-\delta} P_0$, giving the coarse correction $-(1-1/p)R$. For $p > 1$, write $P_0 = u^{m-\delta}\psi(u^m)$. After tangent elimination, the symmetric block has diagonal $v = u^m$ and superdiagonal 1. For $p = 1$, the same follows from $dg_1 = dv + df_1$. The selected inverse row has a simple pole: $\dot{P}_0 = v^{-1}\dot{Z}$, or $\dot{f}_{r-1} = v^{-1}\dot{g}_r$. At a product enlargement there is no normal pole. At a cusp invert the differential in (3.2). Its horizontal rows have pole at most $q^{-\max b_{c,i}}$. The selected trivial character makes their entries functions of $Q = q^{\kappa_c}$, giving ν_c. Equation (3.1) now gives (3.4). □

Whenever a unique degree m line descends, mt is integral. Repeated degrees justify only the greatest common divisor of degrees of lines actually shown to descend.

3.2 Elimination and vector bundle tests

Lemma 3.5. Let the maps under consideration have zero tangent coefficient. Suppose their pullback coefficients $b_i \in H^0(C, QE_i^{-1})$ determine them injectively. At a reduced divisor D, suppose their extension equations come from the upper bidiagonal differential with first row $(a(u), 1, 0, \ldots)$, $a(u) \not\equiv 0$, and remaining diagonal u and superdiagonal 1. At each end c, suppose the differential is $\operatorname{diag}(u^{\nu_i(c)})J(u)$. Assume the exponents are nondecreasing and every leading block at a strict increase is invertible at zero. If

$$\deg QE_i^{-1}(-D - \sum_c \nu_i(c)[c]) < 0 \quad (i < q),$$

and the last such degree is zero, the Hom space has dimension at most one.

Proof. The tangent equation kills the first target coefficient at D. Once preceding source coefficients vanish, the next is divisible by u. The invertible leading blocks at ends give the analogous successive divisibility. Negative degree kills the first $q-1$ coefficients, and degree zero bounds the last by one. □

Lemma 3.6. Suppose $\bigoplus_{i=1}^r \mathcal{O}_{\mathbb{P}^1}(-ij) \to V$, $j \geq 1$, is invertible away from R points and has the bidiagonal matrix just described in its individual invariant frames. Then $V = \bigoplus_{i=1}^r \mathcal{O}(R - ij)$. For $p > 1$, the symmetric block has length $r-1$, and the separate product direction has degree $(R - rj)/p$.

Proof. A map $V \to \mathcal{O}(q)$ has coefficients $b_i \in H^0(\mathcal{O}(q+ij))$ satisfying $b_1 = vc_1$, $b_i = c_{i-1} + vc_i$. Induction kills the coefficients with $q + ij < R$; thereafter evaluation at R points is surjective. Thus its Hom dimension is $\sum_i \max(q + ij + 1 - R, 0)$, which determines the splitting. For the product line add its local simple modification R to the age correction $-(1 - 1/p)R$ and initial degree $-rj/p$. □

Proposition 3.7. Suppose $C = \mathbb{P}^1$ satisfies Proposition 3.4 with every invariant direction preserved. Put $j = mt$, $\rho = \operatorname{lcm}_c \kappa_c$, and $B = \sum_c \max_i b_{c,i}/\kappa_c$. When at most two widths are nontrivial, use the cyclic degree ρ cover $\varrho : C' \to C$ branched at two values containing those ends and avoiding enlargements. Use another end as the second value if possible; let s_0 count auxiliary branch values, and put $B_T = B + s_0(1 - 1/\rho)$. For $\rho = 1$ use the identity cover and $s_0 = 0$. Let $(d_i) = (m, 2m, \ldots, rm)$ for $p = 1$, or $(m, \ldots, (r-1)m, rm/p)$ otherwise. Then $C' = \mathbb{P}^1$, and the following injections have actual integral line degrees:

$$(f\varrho)^* T_W \hookrightarrow \mathcal{O}(\rho B_T + 2) \oplus \bigoplus_i \mathcal{O}(\rho(B_T - d_i t)) \qquad (R = 0), \tag{3.7}$$

$$\varrho^* N_f^{\mathrm{sat}} \hookrightarrow \begin{cases} \bigoplus_{i=1}^r \mathcal{O}(\rho(B + R - ij)), & p = 1, \\ \bigoplus_{i=1}^{r-1} \mathcal{O}(\rho(B + R - ij)) \oplus \mathcal{O}(\rho(B + (R - rj)/p)), & p > 1. \end{cases} \tag{3.8}$$

For $R = 0$, the comparison using the normal bundle allows arbitrarily many nontrivial widths and any finite cover clearing them; its genus need not be zero. The critical divisor of the original map has degree

$$c_f = \sum_c \left(\min_{i : \theta_{c,i}(x_c) \neq 0} b_{c,i}/\kappa_c - 1 \right).$$

If the marked quotient has ℓ coincident labels of common degree $-e < 0$, sort the normalized target degrees as $v_1 \geq \cdots \geq v_q$. A necessary condition is

$$\#\{i : v_i \geq e\} \geq k_0, \qquad A \leq \sum_{i=1}^{k_0} v_i, \tag{3.9}$$

where $(k_0, A) = (n - \ell - 2, a + (\ell - 1)e)$ for the tangent test, and $(k_0, A) = (n - \ell - 3, a - 2 - c_f + (\ell - 1)e)$ for the comparison using the normal bundle. Failure at the least allowed e excludes every larger e for the same record.

Proof. Riemann–Hurwitz makes the two point cover rational. At an end of ramification index e_c, write $q = x^{e_c/\kappa_c}$. Saturating the actual inertia gives symmetric degrees $-\rho ij$ and product degree $-\rho rj/p - \rho(1 - 1/p)R$. Extend through enlargements by (3.6); Lemma 3.6 gives the normal source bundles. For $R = 0$, the full normalizer charts glue the first order tangent bundle, whose extension splits by $H^1(\mathcal{O}(2 + \rho d_i t)) = 0$. The cusp inverse has total pole degree ρB; an auxiliary branch value adds only the tangent pole $\rho - 1$. This proves both injections. For the arbitrary genus option, no interior modification is needed and the actual invariant lines already form a direct sum. A cover clearing widths is the normalization of $y^\rho = \prod_c (Q - a_c)^{\rho/\kappa_c}$; extra branching adds no horizontal pole. Saturation of a primitive derivative commutes with finite pullback, and differentiating the actual cusp image gives c_f. The marked kernel of Proposition 2.3, or its quotient by the saturated tangent, has rank k_0, degree A, and is generated by line bundles of degrees at least e. Maps to target lines of degree below e vanish. A nonzero maximal minor proves the degree inequality in (3.9). Its left requirement and the required degree both increase with e. □

3.3 Two universal geometric inequalities

Lemma 3.8. At a fixed horizontal point the cusp width is $\kappa = \gcd\{b_i : \theta_i(x) \neq 0\}$. If $b_i \in \{1, 3\}$ and $\kappa = 1$, the fixed curve is immersed at the end.

Proof. Its fiber stabilizer is defined by $\lambda^{b_i} = 1$ for all nonzero coordinates. Width one forces a nonzero degree one section, giving a nonzero derivative in $U_i(Q)$. Properness extends the curve to W, and $d\pi\, df_W = df_X$ preserves this immersion. □

Lemma 3.9. A k-dimensional family of distinct complete embedded rational curves in the smooth coarse ball quotient has at least $k + 3$ branch values on a general member, provided it avoids higher stabilizer strata and meets the branch divisor transversely. In particular, a coordinate curve γ_i on W satisfies

$$L \cdot \gamma_i \geq \frac{d - 2}{2(d + 1)} = \frac{n - 5}{2(n - 2)}. \tag{3.10}$$

Proof. On a normal neat cover the pullback families are smooth proper after restricting parameters; choose connected components of fixed degree after finite base change. Each cyclic inertia acts freely on sheets, so every component retains all branch values. With b branch values, finite permutation monodromy tuples leave at most $b-3$ parameters. The cases $b \leq 2$ would give a rational covering curve, impossible in a ball quotient. For a fixed covering curve D, negative bisectional curvature and the integrated Bochner formula give $H^0(D, v^*T) = 0$ for every nonconstant v [18]. Hence $k \leq b-3$. The canonical formula gives $(d+1)L\gamma = -2 + \sum(1 - e_x^{-1}) \geq -2 + b/2$. For an existing coordinate curve γ_i, all pairs containing i are allowable. The open set where the other marks are distinct is therefore the universal smooth projective line bundle over $M_{0,n-1}$. Its fibers are the complete embedded coordinate curves and its base has dimension $d - 1$. The chamber perturbation never allows a pair of original weight at least one, so the family lies in the original stable locus. The finite cusp set has proper image in the base. A stabilizer stratum of codimension at least two has base image of dimension at most $d-2$, so a general fiber avoids it. A branch divisor with nondominant base image is also avoided; otherwise its projection is generically smooth in characteristic zero, making a general fiber transverse. The family thus satisfies the hypotheses already proved, and $H \cdot \gamma_i = 2$ gives the bound. □

Lemma 3.10. Let $Z = \mathbb{P}^k$, $k \geq 2$, be closed in a coarse arithmetic ball compactification, with $Z^\circ = Z \setminus S$ in its smooth interior for a finite set S containing the boundary points. Suppose $N_{Z^\circ} = \mathcal{O}(-1)^{\oplus c}$ and the original automorphic class restricts to $\lambda\mathcal{O}(1)$. Retain the finite maps from the full fixed normalizers in Lemma 2.12 and their restricted automorphic class. Let the full generic fixed group have normal rank $s \leq c$. If $s = c$, its actual fixed compactification is Z. If $s < c$, then $\lambda \geq 1$; equality makes the regular inverse image locally totally geodesic in the generic fixed ball.

Proof. For $s = c$, the finite actual fixed quotient has a unique generic preimage over Z°, so its compactification is finite birational to normal Z, hence isomorphic to it. For $q = c - s > 0$, work on a neat cover of the fixed ball and resolve the Gauss map graph of the inverse image of Z°. Its saturated tangent plane gives a subbundle $\mathcal{S} \subset \rho^* T$ of rank k. The quotient $\mathcal{Q}$ has rank q, maps generically injectively to the pullback of $\mathcal{O}(-1)^c$, and $\mathcal{Q} \otimes \rho^*\mathcal{L}$ is semipositive. Ball curvature in a direction perpendicular to $\mathcal{S}$ is $-\|d\rho(v)\|^2\|\xi\|^2$; the quotient curvature adds the nonnegative square of the second fundamental form [18, Section 1.6]. On a complete inverse line of hyperplane degree $v > 0$, avoiding S, this gives

$$0 \leq \deg \det(\mathcal{Q} \otimes \rho^*\mathcal{L}) \leq q(\lambda - 1)v.$$

Thus $\lambda \geq 1$. At equality the integral of the nonnegative curvature term is zero, so the second fundamental form vanishes along every such curve. General inverse lines cover all tangent directions on the regular immersion locus, proving local total geodesy. This argument uses the actual fixed quotient; it does not lift a coarse tangent splitting through ramification. □

The conclusion at equality is supplementary and is not needed for Theorem 1.1.

Proposition 3.11. Let a fixed curve have full generic group A_r, $r \geq 2$, larger groups only A_{r+1} and $A_r \times \mu_e$, and degree one full horizontal cusp models. Put $q = r + 1$, let R count A_{r+1} points, and put $h = -qt + R + s$. If $h < 0$, the space of homomorphisms to the line bundle associated with the invariant of greatest degree has dimension one. In all cases

$$a \geq \frac{r^2 - 3}{2} t + \sigma. \tag{3.11}$$

Proof. The lines E_j, $2 \leq j \leq q$, associated with the polynomial invariants on the normal space have degrees $-jt$, with their actual combined characters. At A_{r+1}, expand

$$(z + qu)\left((z - u)^q + \sum_{j=2}^{q} f_j (z - u)^{q-j}\right).$$

The differential has tangent diagonal $-q(q+1)u$, normal diagonals $(q+1)u$, and superdiagonal one. Let D_R be the reduced divisor of A_{r+1} points, and let S be the reduced divisor of ends. The line bundle associated with the invariant of greatest degree is $Q = E_q(D_R + S)$. Its tangent coefficient has residual degree $-(q-2)t - \sigma < 0$. Lemma 3.5 kills all $j < q$, with residual degrees $-(q-j)t$; the last is constant. Let $N = N_f^{\mathrm{sat}}$, and put $E = E_{q-1} \oplus E_q$. The inverse normal differential followed by projection onto these two directions defines a morphism $\psi : N \to E(2D_R + S)$. Define $\mathcal{Q} = \operatorname{im} \psi$, without taking its saturation in the target. This image is a vector bundle of rank 2 on C, and $N \to \mathcal{Q}$ is surjective. At an A_{r+1} point, put $\alpha = (q+1)u$; the stalk is generated, in the frames of E_{q-1}, E_q, by $(\alpha^{-1}, 0)$ and $(-\alpha^{-2}, \alpha^{-1})$. At a point with stabilizer $A_r \times \mu_e$, the normal differential is invertible and the stalk is E. At a cusp with parameter v, the invertibility of J_c in (3.2), after removing the radial direction, gives the stalk $v^{-1}E$. Projection onto the second coordinate is consequently surjective onto $Q = E_q(D_R + S)$ at every point, and its kernel is $E_{q-1}(D_R + S)$. Thus

$$0 \longrightarrow E_{q-1}(D_R + S) \longrightarrow \mathcal{Q} \longrightarrow Q \longrightarrow 0, \qquad \deg \det \mathcal{Q} = -(2q-1)t + 2R + 2s = 2h + t.$$

These equalities of stalks exclude any additional quotient of finite length. If both $h < 0$ and $2h + t < 0$, Proposition 2.4 forces at least three coincident labels. Proposition 2.2 then gives at least two independent Hom directions, a contradiction. Thus $h < 0$ implies $2h + t \geq 0$. The determinant and logarithmic identities give

$$qh = a - \frac{(q-2)(q+1)}{2}t - \sigma.$$

Substitute $h \geq -t/2$; for $h \geq 0$ this identity is stronger. □

3.4 Integral classes and local flags

Proposition 3.12. On a smooth coarse quotient of dimension d, let c_i^{orb} denote the average of tangent Chern classes on a neat cover. Then $c_i^{\mathrm{orb}} = (-1)^i \binom{d+1}{i} L^i$. If all stabilizers in codimension at most i have orders dividing M_i, then $M_i c_i^{\mathrm{orb}}$ is integral. It extends over a removed subset of codimension at least $i + 1$. If all reflections have order two,

$$\mathcal{T} = 2NL = 2K_W + B \in \mathrm{Pic}(W)$$

for the reduced branch divisor B. For products of type A, the denominators $M_2 = 12$, $M_3 = 24$ suffice. On a closed $\mathbb{P}^k$, $k \geq 3$, write $\mathcal{T} = b\mathcal{O}(1)$, $b > 0$. Then

$$\frac{3(N-1)b^2}{2N},\ \frac{(N-1)(N-2)b^3}{2N^2} \in \mathbb{Z}, \qquad N^2 \mid b^3, \qquad L \cdot \mathrm{line} \geq \frac{1}{2N^{1/3}}. \tag{3.12}$$

Proof. The following identities do not require a lower bound on the number of labels. Let U be a smooth coarse ball quotient of dimension d, let $p : Y \to U$ be a finite neat cover of degree v, and let $L = c_1(\mathcal{L})$. The flat Euler sequence gives $c(T_Y) = (1 - p^*L)^{d+1}$ in rational cohomology. If G_x is the full stabilizer at $x \in U$, proper pushforward of constructible functions and the normalization of Chern–Schwartz–MacPherson classes give

$$c_{\mathrm{SM}}\big(x \longmapsto |G_x|^{-1}\big) = v^{-1} p_*\big(c(T_Y) \cap [Y]\big), \qquad c_i^{\mathrm{orb}} = (-1)^i \binom{d+1}{i} L^i. \tag{3.13}$$

Here the orbifold classes are defined by the displayed average, and the cohomology equality uses the duality of the smooth U. We use the integral transformation underlying [24]. For a fixed ball, the flat Euler identity holds on its neat cover, with its own dimension and the restricted pullback of the original automorphic class L. Set the constructible function to zero on strata of codimension greater than i. They do not change the dimension $d - i$ component, and multiplying by M_i gives an integral function and integral Chern class. For a removed support of codimension c, local cohomology is Borel–Moore homology in degree $2d - q$, so vanishes for $q < 2c$; this gives the stated extension. The order two canonical formula gives $\mathcal{T}$; the small chamber morphism has no exceptional divisor. The low rank type A group orders give 12 and 24. Substitution into $12c_2^{\mathrm{orb}}$ and $24c_3^{\mathrm{orb}}$ gives the two fractions. For odd $p \mid N$, the cubic fraction forces $3v_p(b) \geq 2v_p(N)$. For $v_2(N) \geq 2$, use $v_2(N - 2) = 1$; for $v_2(N) = 1$, the quadratic fraction forces b even. □

Proposition 3.13. For $R = A_r, D_r, E_6, E_7$, use $A_\tau = Q(R) \otimes (\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}))$ with the polarization associated with the root lattice and Weyl linearization trivial at the origin. Let d_i be the degrees of the polynomial invariants and m_i the corresponding degrees of the generators of the ring of invariant sections, with distinguished generator ϕ_0 of degree one. Then $\phi_0(0) \neq 0$, and the first derivatives of invariant sections of degree at most j, divided by the corresponding powers of ϕ_0, project isomorphically onto $\langle dp_i : m_i \leq j \rangle$. The statement holds at all common fixed points with trivial fiber character by theta group translation, retains higher primitive

components, and respects $z \mapsto -z$. In type D, the product direction is separated by its sign character.

Proposition 3.14. At the inclusions in the following table, use increasing degrees k_i of the polynomial invariants on the normal space, the coarse axis $v = u^e$, and punctured normal frames p_i/u^{ϵ_i}. Target operations alone make the normal differential upper triangular with the indicated successive orders:

$P \subset G$	e	(ϵ_i)	orders
$D_5 \subset E_6$	1	0	$(3,2,3,3,4)$
$E_6 \subset E_7$	2	$k_i \bmod 2$	$(2,2,2,2,3,3)$
$E_7 \subset E_8$	2	$k_i \bmod 2$	$(3,3,3,4,4,5,6)$
$G_{33} \subset G_{34}$	6	$(-2,0,-2,0,0)$	$(1,2,2,3,4)$.

The last row of the inverse differential has pole order at most the last integer in the table; on its kernel, the preceding row has pole order at most the preceding integer.

The source degrees in the four rows are, respectively, $(2,4,5,6,8)$, $(2,5,6,8,9,12)$, $(2,6,8,10,12,14,18)$, and $(4,6,10,12,18)$. The frames and successive orders in the table refer to these same directions, rather than to new source coordinates. Lemma 4.3 uses the first row; Theorem 5.4 uses the second, Proposition 6.5 the third, and Lemmas 4.2 and 4.9 the fourth. At the ends, Proposition 3.13 identifies the same directions with the successive derivative projections; Lemma A.1 supplies the divisions $(1,3,3,4,6)$ for the fourth row. Proofs of both statements, including the intrinsic complete flags that justify elimination without changing source directions, are in Appendix A.

3.5 Compact monomial curves

Lemma 3.15. For an arithmetic realization with $n = r+4$, $r \geq 5$, no fixed curve without ends and with full generic group $G(m,p,r)$ has only the interior enlargements of Proposition 3.4. The ambient lattice may be compact or nonuniform.

Proof. The image of this compact fixed curve lies in the smooth coarse ball interior, where the small chamber morphism is an isomorphism by Lemma 2.1 and Proposition 2.11. Thus all calculations take place on W, even if the ambient quotient has cusps elsewhere. Put $j = mt \in \mathbb{Z}_{>0}$, $A = r(r-1)/2 + r/p$, $\eta = r - 1 + 1/p$, and $k_0 = k/m$. Normalizer preservation and the comparison give

$$(m-1)R + m\sigma = 2j + 2m(1-g), \quad a = 2 - 2g - Aj + \eta R, \quad h = R - k_0 j.$$

For $g \geq 1$, $R \leq 2j/(m-1)$, hence $h < 0$. The tangent coefficient of any map to the target associated with the invariant of greatest degree has negative degree $R/m - (k_0 - 2/m)j - \sigma$. Successive symmetric residual degrees are $-(k_0 - i)j < 0$. For $p > 1$, the product residual degree is at most

$$(1 - 1/p)R - (r - 1 - r/p)j \leq ((r+2)/p - r + 1)j < 0.$$

Thus the Hom evaluation rank is at most 1. Proposition 2.2 makes the marked quotient have exactly two coincident labels, so $a = re + 2\delta > 0$. But

$$a \leq (2 + 2m\eta/(m-1))(1-g) - (A - 2\eta/(m-1))j < 0,$$

since $A - 2\eta/(m-1) \geq (r-1)(r-4)/2 + (r-2)/p > 0$. Consequently $C = \mathbb{P}^1$. The coordinate $g_1 = v$ at enlargements and the unit normal matrix at points with stabilizer $G(m,p,r) \times \mu_e$

give immersion. The symmetric prefixes and product block are complementary at every stalk, so Lemma 3.6 gives

$$N_f = \begin{cases} \bigoplus_{i=1}^{r} \mathcal{O}(R - ij), & p = 1, \\ \bigoplus_{i=1}^{r-1} \mathcal{O}(R - ij) \oplus \mathcal{O}((R - rj)/p), & p > 1. \end{cases}$$

If negative summands occur, Proposition 2.2 makes all of them $\mathcal{O}(-e)$ and every other summand of degree at least e. Two symmetric negative summands have different degrees and are impossible. For $p > 1$, one negative symmetric summand forces a negative product summand of the same degree, hence

$$j = (p-1)e, \qquad R = [r(p-1) - p]e.$$

For $p = 2$, the preceding symmetric degree is zero, a contradiction. For $p \geq 3$, the logarithmic identity gives $ce \leq 2m$, but

$$c = (m-1)[r(p-1) - p] - 2(p-1), \quad c - 2m \geq (4p-7)m - 6p + 7 \geq 4 + (p-3)(4p-1) > 0.$$

The remaining negative possibilities force $(r-1)j < R < rj$. For $m \geq 3$ the logarithmic bound contradicts this; for $m = 2$, it leaves only $r = 5, 6$, $j = 1$, and there is no integral R in that interval. Thus N_f is nef and $R \geq rj$, $[r(m-1) - 2]j \leq 2m$. This forces $m = 2$, $j = 1$, $r = 5$ or 6. For $r = 6$, $R = 6$, $a = 17$, contradicting the parity of the marked degree against $-K_W$ for ten labels. For $r = 5$, the only integral pairs are $(R, a) = (5, 12), (6, 17)$, with $t = 1/2$. Nefness of N_f permits at most one negative marked degree. With none, Proposition 2.8 decomposes the curve into coordinate curves, and (3.10) gives $t \geq a/7$. With one, $a = 7e + 2\delta$; the first pair is impossible, and the second has $e = 1, \delta = 5$, giving $t \geq 2\delta/7 = 10/7$. Both contradict $t = 1/2$. □

4 Nine points

Let $n = 9$, and retain arbitrary weights $0 < \mu_i < 1$ of sum two. Suppose that $X = X_\mu$ is the Baily–Borel compactification of an arithmetic quotient of $\mathbb{B}^6$. Use the small chamber model $\pi : W \to X$ of Lemma 2.1, and write $H = -K_W$. For a curve $f : C \to W$ obtained from the full effective normalizer in the original lattice, put

$$t = \deg f^* L > 0, \qquad a = \deg f^* H, \qquad g = g(C), \qquad m_i = \deg M_i.$$

All basic invariant lines below are the actual coarse extensions, with their finite characters. We allow zero ends and arbitrary genus. The proof uses the curve comparison of Lemma 2.10 repeatedly, without deriving a separate nef bound for each reflection group.

4.1 Two bounds for marked curves in the compact case

Lemma 4.1. Suppose that W is a compact arithmetic realization. For a nonconstant complete curve, if $m_i \geq 0$ for every i, then $a \leq 7t$. If exactly one marked degree is negative, write

$$m_0 = -e, \qquad m_j = e + 2\delta_j \ (j \neq 0), \qquad \delta = \sum_{j \neq 0} \delta_j.$$

Then $a = 7e + 2\delta$ and $7t > 6e + 2\delta = a - e$.

Proof. A general coordinate curve γ_i has at least eight reflection points by Lemma 3.9. Its canonical formula gives $7L\gamma_i \geq -2 + 8/2 = 2$. Apply the effective curve decompositions obtained from the marked degrees in Proposition 2.8. They give the first assertion and reduce

the second to $L\beta_0 > 6/7$. Choose a maximal short subset $J \ni 0$, and set $q = |J| \le 6$. The same decomposition gives

$$[\beta_0] = [\beta_J] + \sum_{j \in J \setminus \{0\}} [\gamma_j].$$

For $q = 1$, Proposition B.2 gives $L\beta_0 \ge 1$. For $q = 2, 3$, Proposition B.4 gives, respectively,

$$L\beta_0 \ge \frac{2}{3} + \frac{2}{7} = \frac{20}{21}, \qquad L\beta_0 \ge \frac{1}{3} + \frac{4}{7} = \frac{19}{21}.$$

For $q \ge 4$, ampleness gives $L\beta_J > 0$, hence $L\beta_0 > 2(q-1)/7 \ge 6/7$. □

Lemma 4.2. A compact arithmetic realization has no full G_{34} point.

Proof. Take its actual G_{33} disk and then its quotient by the full normalizer. Let $R > 0$ count G_{34} points and put $\sigma = \sum(1 - 1/e)$ for the product enlargements. The actual line of degree six gives $j = 6t \in \mathbb{Z}_{>0}$. The formulas for the axis, the determinant, and the row of the inverse differential associated with the invariant of greatest degree give

$$\frac{5}{6}R + \sigma = 2t + 2 - 2g, \quad a = 2 - 2g - 50t + \frac{38}{3}R, \quad h = -18t + 4R. \tag{4.1}$$

The five successive coefficient degrees for a map to a line of degree d are

$$d + 4t - \frac{4}{3}R, \quad d + 6t - 2R, \quad d + 10t - \frac{7}{3}R, \quad d + 12t - 3R, \quad d + 18t - 4R. \tag{4.2}$$

These are the source degrees $(-4t + R/3, -6t, -10t + R/3, -12t, -18t)$ corrected by the successive zeros $(1, 2, 2, 3, 4)$ of Proposition 3.14. If $g \ge 1$, then $a, h < 0$ and the first two coefficients for $d = h$ vanish. Proposition 2.2 gives $\ell \le 4$ and therefore $a = (9 - 2\ell)e + 2\delta > 0$, a contradiction. If $g = 0$ and $j \ge 8$, the four lower coefficients vanish and the Hom space has dimension one. The resulting rank 1 marked quotient requires $a + 5h \ge 0$, but

$$a + 5h \le (402 - 308t)/5 < 0.$$

For $1 \le j \le 7$, the conditions

$$R \le (2j + 12)/5, \quad R \equiv 2j \pmod 3, \quad \sigma = 0 \text{ or } \sigma \ge 1/2$$

leave $(j, R) = (1, 2), (2, 1), (3, 3), (4, 2), (5, 1), (6, 3), (7, 2)$. The five rows with $h < 0$ have $a < 0$ and a one-dimensional Hom space, again impossible. The other rows are

$$(t, R, \sigma, a, h) = (1/6, 2, 2/3, 19, 5), \qquad (1/2, 3, 1/2, 15, 3).$$

Formula (4.2), including the tangent coefficient, excludes every negative line quotient. Thus at most one m_i is negative. If none is negative, Lemma 4.1 contradicts $a \le 7t$. Otherwise $a = 7e + 2\delta$ gives $(e, \delta) = (1, 6)$ or $(1, 4)$, and the weaker inequality $t > 2\delta/7$ already contradicts the displayed values of t. □

4.2 Successive kernels for D_5

The following sequence of kernels is used in both the compact and nonuniform cases.

Lemma 4.3. Let a curve obtained from the full normalizer have generic group D_5, larger irreducible groups E_6, D_6, and other interior groups $D_5 \times \mu_e$ acting trivially on the root space. Allow ends with the actual minimal D_5 polarization, generator degrees $(1^4, 2^2)$, trivial fiber character, and width one. Write R, D, S for the reduced divisors of these E_6 points, D_6 points,

and ends, using the same letters R, D for their degrees and $s = \deg S$. Set $E = f^*T_W$ and $T_f = \mathrm{sat}_E\, df(T_C)$. Then $T_f = T_C(R)$ and E/T_f has successive line quotients of degrees

$$q_2 = -2t + 3R + D + s, \qquad q_4 = -4t + 2R + D + s, \tag{4.3}$$

$$q_5 = -5t + 3R + D/2 + s, \qquad q_6 = -6t + 3R + D + 2s, \tag{4.4}$$

$$h = -8t + 4R + D + 2s. \tag{4.5}$$

In particular $E \twoheadrightarrow Q_h$, and

$$2g - 2 + s + D/2 + \sigma = 2t, \quad a = 4h + 5t + D + \sigma, \quad \deg T_f = (h + D)/4 + \sigma + s/2. \tag{4.6}$$

If $R > 0$, then $h > 0$ and E is nef.

Proof. The source line bundles have degrees $-2t, -4t, -5t - D/2, -6t, -8t$. The half contribution comes from the coarse frame P/u at a D_6 point, with its character included. The full normalizer preserves their flag ordered by increasing degree. Proposition 3.14 gives successive zero orders $(3, 2, 3, 3, 4)$ at E_6. At D_6, put $v = u^2$ and $P = u\xi$; after removing the tangent, the normal matrix is

$$\begin{pmatrix} v & 1 & 0 & 0 & 0 \\ 0 & v & 0 & 1 & 0 \\ 0 & 0 & v & 0 & 0 \\ 0 & 0 & 0 & v & 1 \\ 0 & 0 & 0 & 0 & v \end{pmatrix},$$

so every quotient gains one zero. At an end the nonzero radial section has degree one. After its removal, the matrix is $\mathrm{diag}(q, q, q, q^2, q^2)A(q)$, where the leading 3×3 block of $A(0)$ is invertible. The least orders of its first-column minors are $1, 2, 3, 5, 7$; their differences give $(1, 1, 1, 2, 2)$. These are orders of minors from the initial columns, not merely Smith exponents after an arbitrary source change. The E_6 axis differential has one common zero and the other coarse axes have unit differential. This proves the flag and (4.6).

We record the local obstruction that also treats $h = 0$. Use coordinates $T_2, T_5, T_6, T_8, T_9, T_{12}$. The E_6 axis is

$$T(u) = (-3u^2, 3u^5, -3u^6/2, -3u^8/4, u^9, -3u^{12}/16).$$

The image of its ambient local ring is $A = \mathbb{C}\{u^2, u^5\}$, whose missing nonnegative exponents are $1, 3$. Primitive rows for the descending saturated quotients are

$$\begin{aligned} C_8 &= dT_{12} + u^3 dT_9/2 + u^6 dT_6/4, \\ C_6 &= dT_9 + u\, dT_8 + u^3 dT_6 + u^4 dT_5/2 + u^7 dT_2/4, \\ C_5 &= dT_8 + u^3 dT_5/2 + u^6 dT_2/4, \\ C_4 &= dT_6 + u\, dT_5 + u^4 dT_2, \qquad C_2 = dT_5 + 5u^3 dT_2/2. \end{aligned}$$

They annihilate $T'(u)/u$ and have a unit minor in columns $T_{12}, T_9, T_8, T_6, T_5$. The intrinsic Tjurina flag of Proposition 3.14 identifies their successive kernels. No unit multiple of C_8 is A-valued, since its coefficient ratio is $u^3/2 \notin A$. A regular row on the ambient quotient that annihilates the tangent direction and the quadratic invariant has an extra zero in the coefficient of the invariant of degree four: compare the missing u term in its dT_5 coefficient. If it also kills degree four, comparison of the missing u^3 terms in dT_6, dT_5 puts its value at zero in $\mathbb{C}dT_{12}$.

Suppose $R > 0$ and $h \le 0$. For a row with target degree $q \le h$ the first two coefficient degrees are bounded by

$$q - q_2 \le 3h/4 - 2R - 3D/4 - s/2 < 0, \qquad q - q_4 \le (h - D)/2 \le 0.$$

Equality in the second bound requires $q = h = D = 0$. If $h < 0$, the independent quotients associated with coincident pairs in Proposition 2.2 all have values in $\mathbb{C}dT_{12}$, hence $\ell = 2$. The quotient associated with the invariant of greatest degree factors through this pair quotient with a unit coefficient, contradicting $u^3/2 \notin A$. For $h = 0$, the same argument first excludes negative line quotients. Two negative marks would supply one; a unique negative mark would make the row associated with the invariant of greatest degree zero on the positive outside columns, and infinitesimal projective transformations then kill its remaining column. Thus all marked degrees are nonnegative and the row associated with the invariant of greatest degree only uses marks whose lines have degree zero. Their distinct positions are disjoint. Three positions trivialize the family and make the relevant row constant; one position makes it a constant sum of pair differences. Both contradict the same coefficient ratio. If exactly two positions remain, each position contains at least two marks and gives an independent quotient of degree zero associated with that pair. Their first coefficients vanish by degree. The second vanish by degree when $D > 0$, and by their extra zero when $D = 0$. Both pair quotients therefore have value in $\mathbb{C}dT_{12}$, contradicting independence. Consequently $h > 0$.

Finally (4.6) gives

$$q_2 = h/4 + 2R + 3D/4 + s/2 > 0, \quad q_4 = (h + D)/2 > 0,$$

$$q_6 = 3h/4 + D/4 + s/2 > 0, \quad \deg T_f > 0, \qquad q_5 = 3h/4 + g - 1 + \sigma/2 \geq -1/4.$$

Since h, q_5 are actual integral line degrees, $q_5 \geq 0$. The flag has quotients of nonnegative degree and remains so on every finite cover; hence E is nef. □

Proposition 4.4. In the compact case the curve of Lemma 4.3 cannot have $R > 0$.

Proof. We have $h > 0$, E nef, and at most one negative marked degree. Put $Y = 2t - h - D - \sigma$. If all marked degrees are nonnegative, Lemma 4.1 gives $Y \geq 3h > 0$. With one negative mark $-e$, the eight outside columns generate E, so $e \leq h$ and

$$S' = 7t + e - a > 0, \qquad Y = S' + 3h - e > 0.$$

Every marked column, written in a frame pulled back from W, has coefficients in A. For a column killed by C_8, successive use of the displayed rows and the missing exponents $1, 3$ gives extra zero orders $(R, 2R, 2R, 3R)$ in Q_6, Q_5, Q_4, Q_2. The equation $C_8 v = 0$ gives $v_9(0) = 0$, then $C_6 v = 0$ gives $v_8(0) = 0$; the u^2, u^3 terms of $C_5 v = 0$ give $[v_8]_{u^2} = v_5(0) = 0$. The u^3 terms of $C_6 v = 0$ and $C_4 v = 0$ give $v_6(0) = [v_5]_{u^2} = 0$. The adjusted degrees are

$$q_6 - R = -t - \sigma/2 - Y/2, \quad q_5 - 2R = -5t/2 - \sigma/4 - Y/4,$$

$$q_4 - 2R = -2t - h - \sigma - Y, \qquad q_2 - 3R = -h - \sigma - Y.$$

All are negative, so a nonnegative marked column killed by the row associated with the invariant of greatest degree lies in T_f. Let p count these columns and put $\tau = \deg T_f = (h + D)/4 + \sigma$. At a general point, the kernel of their evaluation consists of projective vector fields vanishing at complementary marked positions. There is at least one such position, hence $p \leq 3$. With a unique negative mark there are two distinct complementary positions, hence $p \leq 2$. Individual columns are nonzero because a nonconstant ordered configuration has at least four distinct positions.

If all marks are nonnegative, put $A_7 = 7t - a \geq 0$. For $\tau \leq h$ use $B_0 = 9h - a \geq 0$; for $\tau > h$ use $B_3 = 6h + 3\tau - a \geq 0$. But

$$5A_7 + 4B_0 = -10t - 9D - 9\sigma < 0, \qquad 2A_7 + B_3 = -t - 21h/4 - 9D/4 < 0.$$

With one negative mark, $T_p = (8 - p)h + p\tau - e - a \geq 0$, but

$$S' + T_p = -3t - 3ph/4 + (p/4 - 2)D + (p - 2)\sigma < 0 \qquad (0 \leq p \leq 2).$$

Every case is impossible. □

4.3 The compact proof closes at $K_W + 10L$

Lemma 4.5. After excluding persistent proper factors and the compact $G(m,p,5)$ curves with $m \geq 4$ or $(m,p) = (3,1)$, the divisor $K_W + 10L$ is nef.

Proof. The remaining full rank 6 groups are $A_6, B_6, D_6, G(3,3,6), E_6, G_{34}$. Their reflections have order two; put B for the reduced branch divisor. Then $14L = 2K_W + B$ is integral and the rank 2 groups are A_1^2, A_2, B_2. Set

$$C_1 = H + 7L = B/2, \quad C_2 = c_2(T_W) + 6HL + 21L^2, \quad A_0 = 3C_1(C_1 + L) - 6C_2.$$

For a normal neat cover $p : Y \to W$ with group G, Burnside's fiber count gives

$$\mathbf{1}_W = |G|^{-1} \sum_{g \in G} p_* \mathbf{1}_{\mathrm{Fix}(g)}.$$

Apply the integral CSM naturality used in Proposition 3.12. A codimension-r fixed ball has Chern polynomial $(1 - Lz)^{7-r}$. The codimension-one and codimension-two triangular identities give respectively C_1, C_2 as defined above, and

$$48C_2 = 12Z_{A_1^2} + 16Z_{A_2} + 18Z_{B_2}, \qquad 4A_0 = 4Z_{A_2} + 9Z_{B_2}. \tag{4.7}$$

Here Z_P are ordinary reduced fixed cycles. For clarity, the first coefficients are 48 times the proportions $1/4, 1/3, 3/8$ of elements with no nonzero fixed vector. For the second identity, the coefficients of $B^2 + 2LB$ are $2, 4, 6$, obtained from $4\nu(\nu-1)/|P|$. The mirror normal class is $-L$, so its diagonal self-intersection cancels with $2LB$. These observations derive both identities from the common CSM formula.

On a closed coincidence space $Z_I = \mathbb{P}^k$, put $q = |I|$, $\alpha = 9 - 2q$, and $L|_{Z_I} = \lambda H_0$. The marked tangent sequence gives $c_1(T_W)| = \alpha H_0$ and $c_2(T_W)| = (\alpha^2 - 7)H_0^2/2$. A plane in Z_I is the intersection of $q-1$ divisors where pairs coincide and $k-2$ outside relative tangent classes. Thus (4.7) pairs integrally with this plane, including excess intersections. For $b = 14\lambda \in \mathbb{Z}$, substitution gives

$$48 \int_{\mathbb{P}^2} C_2 = 24(\alpha^2 - 7) + \frac{36}{7} b(b + 4\alpha), \qquad 4 \int_{\mathbb{P}^2} A_0 = 84 + \frac{6}{7} b(b + 3\alpha).$$

Their difference gives $7 \mid 6\alpha b$ and then $7 \mid 6b^2$, hence $7 \mid b$. For the K_W-negative coincidence lines $q = 2, 3, 4$, their H-degrees are $5, 3, 1$ and their L-degrees are at least $1/2$. The corresponding ratios are at most $10, 6, 2$. Coordinate curves have ratio at most seven by Lemma 4.1. Proposition 2.8 proves the result; for $W = \mathbb{P}^6$ use Proposition B.2. This argument quantizes restrictions to coincidence spaces and does not assert descent of $2L$ on the compact W. □

Theorem 4.6. No nine-point quotient under consideration is a compact arithmetic ball quotient.

Proof. By Proposition 2.11, stabilizers are reflection groups, so the compact coarse quotient is smooth and π is an isomorphism. Increase an irreducible factor through full actual parabolic stabilizers. If enlargement stops below rank 6, Proposition 2.14 gives equal basic degrees δ and a closed image $\mathbb{P}^{6-s}$ with normal bundle $\mathcal{O}(-1)^s$ and $\delta L = \mathcal{O}(1)$. The classification of groups with equal basic degrees first gives $s \leq 2$; no marked antiample statement for four labels is needed. For $s = 1$, $\delta \geq 2$ contradicts the $\mathbb{P}^5$ lower bound in Proposition B.4. For $s = 2$ the groups are $G(m,2,2)$, with even $m \geq 4$, and G_7, G_{11}, G_{19}, all with $\delta \geq 4$, contradicting its $\mathbb{P}^4$ bound. Thus every reflection enlarges to rank 6. Lemma 3.15 excludes $G(m,p,5)$ for $m \geq 4$ or $(m,p) = (3,1)$ before any bound on a/t is used. Lemma 4.5 now gives $a \leq 10t$ on every complete curve. Lemma 4.2 excludes G_{34}, and Proposition 4.4 excludes E_6 through its

full parabolic D_5 disk. Next $B_6, D_6, G(3,3,6)$ give full $B_5, D_5, G(3,3,5)$ disks, whose primitive enlargements have just been excluded; Lemma 3.15 excludes them. Only A_6 remains. The nontrivial lattice generated by reflections has a reflection, so actual enlargement supplies a full A_5 disk. Proposition 3.11 gives $a \geq 11t + \sigma > 10t$, a contradiction. No imaginary quadratic hypothesis or compact descent of $2L$ was used. □

4.4 The nonuniform proof: bounds fixed once

Now suppose that the arithmetic quotient is nonuniform. At a genuine wall vertex the local slice is the cone over $\mathbb{P}^a \times \mathbb{P}^b$, $a + b = 5$, $a, b \geq 1$. The chosen small model retains the smaller factor, so $\dim \mathrm{Exc}(\pi) \leq 2$ and its codimension is at least four. It is an isomorphism near every smooth cusp. Proposition 2.11 applies since $12 > 7$; the coefficient field K is imaginary quadratic and $w_K L$ is an actual ample line bundle on X, $w_K \in \{2, 4, 6\}$. After higher-order reflections are excluded below, it gives actual descent of $2L$ on the original X.

We use Lemma 2.10 in the following form throughout. For an actual ample line bundle $A_X = cL$, put $b = ct$. Then

$$a \leq 7b; \qquad \mathrm{Hom}(E, Q) \neq 0,\ \deg Q \leq 0 \quad \Longrightarrow \quad a \leq 5b. \tag{4.8}$$

If the common negative marked degree is $-e$, the same lemma gives

$$a = (9 - 2\ell)e + 2\delta, \quad e + \delta \leq b,$$

and every target of degree less than e kills the positive marked part F. Thus, after $2L$ descends, the two bounds in (4.8) are $a \leq 14t$ and $a \leq 10t$. These are curvewise statements, valid also for $W = \mathbb{P}^6$; no global nef theorem is invoked.

Lemma 4.7. Every irreducible reflection factor enlarges through full actual point stabilizers to rank 5 or 6. Every irreducible rank 5 disk avoids genuine Segre vertices, so all its ends are smooth ambient cusps.

Proof. If actual enlargement stops in rank $s \leq 4$, the product hypothesis of Proposition 2.14 holds. Its image is a closed $Z_X = \mathbb{P}^{6-s}$ with equal degrees δ, and classification gives $s \leq 2$. The strict transform $Z \to Z_X$ has no exceptional divisor since $\dim Z \geq 4$ and $\dim \mathrm{Exc}(\pi) \leq 2$. A projective small morphism to a factorial target is an isomorphism: push down a relatively ample Cartier divisor and pull it back; their difference would be an exceptional divisor. Thus $Z = Z_X$, $N_{Z/W} = \mathcal{O}(-1)^s$, and $\delta L|_Z = \mathcal{O}(1)$. Its actual fixed Hermitian space is defined over the original imaginary quadratic field and has dimension at least three. Meyer supplies a rational isotropic line in that space, so the induced fixed quotient of Lemma 2.12 is nonuniform. Descent of the original $w_K L$ then gives $\delta \mid w_K$. The lowest invariant term of its ideal gives a generically invertible map

$$L^{-\delta} \otimes U_\delta \longrightarrow N_{Z/W}.$$

After canceling equal degrees it is a map from a finite flat bundle to $\mathcal{O}_Z^s$. On complete curves away from cusps its coefficients become constant after a finite cover. Interior inertia is therefore trivial; simple connectedness and Hartogs make the matrix constant and invertible on Z. At a smooth ambient cusp every vanishing function and its lowest normal coefficient have a central factor, a contradiction. All cusps of Z consequently lie at genuine Segre vertices. Surjectivity on local rings and the generators of degree one in Theorem 2.19 force equality of the full central periods: an additional central index would kill every nonconstant restricted generator. The induced smooth cusps therefore have only generators of degree one and trivial fiber character.

In rank 2, the possibilities are $G(4,2,2), G(6,2,2)$. The two lowest directions require every enlargement to have two basic degrees equal to δ. The lift fixing the point for the full finite group permits this test at a point of a Segre cusp. Standard smooth abelian quotient actions

have point groups that are products of symmetric groups and $G(e,1,k)$, whose essential degrees are distinct. The sole remaining surface action is $G(4,2,2)$, excluding $\delta = 6$. For $\delta = 4$, the induced groups on $\mathbb{P}^4$ are $A_3, B_3, G(4,1,3)$. The generic branch groups are $G(4,2,2) \times \mu_e$, $e = 2, 4$. Their canonical formula is $2B_2 + 3B_4 = 25H_0$, so $B_4 \neq 0$. Meyer in that actual higher-dimensional fixed quotient yields a $G(4,1,3)$ cusp and its full fixed disk. Writing $u = 4t$, its equations are

$$46t = 4s + 15R/4 + \sigma, \qquad -12t + s + R \geq 4t.$$

The first right side is at least $15(s+R)/4 \geq 60t$, impossible.

For rank 1, $\delta = 4, 6$ are excluded by Propositions B.6 and B.5. For $\delta = 2, 3$, put $u = \delta t$ on $Z = \mathbb{P}^5$. All induced cusps have five generators of degree one. The field and cusp classifications leave A_4 and $G(m,1,4)$ with $m = 2, 3, 4$ for $\delta = 2$, and $m = 2, 3$ for $\delta = 3$. Parameter six is excluded by the general rank 4 product inequality in the proof of Proposition B.5:

$$a \geq (17p/6 + 1)h_P + 30t + R + \sigma > 6u \qquad (h_P \geq u,\ t = u/\delta,\ \delta = 2, 3).$$

This uses that inequality itself, rather than the conclusion at slope $1/6$ of the proposition. A G_{32} disk has no ends, since the horizontal actions whose rings of invariant sections are generated in degree one have no such full stabilizer, and has no irreducible rank 5 enlargement. Its axis and determinant formulas give $a = 2 - 2g - 84t < 0$, since $t = u/\delta \geq 1/3$, contrary to $a = 6u > 0$. For $G(m,1,4)$, the canonical formula and Euler give

$$(10m + 2 + 6\delta)t = 5s + (5 - 1/m)R + \sigma, \qquad -4mt + s + R \geq \delta t.$$

The difference between the forced lower coefficient and the left coefficient is

	$m = 2$	$m = 3$	$m = 4$
$\delta = 2$	11	64/3	63/2
$\delta = 3$	19/2	20	absent

and is positive. Curves of types $D_4, G(3,3,4), G(4,2,4), G(4,4,4), F_4$ have no ends, by the order and character test in the actual smooth horizontal action. For F_4, $(28 + 6\delta)t = 2 - 2g$ contradicts $t \geq 1/\delta$. For D_4, with R, M counting G_{33}, D_5 points, the two degree formulas give

$$g = 0, \quad u \leq 2, \qquad (36 + 96/\delta)u - 12 = 28R + 21M.$$

Modulo seven excludes $\delta = 2$; $\delta = 3$ leaves $(u, t, R, M, \sigma) = (1, 1/3, 2, 0, 1)$. Here the image is a line with normal bundle $\mathcal{O}(1)^4$. The four actual source line bundles on the coarse quotient have degree -2, with zero orders $(1,0,2,1)$ and $(1,2,0,1)$ at the two G_{33} points; the other inertia consists of two order-two points acting trivially on the root space. This is exactly the triality configuration in (B.2), whose following calculation uses only these data. Its tensor identity makes the two products of exchanged ratios equal to $1/12$; the determinant of the four linear columns is therefore zero. This excludes D_4, G_{33}, D_5 before the pure A_4 formula is used. For $G(m,p,4)$ with $p > 1$, put $D_0 = 6m + 4m/p$, $k = 3m$, and $\beta = 4 + 1/p - 1/m$. The equations for a curve without ends give

$$(D_0 + 2 + 6\delta)t = \beta R + \sigma, \qquad R \geq (k + \delta)t.$$

For $\delta = 2$, the differences $\beta(k+\delta) - (D_0 + 2 + 6\delta)$ are $8, 27/2, 14$. These correspond respectively to $(m,p) = (3,3), (4,2), (4,4)$. For $\delta = 3$, $(m,p) = (3,3)$ the difference is six. All are positive. Persistence contradicts the ample tangent bundle of $\mathbb{P}^5$, so a full A_4 disk remains. Its target associated with the invariant of greatest degree satisfies $5 \deg Q = (6\delta - 9)t - \sigma < 5\delta t$, contrary to Euler. This excludes every persistent proper factor.

The field column of finite reflection classification now leaves

$$\begin{array}{l|l} \text{rank } 5 & A_5,\ G(m,p,5),\ G_{33} \\ \text{rank } 6 & A_6,\ G(m,p,6),\ E_6,\ G_{34}, \end{array} \qquad m \in \{2,3,4,6\}, \quad p \mid m.$$

Every rank 6 group has an actual rank 5 parabolic. An irreducible rank 5 root space cannot lie in one horizontal factor at a genuine Segre cusp, whose horizontal ranks split as $1+4$ or $2+3$. Thus every rank 5 end is a smooth ambient cusp. □

4.5 Remove the higher parameters before using $2L$

Lemma 4.8. There are no full disks or smooth cusps of types $G(6,p,5)$, $G(4,1,5)$, or $G(3,1,5)$. After $2L$ descends, the same holds for $G(4,2,5), G(4,4,5)$.

Proof. Apply Proposition 3.4 with actual end periods, polarizations, and fiber ages. For $p>1$, the target degrees for the elementary symmetric invariant of greatest degree and for the product invariant are

$$h = -4mt + R + \nu, \qquad h_P = (-5mt + R + \nu)/p, \qquad \nu = \sum_c \nu_c;$$

for $p=1$ the target associated with the invariant of greatest degree is $h = -5mt + R + \nu$. The age of the source line bundle associated with the product invariant is retained separately.

For $m=6$, put $j = 6t \in \mathbb{Z}_{>0}$ and use $A_X = 6L$. The complete normalized end list is

$$(B_c - \nu_c/p, \nu_c) \in \{(5,1),(9,2),(13,3),(23,6)\}, \qquad \mathrm{age}_c = \lceil \nu_c/p \rceil - \nu_c/p.$$

It follows from the raw degrees $(2,3,6/p,6,6,6)$ with their actual character congruences. Set $\gamma = 23/6 + 1/p$. Since $B_c \geq \gamma \nu_c$, the canonical formula gives

$$a \geq 53j/6 + R + \sigma + p\gamma h_P.$$

Formula (4.8) first gives $a \leq 7j$, then $h_P < 0$ and $a \leq 5j$. Every nonzero line target has degree at least $-j$. The logarithmic formula, using $\nu_c \leq 6$, gives

$$(\bar{k} - 3)j + 4R + 6\sigma \leq 12(1-g), \qquad \bar{k} = 5 \ (p=1), \quad \bar{k} = 4 \ (p>1).$$

Thus $g=0$, $j \leq 6$ for $p=1$ and $j \leq 12$ otherwise; also $s \leq \lfloor 2 + j/3 \rfloor$ and $5R/6 + s \leq 2 + j/3$. Here $\sigma = 0$ or $\sigma \geq 1/2$, with no bound on the order of a product factor.

We give the finite check of the remaining cases to specify its scope. Let $V = \sum V_c$ be the initial required cusp zero order: $V_c = 1$, except that the two $p=1$ end sums $11, 16$ require $V_c = 2, 3$. For a negative target $q = \min(h, h_P)$, the successive symmetric bounds are $q + ij - R - V$. The product bound is $q + (5j - R)/p + \sum \mathrm{age}_c - V$. Require $h, h_P \in \mathbb{Z}$, $h, h_P \geq -j$, and integrality of the degree of the source line bundle associated with the product invariant

$$-5j/p - (1 - 1/p)R - \sum \mathrm{age}_c \,.$$

For $-q \leq e \leq j$ and $2 \leq \ell \leq 6$, require $a = (9 - 2\ell)e + 2\delta$, $e + \delta \leq j$, and $\delta \geq 0$ integral. The coefficient bounds must support evaluation rank $\ell - 1$ and dimension $(\ell - 1)(e + q + 1)$. The independent inverse directions on F also impose

$$\#\{0 \leq i < 4 : h + ij < e\} + \mathbf{1}_{h_P < e} \leq \ell - 1$$

for $p > 1$, with five symmetric directions and no product term for $p = 1$. Substitution in this proved finite range gives no row for $p = 1$. All remaining rows have $R = g = 0$, $a = e = j$, $\ell = 4$, $h = -j$, $\delta = 0$; their complete end table is

p	j	$\{B_c\}$	σ	h_P
2	2	$\{26\}$	5/3	-2
2	3	$\{29/2, 26\}$	1	-3
2	3	$\{11/2, 10, 26\}$	0	-3
3	6	$\{29/3, 29/3, 25, 25\}$	0	-4
6	6	$\{24, 24, 24\}$	1	-2
6	9	$\{27/2, 24, 24, 24, 24\}$	0	-3.

This includes zero ends and every sign of a; positivity was not imposed. For $p = 3, 6$, the product is an actual direct summand because $R = 0$. Its degree strictly between $-j$ and zero contradicts $E = \mathcal{O}(-j)^3 \oplus F$, whose other summands have degree at least j. For $p = 2$, every row has an end with sum 26. The actual elliptic sections obey $y^2 = 4x^3 - g_3u^6$, $g_3 \neq 0$, and

$$2y_1/y_0 - 3x_1/x_0 = -g_3u_0^6/y_0^2 \neq 0.$$

The radial and lowest symmetric derivatives are independent. Once the first symmetric coefficient vanishes, all symmetric coefficients are divisible by the sixth central power. Only one symmetric direction and the product can remain, giving evaluation rank at most 2 instead of $\ell - 1 = 3$.

For $m = 4, p = 1$, use $j = 4t$ and $A_X = 4L$. The actual generator degrees $(d_1, d_2, 2d_3, 4, 4, 4)$, $d_1, d_2 \mid 4$, $d_3 \mid 2$, give $B_c \geq 19\nu_c/4$ for every torsor and linearization. Consequently

$$a \geq 19h/4 + 33j/4 + R + \sigma.$$

Starting with $a \leq 7j$ gives $h < 0$, then $a \leq 5j$, $h \leq -13j/19$, and $e \geq 13j/19$. On F the equations $dZ = v\,df_5$, $dg_5 = df_5 + v\,df_4$ reduce the next inverse pole, giving a second independent target of degree $h + j < e$. Thus $\ell \geq 3$ and $a \leq 3j$, but $h \geq -j$ gives $a \geq 7j/2$.

For $m = 3, p = 1$, put $j = 3t$. Its integrality comes from the actual symmetric lines, not from $6L$ alone. After parameter six, the end degrees are $(1^6), (1^2, 3^4), (1, 3^5)$, with counts x, y, z. Set $h = -5j + R + x + 3y + 3z$. The equations give

$$a - 5h = 10j + 2 - 2g - 2y > 4j, \qquad j + R + 3\sigma \leq 6(1 - g).$$

Here $y \leq 2 + 2j/3 - 2g$ proves the strict inequality. Lemma 2.10 gives $g = 0$, $1 \leq j \leq 6$, $h < e \leq 2j$, and factorization through the negative marked quotient. The first symmetric residual degree is $-4j + 2y < 0$; its sole possible exception $j = 1, y = s = 2, R = 0, e = 2$ gives $a = 13 > 10j$. At $(1^2, 3^4)$ the actual local equation $(s_u/s_v)^3 = i(1 + z_i^3)$ makes the radial and first symmetric derivatives independent at every fixed point. At $(1, 3^5)$, the section of degree one for width one is nonzero. Thus all symmetric coefficients acquire a third central power; their residual degrees are $(i - 5)j$. Hom has dimension one, so $\ell = 2$, $h = -e$, and $5e + \delta = 5j + 1 - y$. The only integer case with $2e \leq j$ is $(j, e, y, R) = (2, 1, 3, 0)$, forcing $\sigma = 1/3$, impossible for product contributions. Hence $2e > j$ and the next inverse target $h + j = j - e < e$ gives two independent rows through a rank 1 marked quotient.

These exclusions remove every reflection of order greater than two, including those in smaller horizontal factors by Lemma 4.7 and the lift fixing the point for the full group of Theorem 2.19. The normalized lifts now have determinant ± 1, so a scalar zI_7 in their kernel satisfies $z^{14} = 1$. Its order also divides w_K, hence divides two. Proposition 2.11 gives actual descent of $2L$, with the full cusp group included.

For $m = 4, p = 2, 4$, we may therefore put $j = 4t \in 2\mathbb{Z}_{>0}$, $b = j/2$. The actual product gives $h_P = (h - j)/p$. The raw degrees imply $5\nu_c + 1 - B_c \leq 4$ for $p = 2$ and ≤ 5 for $p = 4$. For

$p = 2$ the stronger bound uses the even degree of the unnormalized product: if its maximum exponent is four, the opposite order-four branch has odd exponent and the sum is at least seventeen. Elimination of s gives

$$a - 5h \geq 11j/2 - 6 + 6g + 5R/2 + 4\sigma \quad (p = 2),$$

$$a - 5h \geq 25j/4 - 8 + 8g + 3R + 5\sigma \quad (p = 4).$$

Both exceed $2b = j$. Thus $h < e \leq j/2$, and the independent product target also has degree less than e; hence $\ell \geq 3$ and $a \leq 3j/2$. For $p = 2$, $h_P \geq -j/2$ gives $h \geq 0$, contradicting the first bound. For $p = 4$, $h \geq -j/2$ gives $a - 5h \leq 4j$, contradicting the second bound for $j \geq 4$. For $j = 2$, $h = -1, 0$ and $(h-2)/4$ is not integral. All corresponding cusp and rank 6 exclusions use the lifts fixing the points for their full finite groups, preserving period and character. □

4.6 The remaining five types

Lemma 4.9. There are no G_{34} points, full G_{33} cusps, or full G_{33} disks.

Proof. The remaining horizontal groups are $A_5, B_5, D_5, G(3,3,5), G_{33}$. Primitivity forces every G_{33} end to have full group G_{33}, trivial fiber character, and width one. Its actual generator degrees are $(1,1,3,3,4,6)$. The four fixed points form the polarization kernel; theta translation preserves the actual characters at all four, by Lemma A.1. The only larger interior groups are $G_{34}, G_{33} \times A_1$. Let R count the former and let σ be the latter contribution, so $2\sigma \in \mathbb{Z}_{\geq 0}$. The equations are

$$2g-2+s+5R/6+\sigma = 2t, \quad a = 36(1-g)-16t-3R/2-17\sigma, \quad h = 12(1-g)-6t-R-6\sigma. \tag{4.9}$$

The basic source degrees are $(-4t + R/3, -6t, -10t + R/3, -12t, -18t)$; their successive cusp divisions are $(1,3,3,4,6)$. If $g \geq 1$, the first two residual degrees for the target associated with the invariant of greatest degree are

$$10(1-g) - 4t - 3R/2 - 5\sigma, \qquad 10(1-g) - 2t - 13R/6 - 5\sigma.$$

They and the tangent coefficient are negative. The marked rank bound gives $\ell \leq 4$, thus $a > 0$, contrary to (4.9). Hence $g = 0$.

For $R > 0$, put $j = 2t$, $R = 3r$, $q = 2\sigma$; actual coarse lines make these integers. Then

$$r + q \equiv 0 \pmod 2, \quad s = 2 + j - (5r+q)/2 \geq 0, \quad a = 36 - 8j - (9r+17q)/2 \leq 7j.$$

If $r = 1$, then q is odd and $j \geq 2$; if $r \geq 2$, then $j \geq 3$. The first two residual degrees are strictly negative, and the marked rank argument gives $a > 0$ when $h < 0$. When $h \geq 0$, use $a = 3h+j+9r/2+q/2 > 0$ instead. For $r \geq 3$, the inequality for s gives $a \leq 52-49r/2-25q/2 < 0$. Positivity leaves exactly

$$(t, R, s, \sigma, a, h) = (3/2, 6, 0, 0, 3, -3), \qquad (1, 3, 1, 1/2, 7, 0).$$

In the first row a third coefficient vanishes, so $\ell \leq 3$, $e \geq 3$, and $a \geq 3e \geq 9$. In the second, the five residual degrees for a target $\mathcal{O}(-1)$ are $(-2,-4,-1,-2,-1)$, so no quotient onto a line of negative degree exists. All marked degrees have the same parity and sum seven. They cannot all be nonnegative, since then all nine would be positive odd integers. Two negative marks give a quotient onto a line of negative degree. With one, $7 = 7e + 2\delta$ gives $e = 1$, $\delta = 0$, and $E = \mathcal{O}(2) \oplus \mathcal{O}(1)^5$, which cannot map nontrivially to Q_0. Thus G_{34} is excluded.

Now $R = 0$. The bound $a \leq 10t$ follows from $a \leq 14t$: it is automatic for $t \geq 3/2$, and for $t = 1/2, 1$ the logarithmic equation leaves respectively $a \leq 0, 3$. The central scalar ω has

characters $(\omega, 1, \omega, 1, 1)$ on the normal directions. Its projector preserves every actual normal lattice and splits the saturated normal bundle as $U \oplus V$ of ranks 2 and 3. We have

$$\deg U = -14t + 4s = 8 - 6t - 4\sigma < 0, \qquad 0 \longrightarrow E_4(S) \longrightarrow U \longrightarrow E_{10}(3S) \longrightarrow 0.$$

The marked splitting forces $U = \mathcal{O}(-e)^2$; a positive marked summand in U would make its determinant nonnegative. Thus $e = 3t + 2\sigma - 4 > 0$ and $t + \sigma \leq 2$. Integrality leaves $(t, s, \sigma, a, h, e) = (1, 3, 1, 3, 0, 1), (2, 6, 0, 4, 0, 2)$. The residual degrees for $V \to \mathcal{O}(-e)$ are $(-4, -1, -1)$ or twice these values, so $\ell = 3$. The second row contradicts $a \geq 3e$; the first gives $V = \mathcal{O}(1)^3$ and a nonzero map associated with the invariant of greatest degree $V \to \mathcal{O}$, also impossible. A disk without ends has $h = -18t$ and $a = 2 - 2g - 50t < 0$, since $2t$ is a positive integer. Only its last coefficient can survive, giving $\ell = 2$ and $a \geq 90t > 0$. □

Lemma 4.10. There are no full B_5 disks or smooth cusps.

Proof. The remaining horizontal groups force full B_5 ends with trivial character and width one. The complete pairs are $(B_c, \nu_c) = (6, 1), (9, 2), (10, 2), (11, 2)$; the only irreducible enlargement is B_6. For $h = -10t + R + \sum \nu_c$ the common formula gives

$$a \geq 9h/2 + 13t + R + \sigma.$$

First $a \leq 14t$ gives $h \leq 2t/9$ and $a - 5h \geq 116t/9 > 4t$. Then Lemma 2.10 gives $h < e$, $\ell \geq 2$, $a \leq 10t$, and $h \leq -2t/3$. The lowest two residual coefficients have degrees $-8t+\nu-s, -6t+\nu-s$, with $\nu - s \leq s \leq 2 + 2t - 2g$. They are negative for $t \geq 1$; for $t = 1/2$ use integrality of $h < 0$ to get $R + \nu \leq 4$. In each of the three actual models with $\nu_c = 2$, the sections of degree one vanishing at a fixed point project injectively to the first two symmetric directions. Their elliptic ratios are $x_c/x_b = \gamma_c(1 + a_c z^2)^{1/2}$ and $x_d/x_b = \gamma_d(1 + a_d z^2)^{1/2}$, with $a_c a_d(a_d - a_c) \neq 0$. Their three product sections have an invertible matrix of values and derivatives in degrees two and four, and this remains injective on the actual descending subspace. Theta translation retains all torsors and linearizations at every fixed point. Thus the two vanished coefficients give a second central division; the $\nu_c = 1$ model needs only the initial division. The five remaining degrees are $(2i - 10)t$, so Hom has dimension one and $\ell = 2$. The marked degree formula then yields

$$h + 2t \leq 2e/3 - 2(R + \sigma)/9 < e.$$

On F, the row associated with the invariant of greatest degree vanishes and the B_6 equations $dg_6 = v\, df_5$, $dg_5 = df_5 + v\, df_4$ leave only a simple pole in the next row. Its target has degree $h + 2t < e$. Two independent rows cannot factor through the rank 1 marked part. The lift fixing the point for the full group excludes the cusps by the same argument. □

Lemma 4.11. There are no full E_6 points.

Proof. Their full D_5 disks have precisely the models of Lemma 4.3: G_{33} has no full-rank reflection subgroup D_5, and other horizontal possibilities have been excluded. Since $R > 0$, that lemma gives $h > 0$ and E nef. If all $m_i \geq 0$, the effective coordinate decomposition and integrality of $(2L)\gamma_i$ give $a \leq 4t$, contrary to $a = 4h + 5t + D + \sigma$. Otherwise nefness leaves one negative mark $-e$, and the outside columns generate E, giving $h \geq e$. We need $(2L)\beta_i \geq 2$. If this degree were one, on the open set where mark i is distinct the polarization would be $\mathcal{O}(1)$ on $\mathbb{P}^6$ minus codimension-two linear subspaces. Represent $2L$ by multidegrees c_i, $\sum c_i = 2q$, so $q - c_i = 1$. Every short pair (i, j) gives $c_j \geq 1$ using its noncontracted coordinate curve. For the other pairs, the bracket $[ij]$ is a unit on the stable locus; adding its multidegree makes all $c_j \geq 1$, preserving $q - c_i = 1$. For $j, k \neq i$, the global invariant

$$[jk] \prod_{\ell \neq i} [i\ell]^{c_\ell - \delta_{\ell j} - \delta_{\ell k}}$$

restricts to $z_j - z_k$. Normality and multiplication identify the section ring with a polynomial ring on seven generators of degree one. Hence $(X, 2L) = (\mathbb{P}^6, \mathcal{O}(1))$, excluded independently by Proposition B.2. The small projective map to factorial $\mathbb{P}^6$ is an isomorphism, so this exclusion applies to X as well as W. The marked decomposition now gives $t \geq e + \delta/2$, and

$$4h = 7e + 2\delta - 5t - D - \sigma \leq 2e - \delta/2 - D - \sigma < 4e,$$

contrary to $h \geq e$. □

Lemma 4.12. There are no full D_5 disks or smooth cusps.

Proof. Only D_6 and product enlargements remain, and every cusp has full group D_5 and width one. The source line bundle E_P associated with the product invariant has degree $-5t - D/2$ and defines a direct summand

$$E = N_P \oplus E', \qquad N_P = E_P(D_D + S), \qquad p = \deg N_P = -5t + D/2 + s.$$

At D_6, the product invariant pulls back to $Z = v\psi$ in the frame P/u on the coarse quotient. At an end its column is the odd section $q\theta_P$. Divide that column and its inverse row by the same parameter; their composite is the identity. Theta translation proves this at all sixteen actual fixed points, so the splitting is global. The formulas become

$$2g - 2 + s + D/2 + \sigma = 2t, \quad p = 2(1-g) - 3t - \sigma = h/2 - t, \quad a = 8p + 13t + D + \sigma.$$

Every nonzero target has degree at least $-2t$, so $a - 5p \geq 7t > 4t$. Lemma 2.10 gives $p < e$, $\ell \geq 2$, $a \leq 10t$, and $p < 0$. The conclusion on direct summands in Proposition 2.3 gives $p = -e$. Therefore $g = 0$ and $k = 2 - t - \sigma = 2t - e \in \{0, 1\}$. For $k = 0$, $e = 2t$, $\delta = 0$, $h = -e$, and $s + D/2 = 3t$. The first two symmetric residual degrees $(2i - 8)t + s$, $i = 1, 2$, are negative. The actual derivative projections give the second cusp division; the four symmetric degrees become $(2i - 8)t$. Product and symmetric directions contribute at most two Hom dimensions, hence $\ell = 3$. On F, a further symmetric inverse row has only simple D_6 poles and target $h + 2t = 0 < e$. Three independent rows would factor through the rank 2 marked part. For $k = 1$, $(t, \sigma, e, p, h) = (1, 0, 1, -1, 0)$. The independent rows associated with the product invariant and the elementary symmetric invariant of greatest degree give $\ell \geq 3$. The equations $D = 8 - 2s$, $a = 13 - 2s = (9 - 2\ell) + 2\delta$, $\delta \leq 1$, leave $(s, D, \ell, \delta) = (4, 0, 3, 1)$. For target $N_P = \mathcal{O}(-1)$ the first residual degrees are $-3, -1$ and those after the second division are $-7, -5, -3, -1$. Only the product contributes to Hom, of dimension one; the rank 2 marked part requires dimension two. The lift fixing the point for the full group again excludes all cusps. □

Lemma 4.13. There are no full $G(3, 3, 5)$ disks or smooth cusps.

Proof. Only $G(3, 3, 6)$ and product enlargements remain; the cusps have width one and generator degrees $(1^3, 3^3)$ for the full group. The actual coarse lines give $3t \in \mathbb{Z}$, $-5t - 2R/3 \in \mathbb{Z}$. Together with $2t \in \mathbb{Z}$, they imply $t \in \mathbb{Z}_{>0}$ and $3 \mid R$; the logarithmic equation makes σ integral. The formulas are

$$2g - 2 + s + 2R/3 + \sigma = 2t, \quad a = 24(1-g) - 13t - 3R - 11\sigma, \quad h = 6(1-g) - 6t - R - 3\sigma.$$

Since $h \geq -2t$, we have $4t + R + 3\sigma \leq 6(1 - g)$. Only $(g, t, R, \sigma, s, a, h) = (0, 1, 0, 0, 4, 11, 0)$ remains. The nonzero map associated with the invariant of greatest degree to a line of degree zero gives $a \leq 10t$, contrary to $a = 11$. □

Theorem 4.14. No nine-point quotient under consideration is the Baily–Borel compactification of a nonuniform arithmetic ball quotient.

Proof. The nontrivial lattice is generated by reflections by Proposition 2.11. Start with a reflection and enlarge by Lemma 4.7. The preceding exclusions force A_5 or A_6, and the latter has a full parabolic A_5. All ends of this full A_5 disk are smooth, full A_5 ends, with width one, trivial fiber character, and six generators of degree one. Proposition 3.11 applies without genus or immersion assumptions. Its highest nonzero target satisfies $6h = a - 14t - \sigma$. First $a \le 14t$ gives $h \le 0$, and then (4.8) gives $a \le 10t$. The same local rank 2 image quotient gives $a \ge 11t + \sigma$, a contradiction. Combining this with Theorem 4.6 proves the case of nine points with arbitrary weights. □

5 Ten points

Here the ball has dimension seven. We use one argument for the three arithmetic cases, separating them only before their stabilizers become ADE. For a curve obtained from the full normalizer put $a = -K_W \cdot C$, $t = L \cdot C > 0$, and $g = g(C)$. Propositions 2.2 and 2.3 give, whenever a negative inverse row exists,

$$a = (10 - 2\ell)e + 2\delta, \qquad e \in \mathbb{Z}_{>0}, \quad \delta \in \mathbb{Z}_{\ge 0}, \qquad f^*T_W \twoheadrightarrow M^{\oplus(\ell-1)}, \quad \deg M = -e. \tag{5.1}$$

Moreover a is even, since the anticanonical class descends from $\mathcal{O}(2, \dots, 2)$. We retain $a \le 0$, $s = 0$, and every integer e allowed by (5.1) in the finite comparisons.

Lemma 5.1. In the compact case $K_W + (32/5)L$ is nef. In the nonuniform Gaussian case $K_W + 8L$ is nef. In the remaining nonuniform cases $K_W + 4wL$ is nef, where $w = |\mu(K)| = 2$ or 6.

Proof. The comparison for projective space in Proposition 6.1 excludes $W = \mathbb{P}^7$. In that case $\pi^* A_X = \mathcal{O}(k)$ with $k > 0$ is ample, so π is finite birational onto normal X_μ and hence is an isomorphism. The K_W-negative rays of Proposition 2.8 have respective degrees against $-K_W$ $2, 6, 4, 2$: a coordinate curve, and a line where $q = 2, 3, 4$ labels coincide. Their collision spaces are $\mathbb{P}^{8-q}$ with normal bundle $\mathcal{O}(-1)^{q-1}$. Relative nefness prevents their contraction, and their maps to the normal collision images are isomorphisms. Lemma B.3 and Proposition B.2 give L-degree at least one for $q = 2$. For compact groups Proposition B.4 gives the bounds $1, 2/3, 1/3$ on the three collision rays. Their ratios are at most six, but Lemma 3.9 gives $L \cdot \gamma \ge 5/16$ on a coordinate curve, hence ratio at most $32/5$. For nonuniform groups wL is an actual integral bundle. This gives ratios at most $2w, 6, 4w, 2w$. When $w = 4$, a slope $1/4$ on the $\mathbb{P}^5$ collision would induce a quotient by Lemma B.3. The induced quotient is nonuniform: its actual fixed Hermitian space over the original imaginary quadratic field has dimension at least three, so Meyer supplies a rational isotropic line. Proposition B.6 excludes it. Its slope is therefore at least $1/2$, and all four ratios are at most eight. □

Lemma 5.2. In a nonuniform ten-point realization, all point stabilizers are products of ADE groups, all ends relevant to D_6 and E_6 have width one, and $4L$ is integral. If the field is not Gaussian, $2L$ is integral.

Proof. Apply the monomial comparison of Proposition 3.4, using Lemma 5.1, before using any stronger nef inequality. The domains are given below for rank 6 and in Appendix C.6 for the auxiliary projective spaces; the shared lists of cusps and the conditions on the marked degrees are in Appendices C.1–C.2. For clarity, we give the reductions determining those domains. The primitive group G_{34} is treated separately before the ordinary monomial comparisons. Its polynomial degrees are $(6, 12, 18, 24, 30, 42)$ and its section degrees are $(1, 3, 4, 6, 7, 9, 12)$, with no torsor and width one. Its principal theta divisor is irreducible, and its multiplicity is at most four [21, Theorem 1.4], below the first invariant degree six; hence its invariant radial section is nonzero. Three end zeros leave coefficient degrees $(d - 42)t + 9s$ and $h = -42t + 12s$,

$a = 84(1-g) - 50t - 41\sigma$, with even s and $j = 6t$ integral. The weak inequality $a \le 24t$ gives $g = 0$, $j \le 8$ for $h \ge 0$ and $j \le 11$ for $h < 0$. The logarithmic equation and (5.1) exclude every row; the sole remaining value $1/3$ for the contribution of disk inertia is not an allowed sum of $1 - 1/e$. For parameter six set $j = 6t$, $h_P = (-36t + R + \nu)/p$ and $h = h_P$ for $p = 1$, $h = ph_P + 6t$ otherwise. The product direction is preserved since $6^5 > 6p$. The degree and rank bounds reduce the calculation to $g = 0$, $1 \le j \le 9$, $s \le \lfloor 2 + j/3 \rfloor$, $R \le \lfloor (6/5)(2 + j/3 - s) \rfloor$. The twelve residual rows have respectively $p = 1$ (two rows), $p = 2$ (five), or $p = 3, 6$ (five). The first two violate $a \ge 6(-h)$; the five $p = 2$ rows require $a = 4j$, with three further contradicted by a kernel target of degree zero below $e = j$; the last five have $h < h_P < 0$, contrary to the splitting of Proposition 2.3. The tests for parameter three use $G(3, 1, 6)$ with end degrees $(1^7), (1^2, 3^5), (1, 3^6)$ and $1 \le 3t \le 4$, followed by $G(3, 3, 6)$ with $(1^3, 3^4)$ and $1 \le 3t \le 3$. They have no residual row. In the Gaussian case use $G(4, p, 6)$, $p = 1, 2, 4$, and $j = 4t$. The negative coefficient rank bound gives $j \le 5, 6, 8$, respectively. For $p = 4$, it suffices to retain the following end types, where α is the section tuple and κ the actual width:

type	α	κ
A	$(1^3, 2^4)$	$1, 2$
B	$(1^2, 2^5)$	$1, 2$
C	$(1, 2^2, 4^4)$	$1, 2$
D	$(1, 2, 4^5)$	$1, 2$
E	$(1^2, 2, 4^4)$	1
X	$(1^2, 4^5)$	$1.$

The type X is the nontrivial torsor; width two would require a full fixed point and is therefore excluded. For each multiset of s ends, put $B = \sum_c \sum_i \alpha_{c,i}/\kappa_c$ and $\nu = \sum_c \max_i \alpha_{c,i}/\kappa_c$. The complete necessary numerical conditions are

$$1 \le j \le 8, \quad 0 \le s \le \lfloor 2 + j/2 \rfloor, \quad 0 \le R \le \lfloor (4/3)(2 + j/2 - s) \rfloor,$$
$$\sigma = 2 + j/2 - s - 3R/4 \in \{0\} \cup [1/2, \infty),$$
$$a = B - s + 2 - 33j/2 + 21R/4 \in 2\mathbb{Z}, \quad a \le 2j, \qquad h = -5j + \nu + R.$$

There is no solution with $h \ge 0$. For $h < 0$, put $u = \max(0, \min(5, \lfloor (\nu - s)/j \rfloor + 1))$, and use $2 \le \ell \le \min(7, 1 + u + \epsilon_P)$. Here ϵ_P is the number of possible product coefficients, so $\epsilon_P \le 1$. When all widths are one, the degree $E_P = -3j/2 - 3R/4$ must be integral and $h - E_P - s < R$ forces $\epsilon_P = 0$. Finally require $e \ge \max(1, -h)$ and $a - (10 - 2\ell)e \in 2\mathbb{Z}_{\ge 0}$. Substitution in these bounded sets leaves only the following signatures; all have $R = 0$, $\ell = 5$, and at most two ends of width two:

j	s	ν	h	a	σ
2	3	7	-3	$2, 4$	0
3	3	9	-6	0	$1/2$
3	3	10	-5	$0, 2, 4, 6$	$1/2$
4	4	12	-8	0	0
4	4	13	-7	$0, 2, 4, 6$	$0.$

At such an end the least nonzero generator degree is the actual width; thus the curve is immersed in the coarse coordinate, with no fiber correction. The test in Proposition 3.7 using the normal bundle, for degrees $(4, 8, 12, 16, 20, 6)$, requires

$$2a - 4 + 45j - 12\nu \le 0.$$

The five rows give $2a + 2$, $2a + 23$, $2a + 11$, $2a + 32$, $2a + 20$, all positive. This excludes the remaining groups and cusps with parameter four. For a persistent factor of rank at most 5, apply Meyer to its actual fixed Hermitian space of dimension at least three; the induced

quotient isomorphic to projective space is nonuniform over the same field. The reduction for persistent factors, Proposition B.2, and Propositions B.5–B.6 now reduce the possibilities to parabolics of A_7, B_7, D_7, E_7. For B_6, the four tuples of generator degrees at cusps of width one are $(1^7), (1^3, 2^4), (1^2, 2^5), (1, 2^6)$. With $\nu = \sum_c \max_i \alpha_{c,i}$, its equations are

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = \sum_c (B_c - 1) + 2 - 2g - 42t + 6R, \quad h = -12t + \nu + R.$$

The integrality of $42t$ and $4t$ gives $2t \in \mathbb{Z}_{>0}$. For $h \geq 0$ one obtains $t = 1/2$, $g = \sigma = 0$, and $a \geq 17 - 2s \geq 11 > 8t$. For $h < 0$ only $g = 0$, $2t = 1, 2$ remain, and substitution of the four tuples in (5.1) gives no row. The stabilizers are therefore ADE products. Normalized reflection lifts in dimension eight have determinants ± 1; their scalar kernel has order dividing $\gcd(16, w)$. The descent at cusps uses the invariant corresponding power of the frame $h(v, \ell) = 1$. It gives $2L$ outside the Gaussian case and $4L$ in that case. □

Lemma 5.3. Once the ambient stabilizers are ADE products, $K_W + 6L$ is nef.

Proof. A slope below one on the closed locus $Z = \mathbb{P}^5$ where three labels coincide induces a quotient by Lemma B.3. Its pointwise group is A_2 or $2A_1$. The effective normalizer restrictions are products of classical A, B, D groups; the exceptional restrictions are

J	F	$N_J(F)/F$
E_7	A_2	$A_5 \times \mathbb{Z}/2$
E_7	$2A_1$	$A_1 + B_4$
E_6	A_2	$A_2 \wr (\mathbb{Z}/2)$
E_6	$2A_1$	B_3

and $E_6/A_1 = A_5$ when A_1 is a separate pointwise factor. Smoothness shows that the effective restrictions are generated by reflections. The cusp character kernel is a finite group of this type and rank at most 4, hence order at most 384. This excludes the nonclassical cusps: the Gaussian $G(4, 4, 4)$ lift has order 1536; parameter three or six has nonreal character; F_4 has kernel order at least 576; G_{32} lifts in its entirety; and the G_{29}, G_{31} kernels have order at least $480, 2880$, respectively. For the G_{31} torsor the residual orbit has size six and fiber character kernel order at least 480. Thus only A_4, B_4, D_4 cusp types remain. The integrality of $2L$, or of $4L$ and $24c_2^{\mathrm{orb}} = 360L^2$, makes $1/2$ the only possible slope below one. At that slope put $t = u/2$ and $a = 6u$. For B_4, Euler and $h \leq -4t + 4(1 - g) - 2\sigma$ leave $g = 0, u = 1$; the four end tuples give $h = 0 < u$. For D_4 the equations give

$$7u + 2s + 7\sigma = 16(1 - g), \qquad (u, t, s, R, \sigma) = (1, 1/2, 1, 2, 1), (2, 1, 1, 6, 0).$$

The curve is a line or smooth conic, whose normal bundle has largest summand 1 or 4. The column of the quotient differential associated with the quadratic invariant gives a nonzero map of degree $-2t + R + s = 2$ or 5 into that normal bundle, a contradiction. Finally A_4 gives $5h = 3t - \sigma < 5u$. Consequently $L|_Z \geq H$. On a Gaussian collision $\mathbb{P}^4$ of four labels, the rank 3 pointwise ADE group has real normalizer restriction. The induced group is generated by reflections and its lifts in dimension five have scalar kernel dividing $\gcd(10, 4) = 2$. The same frame at the cusp therefore proves that the restriction of the original $2L$ is integral. Its slope is at least $1/2$. The coordinate bound $5/16$, combined with the actual integral $2L$ or $4L$, gives coordinate degree at least $1/2$. The four degrees against $-K_W$ $2, 6, 4, 2$ now have L-degree bounds $1/2, 1, 1, 1/2$, so Proposition 2.8 proves the assertion. □

Theorem 5.4. For every ten weights satisfying (1.1), the quotient is not the Baily–Borel compactification of an arithmetic ball quotient.

Proof. For compact groups the reduction for maximal persistent factors and Lemma 3.15 remove the monomial factors outside the parabolics of $A_7, B_7, D_7, G(3,3,7), E_7, G_{34}$. Factors of rank at most 5 are excluded by Propositions B.2 and B.4 and equality of basic degrees. The compact B_6 comparison is included in Lemma 3.15. At this stage $a \le 32t/5$ by Lemma 5.1. In the nonuniform cases, Lemmas 5.2 and 5.3 give ADE stabilizers and $a \le 6t < 32t/5$. The common E_6 comparison in Lemma B.7 below excludes E_7, then the remaining E_6 curves. For compact groups G_{34} gives $a \ge 252t$, and the ordinary monomial comparison then gives $a \ge 74t/5$ for $G(3,3,6)$ and $a \ge 44t/5$ for D_6. Each contradicts the compact bound. For nonuniform D_6, the cusps have width one and generator degrees $(1^4, 2^3)$, and the common equations are

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = 9s + 2 - 2g - 36t + 11R/2, \quad h = -10t + R + 2s.$$

They reduce to $g = 0$, $j = 2t = 1, 2, 3$. The full GIT intervals leave only

$$(j, s, R, \sigma, a, h) = (1, 2, 0, 1, 2, -1), (2, 2, 4, 0, 6, -2), (2, 3, 2, 0, 4, -2), (2, 4, 0, 0, 2, -2).$$

The product contributes one map. The flag for the minimal polarization of Proposition 3.13, after two end divisions, leaves symmetric degrees $h + ij - 2s$, less than R for $i < 5$ and equal to R for $i = 5$. The interior zeros therefore leave at most one symmetric map. But (5.1) requires $\ell \ge 4$ and at least three independent maps, a contradiction. Finally Proposition 3.11 gives $a \ge 33t/2 + \sigma$ for the last A_6 disk. This contradicts either available nef bound and completes the ten-point case without any assumption on branch support or cusp width. □

6 Eleven or more labels

Write $d = n - 3$, $r = d - 1$, and $N = d + 1$. On the smooth small model $\pi : W \to X_\mu$, put $L = \pi^*\mathcal{L}$. For a fixed curve $f : C \to W$ of the full arithmetic normalizer, put

$$a = -K_W \cdot f_*C, \qquad t = L \cdot f_*C > 0, \qquad g = g(C).$$

No sign condition on a will be used before it follows from the tangent comparison. We use strong induction on the number of labels; the cases of nine and ten labels are proved in Sections 4 and 5. If an irreducible factor of rank at most $d - 2$ persists, Proposition 2.14 identifies its fixed compactification with the original merged weighted quotient on $n - s$ labels, where $s \le 2$. Thus $9 \le n - s < n$, and the induction hypothesis excludes every such persistent factor. This uses the original weights and their actual fixed quotient; no ball structure is transferred to the small model. All sharper nef bounds below are established only after their required stabilizers have been excluded.

6.1 The common monomial calculation

Proposition 6.1. For $7 \le k \le 9$, projective space $\mathbb{P}^k$ is not the Baily–Borel compactification of an arithmetic quotient of $\mathbb{B}^k$.

Proof. For a nonconstant curve in $\mathbb{P}^k$, set $u = \deg f^*\mathcal{O}(1) > 0$. The Euler sequence makes every nonzero line quotient of $f^*T_{\mathbb{P}^k}$ have degree at least $u = a/(k+1)$. The argument in Proposition 2.11 proving generation by reflections and arithmeticity applies to this smooth simply connected rational compactification. In particular, a persistent reflection factor cannot occur: on a complete curve in its fixed quotient, away from the finite boundary, an inverse invariant differential gives a quotient of negative degree. Parabolic classification therefore leaves corank 1 monomial curves, the exceptional groups G_{34}, E_6, E_7, E_8 in the indicated low ranks, and finally type A. Appendix C.6 performs these comparisons with all full cusp records.

The decisive Chern bounds are $a/t \le 16$ for $k = 7$, $a/t \le 18$ for $k = 8$, and $a/t \le 20$ for $k = 9$. They make the last type A_{k-1} inverse differential negative, contrary to the Euler sequence. All compact monomial parameters $m \ge 2$ are included; no restriction on their CM field is made. □

For $n = 11, 12$, Proposition 6.1 excludes $W = \mathbb{P}^d$. If the source W is projective space, then the pullback of an ample line bundle A_X is $\pi^* A_X = \mathcal{O}(k)$ with $k > 0$, hence is ample. Thus π has finite fibers, and a finite birational morphism onto normal X_μ is an isomorphism. Corollary 2.9 then gives $a \le rwt$ in the nonuniform case. For $n \ge 13$, we retain the weaker curvewise bound $a \le (r+2)wt$ of Lemma 2.10 until a nonpositive tangent quotient has been found. Thus the argument for large n needs no auxiliary theorem for projective space.

For the disk group $G(m, p, r)$, use

$$D = \frac{mr(r-1)}{2} + \frac{mr}{p}, \quad \eta = r - 1 + \frac{1}{p}, \quad k = \begin{cases} mr & p = 1, \\ m(r-1) & p > 1. \end{cases}.$$

At an end let α_c be its full normalized section tuple, κ_c its actual fiber character order, $B_c = \sum_i \alpha_{c,i}/\kappa_c$, and $\nu_c = \lfloor \max_i \alpha_{c,i}/\kappa_c \rfloor$. Write R for irreducible enlargements, s for ends, and σ for the sum of the remaining disk inertia corrections. The common comparison gives

$$2g - 2 + s + (1 - 1/m)R + \sigma = 2t, \tag{6.1}$$

$$a = \sum_c (B_c - 1) + 2 - 2g - Dt + \eta R, \tag{6.2}$$

$$j = mt \in \mathbb{Z}_{>0}, \qquad h = -kt + \sum_c \nu_c + R. \tag{6.3}$$

Here h is the degree of an actual target of the inverse differential, with its finite character and coarse residue retained. These equations are used together with the homomorphism from the tangent bundle to a line of negative degree

$$a = (n - 2\ell)e + 2\delta, \qquad e \ge -h, \quad e > 0, \quad \delta \in \mathbb{Z}_{\ge 0}. \tag{6.4}$$

The coefficient count, raw cusp table, and exact integer ranges are recorded once in Appendix C.2.

Proposition 6.2. If $r = n - 4 \ge 9$, no monomial fixed disk or irreducible smooth monomial cusp occurs.

Proof. Disks without ends are excluded by Lemma 3.15. For a nonuniform realization, the arithmetic trace restricts m to $2, 3, 4, 6$. Descending in m excludes smaller end parameters, and every end has group $G(m, q_c, r)$, $q_c \mid p$, with $\kappa_c \mid p/q_c$. The cusp bounds

$$(r-1) \max \alpha_c \le \sum_i \alpha_{c,i} \le rm + 1$$

and the weak bound $a \le (r+2)wt$ give

$$\begin{aligned} (r-1)h &\le \{(r+2)w + D + 2 - (r-1)k\}t - (1 + 1/p - 1/m)R - \sigma, \\ a &\le (2rm+2)(1-g) - \{r(r-5)/2 + r/p\}j + \{r - 1 + 1/p - r(m-1)\}R. \end{aligned}$$

For $m \ge 3$, the first coefficient is at most $-3r^2/2 + 21r/2 + 14$ if $p = 1$, or $-3r^2/2 + 12r + 11$ otherwise. For $m = 2$, $h \le -(r-3)j + 4(1-g) \le -2$. Thus $h < 0$ without a prior assumption on a or $W \ne \mathbb{P}^d$. Lemma 2.10 now strengthens the bound to $a \le rwt$, so the following finite domain is unchanged. The coefficient count gives $\ell \le 4$ for $g \ge 1$, or $\ell \le 6$ for $j \ge 6$, and then (6.4) gives $a > 0$. The displayed upper bound excludes $g \ge 1$ and $j \ge 6$, the latter by

$3r^2-26r-1 \geq 8$. For $r \geq 29$, the remaining bound $\ell \leq 14$ contradicts $a \leq -r^2/2+43r/3+2 < 0$. For $r \geq 10, j \geq 5$, the contradiction is $(15r^2-142r-6)/6 > 0$. The complete residual domain is therefore $9 \leq r \leq 28$, with $j \leq 5$ at $r = 9$ and $j \leq 4$ otherwise. The 170 rows in Appendix C.3 are excluded by the common bundle tests, the exact Eisenstein width, and the absent full fixed points on the Gaussian torsor. The cusp lifts in Appendix C.1 then exclude the corresponding cusps. □

6.2 The terminal comparison for $n \geq 13$

Proposition 6.3. The nonexistence theorem holds for every $n \geq 13$.

Proof. By the induction hypothesis, no irreducible factor of rank at most $d-2$ persists. A maximal factor outside type A must therefore lie in a monomial group of rank $d-1$ or d, and Proposition 6.2 excludes it. Parabolic classification now gives an A_{d-1} disk, with larger groups A_d and products, and cusps of width one whose section degrees for the full group are 1^d. All finite stabilizers are products of type A. Put

$$b_N = \prod_{q|N,\ q \text{ prime}} q^{\lceil 2v_q(N)/3 \rceil}, \qquad C_N = \frac{2N^2}{b_N}.$$

For the integral $T = 2NL$, Chern integrality gives $N^2 \mid b^3$ on every collision space $T|_Z = bH$ of dimension at least three. Thus $b_N \mid b$, and every K_W-negative collision line has

$$\frac{-K_W \cdot \ell}{L \cdot \ell} \leq \frac{2N^2}{b_N} = C_N.$$

This includes the singleton collision $Z = W = \mathbb{P}^d$, if it exists. Every negative collision space here has dimension at least five, so these Chern restrictions are legitimate. For coordinate curves, Lemma 3.9 gives

$$L \cdot \gamma \geq \frac{N-3}{2N}, \qquad \frac{-K_W \cdot \gamma}{L \cdot \gamma} \leq \frac{4N}{N-3} < C_N,$$

where the last inequality follows from $b_N \leq N$. The Mori generators in Proposition 2.8 therefore give $K_W + C_N L$ nef. There is no preliminary exclusion of $W = \mathbb{P}^d$ in this argument.

The type A_{d-1} comparison, Proposition 3.11, gives

$$a \geq \frac{N^2-4N+1}{2}t + \sigma.$$

For $N = 11, 12, 13, 14$, one has $b_N = N$, and $C_N = 2N < (N^2-4N+1)/2$. For $N \geq 15$, the inequalities $b_N \geq N^{2/3}$ and $N^{1/3} < N/6$ give

$$C_N \leq 2N^{4/3} < \frac{N^2}{3} < \frac{N^2-4N+1}{2}.$$

The last strict inequality is $N^2 - 12N + 3 > 0$, valid for $N \geq 15$. This contradicts $a \leq C_N t$ for every $n \geq 13$. □

6.3 The two remaining high-dimensional cases

Proposition 6.4. The nonexistence theorem holds for twelve labels.

Proof. Here $d = 9, r = 8$, and descent of $\mathcal{O}(1, \dots, 1)$ gives $a \in 2\mathbb{Z}$. An E_8 disk has polynomial degrees $(2, 8, 12, 14, 18, 20, 24, 30)$, section tuple $(1, 2, 2, 3, 3, 4, 4, 5, 6)$, and

$$2g - 2 + s + \sigma = 2t, \quad a = 29s + 2 - 2g - 128t, \quad h = -30t + 6s, \quad 2t \in \mathbb{Z}_{>0}.$$

These equations and parity give $h + 2t < 0$. Thus the projection of the inverse differential to the coordinate of greatest invariant degree kills both df and the quadratic invariant. The unique theta section of degree one of E_8 satisfies

$$\Theta = E_4 + c(E_2E_4 - E_6)p_2 + O(z^4), \qquad c \neq 0.$$

If $E_4 = 0$, then $E_4^3 - E_6^2 = 1728\Delta \neq 0$ makes its quadratic coefficient nonzero. Consequently the radial or the quadratic equation forces a second cusp zero in every horizontal coefficient. Their residual degrees are $(d_i - 30)t + 4s$. For $g \geq 1$ at most two directions remain; for $g = 0, t \geq 1$ at most five remain. Thus $\ell \leq 3$ or 6, contradicting (6.4) and $a \leq 60(1 - g) - 70t$. At $t = 1/2$, the possible $s = 0, 2$ give $a = -62, -4$ and again too few directions. This excludes E_8 points, disks, and cusps before the monomial reduction.

Lemma 3.15 excludes monomial disks without ends. For nonuniform monomial disks, (6.1)–(6.3), $a \leq 8wt$, and the coefficient count imply $g = 0, h < 0, 1 \leq j \leq 7$. Appendix C.4 gives the complete 117-row exclusion. The induction hypothesis excludes persistent factors of smaller rank, so every remaining finite stabilizer is a product of type A groups. For $T = 20L$ restricted as bH to a collision $\mathbb{P}^k$, $k \geq 3$, Chern integrality gives

$$\frac{27b^2}{20}, \ \frac{9b^3}{25} \in \mathbb{Z}, \qquad 10 \mid b.$$

Thus its lines have $t \geq 1/2$; their degree against $-K_W$ is at most eight. Axes have $a = 2, t \geq 7/20$ by Lemma 3.9. The Mori generators therefore give $K_W + 16L$ nef, including balanced crepant rays. For the resulting A_8 disk, Proposition 3.11 gives $a \geq 61t/2 + \sigma > 16t$, the required contradiction. □

Proposition 6.5. The nonexistence theorem holds for eleven labels.

Proof. Here $d = 8, r = 7$. First use only the preliminary bound $a \leq 7wt$. The monomial equations give $g = 0, h < 0, 1 \leq j \leq 7$; Appendix C.5 records the full 413-row domain. Its preliminary tests, the compact monomial lemma except for D_7 with possible primitive enlargements, and the induction hypothesis for persistent factors restrict every irreducible finite factor to the parabolic class generated by

$$A_8, \ B_8, \ D_8, \ G(4, 2, 8), \ G(4, 4, 8), \ E_8.$$

In this class all reflections have order two, and products of total rank at most 3 have orders dividing 384. Thus $T = 18L = 2K_W + B$ is integral and $384c_3^{\mathrm{orb}} = -32256L^3$ extends across the cusps and the small exceptional locus. On a collision $\mathbb{P}^k$, $k \geq 3$, with $T = bH$, this says

$$\frac{448b^3}{81} \in \mathbb{Z}, \qquad 9 \mid b, \qquad L \cdot \ell \geq \frac{1}{2}.$$

The closed divisor where two labels coincide is $E = \mathbb{P}^7$. Relative nefness of K_W makes $\pi|_E$ finite, and normality of its original merged GIT image makes this restriction an isomorphism. It cannot be a branch component, by Proposition 6.1; its normal line is $\mathcal{O}(-1)$, so Lemma 3.10 gives $L|_{\mathbb{P}^7} \geq H$. The remaining collision rays have degree against $-K_W$ at most five, and axes have $t \geq 1/3$. The Mori generators therefore give

$$K_W + 10L \text{ nef.} \tag{6.5}$$

This argument precedes the exclusion of E_8.

At an E_8 point take its E_7 parabolic disk and let $R > 0$ count its E_8 points. Its cusps of width one and the characters of the full groups give

$$2g - 2 + s + R/2 + \sigma = 2t, \quad 2t \in \mathbb{Z}_{>0}, \quad a = 18s + \frac{57}{2}R + \sigma - 72t.$$

The target associated with the invariant of greatest degree has $h = -18t + 6R + 4s$, but (6.5) gives

$$\frac{57}{2}R + 18s + \sigma \le 82t. \tag{6.6}$$

If $h \ge 0$, these inequalities give $5t/2 - 2 \le R \le 2t/3$, hence $t \le 12/11$. If $R + s \ge 4t$, they give $4t - 4 \le R \le 20t/21$, hence $t \le 21/16$. In either case $2t \in \mathbb{Z}$ leaves $t = 1/2, 1$, forcing $R < 1$. Therefore $h < 0$ and $R + s < 4t$. The intrinsic flags and derivatives of the theta sections associated with the root lattice give the following residual degrees:

d_i	deg	d_i	deg
2	$-16t + 3R + 3s$	10	$-8t + 2R + 2s$
6	$-12t + 3R + 2s$	12	$-6t + 2R + s$
8	$-10t + 3R + 2s$	14	$-4t + R + s$

All are negative; use $3R + 2s \le 82t/9$ and $2R + s \le 328t/57$ for the middle entries. The coefficient of the invariant of degree eighteen is constant, so the Hom space is one-dimensional. The GIT kernel K_0 consequently has no quotient of degree less than $e \ge -h$. But its projection to the invariant of degree fourteen has degree $h_{14} = -14t + 5R + 3s$, and

$$h_{14} + h = 11R + 7s - 32t \le \frac{7}{18}\left(\frac{57}{2}R + 18s\right) - 32t \le -\frac{t}{9} < 0.$$

This contradiction excludes E_8 in every genus and CM field.

Now the terminal tests in Appendix C.5, using (6.5), exclude all monomial factors of ranks 7 and 8 and their full cusps. For D_7, no E_7 enlargement is possible, since $\det D_7 / \det E_7 = 2$ is not an integral index square; its possible E_8 enlargements have just been excluded. Its last nonuniform row has $a = 6, t = 1/2$, contrary to $a \le 10t$, and the compact lemma now applies. An E_7 disk has no enlargement left. Without ends its projection of the inverse differential to the coordinate of greatest invariant degree gives $a \ge 126t$. With ends,

$$a = 36(1 - g) - 36t - 17\sigma, \quad h = -18t + 4s, \quad 2t \in \mathbb{Z}_{>0}.$$

Two cusp divisions leave at most two directions for $g \ge 1$, or four for $g = 0, t \ge 1$, forcing $a > 0$ when the displayed formula gives $a \le 0$. At $t = 1/2$, $s = 1, 2, 3$ give $(a, h) = (-16, -5), (1, -1), (18, 3)$. Only $s = 2$ survives positivity and (6.5). Then (6.4) forces $\ell = 5, e = 1, \delta = 0$. Four independent maps to $\mathcal{O}(-1)$, after two divisions, have coefficient degrees $(-4, -2, -1, 0, 1, 2, 4)$. After the constant coefficient of the invariant of degree ten vanishes, three independent covectors remain; the next projection defined by the theta sections associated with the root lattice forces a third division and leaves only degrees fourteen and eighteen, a contradiction. All persistent smaller factors are excluded by the induction hypothesis, including type A. Thus an A_7 disk exists. Proposition 3.11 gives $a \ge 23t + \sigma > 10t$, contradicting (6.5). □

Proof of Theorem 1.1. The case of nine labels is proved by Theorems 4.6 and 4.14, and the case of ten labels by Theorem 5.4. Assume the result for every smaller number of labels, starting at nine. Proposition 2.14 excludes each persistent irreducible factor of rank at most $d - 2$, because its fixed compactification is the original quotient with $s = 1$ or 2 labels merged. Propositions 6.5, 6.4, and 6.3 exclude the remaining cases $n = 11$, 12, and $n \ge 13$, respectively. The small model of Lemma 2.1 and the arguments avoiding the finite boundary include all stability walls. □

A Theta derivatives and quotient differentials

This appendix proves Propositions 3.13 and 3.14. The degrees of polynomial invariants, the degrees of generators of the ring of invariant sections, and the weights of Jacobi forms remain distinct throughout.

A.1 The root theta projection

Proof of Proposition 3.13. Write the Wirthmuller generators as $\phi_0, \dots, \phi_r$, of Jacobi weights $0, -d_1, \dots, -d_r$ and indices $1, m_1, \dots, m_r$. The relative and fiber statements of [30, Theorem 3.6(ii)–(iv)], at level three, identify the section ring on each elliptic fiber with the specialization of the polynomial Jacobi algebra. For linear coordinates z_a, set $\mathcal{J}_{0i} = m_i\phi_i$, $m_0 = 1$, and $\mathcal{J}_{ai} = \partial_{z_a}\phi_i$. The Jacobian formula is

$$\det \mathcal{J} = c \prod_{\alpha \in R^+} \frac{\vartheta(\tau, (\alpha, z))}{\eta(\tau)^3}, \qquad c \neq 0.$$

[29, Propositions 2.2–2.3, Theorem 3.1, Section 4]. Induction on Taylor degree in the Jacobi transformation law kills terms of ϕ_i below degree d_i. The inhomogeneous term only uses lower terms already zero, leaving a holomorphic modular form of negative weight. If $\phi_0(\tau, 0) = 0$, the Jacobian would have order at least $|R^+| + 2$, contradicting the displayed root product. Put $r_i = \phi_i/\phi_0^{m_i}$, and write $r_i = \sum_k a_{ik}(\tau)p_k$ modulo the square of the invariant maximal ideal. The coefficient a_{ik} has modular weight $d_k - d_i$. Negative weights vanish, weight zero coefficients are constant, and the leading Jacobian makes each equal degree diagonal block invertible. For type D_r, the index one form $\prod_i \vartheta(\tau, z_i)/\eta(\tau)^{3r}$ is odd under one coordinate sign change. Averaging separates the product P from a symmetric invariant of the same degree. Thus the indicated projections are triangular with invertible diagonal blocks. Polynomial generation shows that degree at most j derivatives are spanned by the dr_i with $m_i \leq j$. This proves the projection statement while retaining higher primitive components.

For the translation assertion, put $\lambda_\Theta(x) = t_x^*\Theta \otimes \Theta^{-1}$. Primitive simple roots give $A_\tau^{W(R)} = \ker \lambda_\Theta$. At a common fixed point with trivial fiber character, a theta group lift of t_x commutes with the Weyl linearization: its scalar commutator is the ratio of the characters at x and zero. It therefore transports the actual section spaces and their projections. The same comparison preserves eigenspaces of any further automorphism fixing both points with trivial fiber action. The modular action of $-I$ gives parity $(-1)^{d_i}$. □

For the D_v cusp models used below, $v = 5, 6$, the actual polarization and trivial fiber character can be recovered without choosing an isogenous substitute. Adjacent simple roots give $Q(D_v)_{W(D_v)} = 0$ and $\mathrm{Pic}^0(A)^{W(D_v)} = 0$. Invariant positive integral alternating forms are integral multiples of the root form tensored with the elliptic form. Connected reflection divisors through the origin, together with generation of the entire group of lifts by reflections, force trivial fiber character. The minimal section degrees are $(1^4, 2^{v-3})$ [25]. Every higher Veronese has a singular vertex: no nonidentity scalar is a pseudoreflection on the four degree one coordinates. Consequently the actual smooth cusp uses the minimal bundle. Proposition 3.13 gives

$$F_1 = \mathbb{C}\, dP \oplus F_1^{\mathrm{sym}}, \qquad F_1^{\mathrm{sym}} \longrightarrow \langle df_1, df_2 \rangle$$

isomorphically, uniformly in the elliptic parameter and at all relevant translated fixed points.

A.2 Why the algebra comparison preserves the flags

Let G be a finite complex reflection group with well-generated irreducible factors and reflections of order two, and let $\pi : V \to V/G$. The natural Saito product makes $\operatorname{coker} d\pi$ a finite birational unital algebra on the reduced reflection arrangement. Here is the localization argument

needed to use that product. Euler multiplication has matrix $U = x_r I + U_0(x_1, \dots, x_{r-1})$ and determinant the discriminant [17, Theorem 5.6 and its proof]. Its characteristic polynomial is the discriminant with x_r translated by the spectral variable. Generic roots on every discriminant component are distinct [5, Theorem 5.3]. There the product is diagonal in canonical coordinates and the quotient is a fold. The square differential matrix has nonzero determinant, so its image is isomorphic to its locally free source. Multiplication preserves this image at every point of codimension one by the fold description. The coefficients in a basis of the image therefore extend by Hartogs, as in [12, Proposition 2.3]. Its square presentation has reduced determinant, so its cokernel is Cohen–Macaulay and torsion-free of rank 1 on each arrangement component.

For a finite birational unital S-algebra M, one has $\mathrm{End}_S(M) = M$. An endomorphism is multiplication in the total quotient ring, and its value at one belongs to M. An isomorphism of the underlying modules thus conjugates their endomorphism algebras; the original map differs from the resulting algebra isomorphism by a unit. At a point with stabilizer P, the quotient germ is the product of its fixed space and the normal P-quotient. The fixed factor contributes zero to the cokernel, and parabolic subgroups remain well-generated [5, Lemma 2.7]. This proves localization, up to a unit, for the actual normal quotient.

For the Weyl inclusions, let $F(x,t)$ be the corresponding semiuniversal deformation with base B, discriminant D, and critical incidence map $\rho : \{F = d_x F = 0\} \to B$. Kodaira–Spencer identifies $\mathcal{T}_B / \operatorname{Der}_B(-\log D)$ with $\rho_* \mathcal{O}_{\{F = d_x F = 0\}}$ [12, Introduction, (0.1)]. Euler multiplication has the same image [17, Lemma 4.6]. Apply the preceding endomorphism argument before taking fibers to compare these two products. Apply it also on the reflection arrangement to compare the GMS product with the natural Saito product. Only then use the algebra fiber comparison of [12, Proposition 4.21, Theorem 4.23]. Thus the actual quotient differential fiber is the local Tjurina algebra up to a unit. The sheaves on the deformation base and on the reflection space have not been identified as sheaves on the same space.

The complete flags are intrinsic ideals, with $\mathfrak{m}$ the maximal ideal:

$$\begin{array}{r|l} D_5 & \mathfrak{m}^3 \subset \mathfrak{m}^2 \subset \mathrm{Ann}(\mathfrak{m}^2) \subset \mathfrak{m} \subset A \\ E_6 & \mathfrak{m}^3 \subset \mathfrak{m}J \subset \mathfrak{m}^2 \subset J + \mathfrak{m}^2 \subset \mathfrak{m} \subset A \\ E_7 & \mathfrak{m}^4 \subset \mathfrak{m}^3 \subset \mathfrak{m}I \subset \mathfrak{m}^2 \subset I \subset \mathfrak{m} \subset A \\ G_{33} & \mathfrak{m}S \subset \mathfrak{m}^2 \subset S \subset \mathfrak{m} \subset A. \end{array}$$

The first three algebras and their ordered bases are

$$\begin{array}{r|c|c} D_5 & \mathbb{C}[x,y]/(xy, x^2 + y^3) & (y^3, y^2, x, y, 1) \\ E_6 & \mathbb{C}[x,y]/(x^2, y^3) & (xy^2, xy, y^2, x, y, 1) \\ E_7 & \mathbb{C}[x,y]/(x^2 + y^3, xy^2) & (y^4, y^3, xy, y^2, x, y, 1). \end{array}$$

For E_6, J is the closure of square-zero elements in $\mathfrak{m} \setminus \mathfrak{m}^2$, here the ideal (x); for E_7, $I = \mathrm{Ann}(\mathfrak{m}^3)$. Direct multiplication shows each flag is the sequence of initial spans of its ordered basis. It is therefore preserved both by algebra isomorphisms and by unit multiplication.

Brieskorn identifies the weighted deformation base with the Weyl quotient [8, p. 282]. For $E_7 \subset E_8$ and $E_6 \subset E_7$, after splitting off a square use respectively

$$\begin{array}{c|c} F_a & (h_i) \\ \hline y^3 + ayz^3 + z^5 & (z^4, z^3, yz, z^2, y, z, 1) \\ y^3 + az^4 + yz^3 & (yz^2, yz, z^2, y, z, 1), \end{array} \qquad a \neq 0.$$

The same ideals give their initial spans; in the second row $J = (y - z^2/(8a))$. Normal parameter monomials differ only in their first entries, yz^2 and z^3. Their classes have diagonal coefficients $-5/(3a)$ and $-3/(4a)$, respectively, and all other diagonal coefficients one.

For $D_5 \subset E_6$, use

$$F_T = y^3 + z^4 + T_2yz^2 + T_5yz + T_6z^2 + T_8y + T_9z + T_{12},$$
$$T(u) = (-3u^2, 3u^5, -3u^6/2, -3u^8/4, u^9, -3u^{12}/16). \tag{A.1}$$

At its D_5 point put $Z = z - u^3/2$, $w = y - uZ$. Then $F_{T(u)} = w^3 + 3uw^2Z + Z^4$, with Tjurina basis $(Z^3, Z^2, w, Z, 1)$ and relations

$$Z^4 = 0, \qquad wZ = 2Z^3/(3u^2), \qquad w^2 = -2uwZ.$$

Holding T_2 constant, the classes of $(yz, z^2, y, z, 1)$ are lower triangular in this basis, with diagonal $(2/(3u^2), 1, 1, 1, 1)$. At $u = 1$, comparison with the intrinsic flag is upper triangular with nonzero diagonal. The resulting normal matrix is $K_0^{-1}U$, with K_0 lower triangular invertible and U upper triangular invertible, so all its leading principal minors are nonzero.

A.3 The differential for $G_{33} \subset G_{34}$

Use flat G_{34} basic degrees $(6, 12, 18, 24, 30, 42)$ and $e_i = \partial_{x_i}$. The degree six invariant is nonzero on the axis [26, Theorem 3, Case ST34 (3)], so put $\lambda = x_1 = u^6$. The Saito algebra pulled back over $R = \mathbb{C}[\lambda]$ has shifted basis weights $(6, 5, 4, 3, 2, 0)$. The quadratic potentials in [16, Section 9], identified as G_{34} in [17, Remark 5.11], give the special algebra

$$A_{34} = \mathbb{C}[a, b]/(a^4, a^2b, b^2 - a^3), \quad a = e_5,\ b = e_4, \quad (a^3, ab, a^2) = (2e_1, e_2, 2e_3).$$

For G_{33} the algebra is

$$A_{33} = \mathbb{C}[A, B]/(A^3, A^2B, B^2), \quad (\partial_{p_4}, \partial_{p_6}, \partial_{p_{10}}, \partial_{p_{12}}, \partial_{p_{18}}) = (AB, A^2, B, A, 1).$$

Saturate the axis ideal in the pulled back algebra and let Q be the rank 5 free quotient. Subtract support coordinates from a, b; they belong to R because each is one fifth of the multiplication trace. The augmentation ideal satisfies $\mathfrak{m}^3 = 0$ generically and hence everywhere by torsion-freeness. The special kernel contains nonzero a^3 and has dimension one by flatness, so $Q/\lambda Q = A_{33}$. Graded Nakayama gives the basis $(1, a, b, a^2, ab)$. Weights and $\mathfrak{m}^3 = 0$ imply

$$b^2 = c\lambda^2a^2 + d\lambda ab, \qquad d^2 + 4c = 0.$$

The second equation follows from the rank 1 generic quadratic relation. Replace b by $b - d\lambda a/2$; then $b^2 = 0$. The square-zero ideal is $S = \langle b, a^2, ab\rangle$. The inverse images of $\mathfrak{m}S, \mathfrak{m}^2, S, \mathfrak{m}, Q$, together with the axis ideal, are free direct summands, with special fibers the initial spans of $(e_1, \ldots, e_6)$. Their homogeneous lifts can mix an e_i only with lower weight basis elements multiplied by nonnegative powers of λ. Consequently each initial block remains invertible on the generic fiber. This graded step proves all augmented leading minors are nonzero; removing the axis proves all five normal leading minors. The center contains μ_6, and the nonzero degree six invariant bounds the axis scalar order by six, so that order is exactly six. Coarse source coordinates are $u^2p_4, p_6, u^2p_{10}, p_{12}, p_{18}$, of effective weights $(1, 1, 2, 2, 3)$, against target weights $(2, 3, 4, 5, 7)$.

Completion of Proposition 3.14. For each inclusion, let D_i be the increasing ambient degrees after deleting the lowest. The nonzero leading minors just proved and homogeneity give

$$M(v) = \operatorname{diag}(u^{D_i})B \operatorname{diag}(u^{\epsilon_i - k_i}),$$

where B is constant with every leading principal minor nonzero. Tangent elimination uses ratios of derivatives of restricted invariants, holomorphic in v. Elimination below the diagonal uses $u^{D_i - D_j}$, $i > j$, also holomorphic in v, and changes only target coordinates. The diagonal orders are $(D_i - k_i + \epsilon_i)/e$, giving the four displayed tuples. The last inverse row uses only the last transformed target coordinate. On its kernel that coordinate vanishes, so the preceding row has the asserted preceding pole. □

A.4 The D_5 image ring and its tangent obstruction

The image of the ambient local ring along (A.1) is exactly $\mathcal{A} = \mathbb{C}\{u^2, u^5\}$; its only missing nonnegative exponents are one and three. In the local Tjurina algebra, the two lowest normal directions span $\mathfrak{m}^2 = \langle Z^2, Z^3 \rangle$. Reduction of the ambient parameter monomials modulo this intrinsic ideal, with basis $(w, Z, 1)$, gives the three-plane containing the tangent axis and those two directions:

$$dT_8 = -u^6 dT_2/4 - u^3 dT_5/2, \qquad dT_9 = -u^3 dT_6, \qquad dT_{12} = u^6 dT_6/4. \tag{A.2}$$

There is no smooth threefold through zero containing the axis and with these tangent spaces away from zero. Its limiting projection onto (T_2, T_5, T_6) would be invertible, making it a holomorphic graph. Its derivative coefficients along the curve would then lie in $\mathcal{A}$, but the coefficient $-u^3/2$ in (A.2) does not. This obstruction applies to derivatives of actual ambient holomorphic functions; arbitrary covectors on the normalization need not have coefficients in $\mathcal{A}$.

A.5 The actual G_{33} theta characters

Lemma A.1. The generators of the ring of invariant sections in the minimal full G_{33} cusp model have degrees $(1, 1, 3, 3, 4, 6)$. For a scalar ω of order three, their characters are $(1, \omega, 1, \omega, 1, 1)$. Its four common horizontal fixed points equal the polarization kernel. At each relevant point with trivial full fiber character, theta translation transports these characters and the derivative projections. In the increasing polynomial directions $(4, 6, 10, 12, 18)$, the successive cusp divisions are $(1, 3, 3, 4, 6)$.

Proof. Use the actual polarized model [25, Sections 4.2 and 7.6]:

$$E : y^2 = x^3 + 1, \quad b = (0, 1), \quad A_0 = \ker(E^6 \xrightarrow{\sum} E), \quad A = A_0/\langle (b, \dots, b) \rangle.$$

On $A_0/S_6 = |6O|$, use coefficient coordinates $(c_0, c_x, c_y, c_{x^2}, c_{xy}, c_{x^3})$. The degree one sections are $\theta_0 = c_0 + 2c_{x^3}$, $\theta_1 = c_{x^2}$, with $\theta_0(0) \neq 0$. Translation by b is $(x, y) \mapsto (-2x/(y+1), (y-3)/(y+1))$ and multiplies both sections by four, so the normalized fiber action fixes them. The scalar has coefficient weights $(0, 2, 0, 1, 2, 0)$ modulo three.

Here are explicit nonzero minors for the two new characters of degree three. Work over $\mathbb{F}_{1009}$, with $\omega = 374$, and use the following five points for each tuple; the sixth point is the negative of their sum in E.

a	P_1	P_2	P_3	P_4	P_5
1	$(43, 657)$	$(773, 520)$	$(318, 858)$	$(690, 511)$	$(417, 539)$
2	$(227, 130)$	$(228, 395)$	$(142, 879)$	$(864, 669)$	$(981, 805)$
3	$(67, 456)$	$(655, 43)$	$(62, 397)$	$(350, 949)$	$(224, 424)$
4	$(394, 779)$	$(165, 206)$	$(671, 577)$	$(653, 869)$	$(871, 634)$
5	$(695, 1007)$	$(243, 824)$	$(734, 374)$	$(792, 43)$	$(370, 183)$
6	$(461, 528)$	$(350, 60)$	$(773, 520)$	$(543, 63)$	$(492, 183)$
7	$(572, 539)$	$(231, 649)$	$(797, 632)$	$(313, 399)$	$(829, 358)$
8	$(297, 479)$	$(228, 395)$	$(166, 422)$	$(67, 456)$	$(133, 267)$
9	$(133, 742)$	$(864, 340)$	$(717, 489)$	$(224, 585)$	$(734, 374)$.

Every point lies in $E[316]$, where division by three means multiplication by 211. Thus the reflection ρ in $v = (1, 1, \omega, \omega, \omega^2, \omega^2)$ acts on each tuple by

$$\rho(P)_i = P_i - [\omega^{e_i}][211] \sum_k [\omega^{-e_k}] P_k, \qquad (e_1, \dots, e_6) = (0, 0, 1, 1, 2, 2).$$

Here $[\omega](x,y) = (\omega x, y)$, and put $\tau(P)_i = P_i + b$. For any of these tuples or their transforms, let $c(P)$ be the kernel vector of the evaluation matrix on $(1, x, y, x^2, xy, x^3)$, normalized by $c_0 + 2c_{x^3} = 1$. The kernels have dimension one and the normalizing denominators are nonzero. For $j = 0, 1, 2$, order lexicographically the exponent vectors

$$\mathcal{M}_j = \{u \in \mathbb{Z}^6_{\geq 0} : \textstyle\sum_i u_i = 3,\ (0,2,0,1,2,0) \cdot u \equiv j \pmod 3\}.$$

Define the matrix M_j by the rows

$$(M_j)_{2a-1,u} = c(P^{(a)})^u - c(\rho P^{(a)})^u, \qquad (M_j)_{2a,u} = c(P^{(a)})^u - c(\tau P^{(a)})^u.$$

The following minors use the first indicated rows and all columns except the listed ones; the last column gives their determinants modulo 1009.

j	$\lvert\mathcal{M}_j\rvert$	number of rows	omitted columns	det
0	21	18	$4, 19, 21$	775
1	17	15	$13, 17$	457
2	18	17	13	523.

The chord and tangent law on $y^2 = x^3 + 1$ and ordinary determinant expansion give these residues. The points and their transforms have torsion of order dividing 948, prime to 1009. Thus the minors lift to characteristic zero, since every denominator is a unit. Rains's section degrees give total dimension six, so the upper character bounds $(3, 2, 1)$ are equalities. Cubics in θ_0, θ_1 account for $(2, 1, 1)$, leaving new characters $1, \omega$. The polynomial cotangent degrees then give the asserted characters in degrees four and six as well.

For completeness, the common fixed subgroup is the diagonal image of $E[2]$. The lift $(\lambda/6, \dots, \lambda/6, -5\lambda/6)$ changes under the extra reflection by $(\omega\lambda/3)v$. It vanishes modulo the diagonal isogeny kernel exactly when $1 - \omega \mid \lambda$; the Weil pairing gives the same polarization kernel, of order four. At a point with trivial fiber character, a theta group lift of its translation commutes with G_{33}, by the argument comparing the characters used above. Conjugate the scalar by that lift to transport its eigenspaces. The derivative spaces, with higher primitive terms retained, therefore have the same successive projections at all relevant points. Their first appearance degrees are $1, 3, 3, 4, 6$, which proves the divisions. □

B The projective comparisons needed in dimensions six and seven

These comparisons concern the original automorphic class, including its finite characters. They make no assumption about the ambient ten-point nef bound. Write $H = c_1(\mathcal{O}(1))$.

Lemma B.1. If $f : C \to \mathbb{P}^k$ is nonconstant, $u = \deg f^*\mathcal{O}(1) > 0$, and $f^*T_{\mathbb{P}^k} \to Q_h$ is nonzero, then $h = \deg Q_h \geq u$. The corresponding hyperplane map $\kappa : C \to (\mathbb{P}^k)^*$, after common zeros are removed, has degree at most $h - u$. If κ is nonconstant and its derivative vanishes at R distinct points, then

$$R \leq 2(h - u) + 2g(C) - 2. \tag{B.1}$$

A constant κ is a hyperplane containing $f(C)$.

Proof. The Euler sequence represents the map by incidence sections of $Q_h \otimes f^*\mathcal{O}(-1)$. Their span, which has no common zero, defines κ and proves the degree statements. A general projection of its image to $\mathbb{P}^1$ preserves its degree and every prescribed zero of its derivative. Riemann–Hurwitz gives (B.1). □

Proposition B.2. If the Baily–Borel compactification of an arithmetic ball quotient of dimension six is $\mathbb{P}^6$ and $L = \lambda H$, then $\lambda \geq 1$. The group may be compact or nonuniform.

Proof. Assume $0 < \lambda < 1$. On every fixed curve $a = 7u$, $t = \lambda u$, and each nonzero inverse row has $h \geq u > t$. The classification of parabolic subgroups [19, Tables 10–12], together with Proposition 2.14, leaves parabolics of $A_6, B_6, D_6, G(3,3,6), E_6, G_{34}$ and the disks $G(m,p,5)$ with $m \geq 4$ or $(m,p) = (3,1)$. A persistent factor of rank at most 4 contradicts the ample tangent bundle on a complete curve outside the boundary. For a compact monomial disk, the product line and the axis equation give

$$(3m-5)j + (m-1)pu + 2mg + m\sigma \leq 2m, \qquad j = mt \in \mathbb{Z}_{>0}.$$

For $m \geq 4$ this is impossible already with $j, u \geq 1$. The exceptional $(3,1)$ equality has $h = -1$. For a nonuniform disk, the actual integral bundle wL gives $t \geq u/w$. The product computation for parameter six, with all polarizations and widths retained, gives

$$a \geq (23p/6 + 1)h_P + 53t + R + \sigma \geq (23p + 59)u/6 > 7u.$$

For the groups $G(3,1,5)$ and $G(4,1,5)$ the same computation gives

$$a \geq 23t + 14h/3 + R + \sigma \geq 17u/2, \qquad a \geq 33t + 19h/4 + R + \sigma \geq 13u.$$

Here respectively $6L$ and $4L$ are actual integral bundles, and the relevant highest or product rows are nonzero. The remaining Gaussian product rows give a shorter argument than a finite enumeration. After the group $G(4,1,5)$ is excluded, the actual $G(4,2,5)$ ends have width one and types $A = (1^3, 2^3)$, $B = (1^2, 2^4)$, $C = (1, 2^2, 4^3)$, $D = (1, 2, 4^4)$. Their sums B_c and product poles b_c are $(9,1), (10,1), (17,2), (19,2)$. The torsor X has no full $G(4,2,5)$ fixed point, so is not an end of this disk. We justify the product projector in the actual Gaussian model $Y^2 = X^4 - Z^4$. Write $\mathsf{X} = \prod_i X_i$, $\mathsf{Y} = \prod_i Y_i$, $\mathsf{Z} = \prod_i Z_i$, $z_i = Z_i/X_i$, $f_j = e_j(z_1^4, \dots, z_5^4)$, and $P = (\prod_i z_i)^{4/p}$, where $p = 2$ or 4. The symmetric theta ratios and the product ratio satisfy

$$\frac{\mathsf{Y}}{\mathsf{X}^2} = (1 - f_1 + f_2 - f_3 + f_4 - P^p)^{1/2}, \qquad \frac{\mathsf{Z}^{4/p}}{\mathsf{X}^{4/p}} = P.$$

Since $p > 1$, their linear parts separate dP from every symmetric direction. Thus the actual coarse conormal lattice, after radial removal, is $q^{b_c}\mathcal{O}\, dP \oplus \Lambda_{\mathrm{sym}}$. Changing to additive elliptic coordinates is triangular on the symmetric invariants and a nonzero scalar on P, so this is the original root representation. At an interior enlargement, $P = u^{4-4/p}w$, $v = u^4$, and the ambient product is $Z = vw$; the symmetric normal block is $vI + J_4$. The equation $g_1 = v + f_1$ and the radial generators of degree one make the curve immersed at enlargements and ends. There is no first-order product mixing, and product inertia is trivial on the root space. The same rational idempotent therefore preserves every actual local lattice. At an enlargement, the source line bundle on the coarse quotient contributes $-(1-1/p)$, and the pullback of the product invariant has one zero, giving net correction $1/p$. Consequently the nonzero actual product quotient has degree $h_P = -20t/p + R/p + \sum b_c$. For $p = 2$, put $\beta = \sum b_c$, so $h_P = -10t + R/2 + \beta$. The bounds $B_c \geq 8b_c + 1$ give

$$a \geq 8h_P + 28t + s + 5R/4 + \sigma > 7u,$$

because Euler gives $h_P \geq u$. At the full fixed point over $Z = 0$, the invariant nonzero section of actual degree one is X^2 for A, B, and X for C, D. The fiber character is trivial there, so Lemma 2.17 supplies the excluded full $G(4,2,5)$ disk. This removes the entire A–D cusps, including the opposite side of C, D where the kernel is $G(4,4,5)$ and the width is two. Only $E = (1^2, 2, 4^3)$ and the torsor $X = (1^2, 4^4)$ remain for $G(4,4,5)$, both with actual width one. At every relevant point their invariant section X is nonzero of actual degree one, giving trivial fiber character. Here $b_c = 1$, $B_c \geq 16$, and the same product quotient has $h_P = -5t + R/4 + s$ and

$$a \geq 16h_P + 33t + R + \sigma > 7u.$$

These local product computations use neither descent of $2L$ nor a nef bound. They retain the original horizontal actions, scalar characters, and central periods. In particular the preliminary finite tests for the larger values of the parameter are unnecessary.

All remaining reflections have order two, and rank 2 stabilizer orders divide 24. Thus $14L = bH$ is integral, and Proposition 3.12 gives $24c_2^{\mathrm{orb}} = 504L^2 = (18b^2/7)H^2$ integral. It follows that $7 \mid b$ and $\lambda = 1/2$. Consequently $u = 2t$, $a = 14t$, and $t \in \frac{1}{2}\mathbb{Z}_{>0}$. For G_{33}, lifting the entire group at a cusp of width one gives

$$51s + 38R = 192t - 6, \quad s + 5R/6 \le 2 + 2t, \quad t = 1/2, 1.$$

Neither value has an integral solution. For B_5, the four end pairs are

$$(B_c, \nu_c) = (6,1), (9,2), (10,2), (11,2), \qquad B_c \ge 9\nu_c/2.$$

The canonical formula and $h = -10t + R + \sum \nu_c$ therefore give

$$a \ge 13t + 9h/2 + R + \sigma \ge 22t > 14t.$$

If an E_6 point exists, take a D_5 curve through it. Let R, D count its E_6, D_6 points. The primitive row associated with the invariant of greatest degree, in the image ring $A = \mathbb{C}\{\xi^2, \xi^5\}$, has direction

$$\nu = dT_{12} + \xi^3 dT_9/2 + \xi^6 dT_6/4.$$

The ratio $\xi^3/2 \notin A$ excludes a constant hyperplane map; this remains true under $\mathrm{GL}(6, A)$ coordinate changes. Both the image and the row have zero derivative at $\xi = 0$, so (B.1) applies at all R points. The degree equations give

$$16R + 8s + 5D + \sigma = 41t, \quad h - u = (t - D - \sigma)/4,$$

and their excess over (B.1) is $(t + 8s + 11D + 23\sigma)/16 > 0$. After these exclusions, the full $G(3,3,5)$ ends have width one and degrees $(1^3, 3^3)$. The nonzero actual product target has degree $z = -5t + R/3 + s$. The canonical and logarithmic identities give

$$a = 12z + 23t + R + \sigma \ge 47t > 14t,$$

using Euler's inequality $z \ge u = 2t$. For D_5 there are now no E_6 enlargements; its cusps have width one and generator degrees $(1^4, 2^2)$ for the full group. The actual product quotient, obtained by the separated odd theta direction and the coarse frame P/u at D_6, has degree $p = -5t + D/2 + s$. Here D counts D_6 points. The canonical and logarithmic identities give

$$a = 8p + 13t + D + \sigma \ge 29t > 14t,$$

again by Euler, $p \ge u = 2t$. Finally A_5 gives $6h = a - 14t - \sigma \le 0$, contrary to $h \ge u$. □

Lemma B.3. Let $Z = \mathbb{P}^k$, $k \ge 2$, be a closed collision in a smooth GIT model, with $N_{Z/W} = \mathcal{O}(-1)^c$. Assume $\pi|_Z$ is an isomorphism and its image meets the boundary in a finite set. If $L|_Z = \lambda H$ and $\lambda < 1$, then Z is the compactification of the induced arithmetic quotient of the fixed ball, with the same automorphic class.

Proof. Since $\pi|_Z$ is an isomorphism, the image of Z has the same normal bundle away from its finite boundary set. Lemma 3.10 applies with the original restricted automorphic class. The inequality $\lambda < 1$ forces $s = c$, so its full fixed compactification is Z. □

Proposition B.4. For a compact arithmetic ball quotient with coarse space $\mathbb{P}^k$ and $L = \lambda H$, one has $\lambda \ge 1/3$ if $k = 4$ and $\lambda \ge 2/3$ if $k = 5$.

Proof. For every complete fixed curve use $a = (k+1)u$ and $h \geq u$. In dimension five a $G(m,p,4)$ product direction yields

$$(4m-6)j + (m-1)pu + 2mg + m\sigma \leq 2m.$$

It excludes $m \geq 4$ and $(m,p) = (3,1)$. The remaining groups are $A_5, B_5, D_5, G(3,3,5), G_{33}$. Their rank 2 orders divide 24. The integral classes $12L$ and $24c_2^{\mathrm{orb}} = 360L^2$ imply $\lambda \in \frac{1}{6}\mathbb{Z}_{>0}$. For $\lambda = 1/6, 1/3, 1/2$, a D_4 curve, with R, M counting G_{33}, D_5 points, gives

$$6u + 18t = 11R/2 + 4M + \sigma, \qquad 5R/6 + M/2 + \sigma = 2t + 2 - 2g.$$

Thus $g = 0, u \leq 2$ and $28R + 21M = (36 + 96\lambda)u - 12$. Modulo seven excludes the first and third slopes. The sole remaining row is

$$\lambda = 1/3, \quad u = 1, \quad R = 2, \quad M = 0, \quad \sigma = 1. \tag{B.2}$$

We exclude it below, retaining the triality characters. Then a B_4 curve obeys $(1+4\lambda)u + 4g \leq 4$; integrality of $2t$ excludes $\lambda = 1/6, 1/3$, and $\lambda = 1/2$ requires $4R = 16u - 2$. With G_{33} absent the $G(3,3,4)$ product comparison gives $u + j \leq 1$. Finally A_4 gives $5h = 6u - 9t - \sigma < 5u$.

For $k = 4$ the analogous inequality $(3m-5)j + (m-1)pu + 2mg + m\sigma \leq 2m$ excludes $m \geq 5$ and $G(4,1,4)$. The remaining full groups are

$$A_4, B_4, D_4, G(3,1,4), G(3,3,4), G(4,2,4), G(4,4,4),$$
$$G_{28}, G_{29}, G_{30}, G_{31}, G_{32}.$$

Now $30L = bH$ and $25920c_3^{\mathrm{orb}} = -259200L^3$ give $5 \mid b$, hence $\lambda \in \frac{1}{6}\mathbb{Z}_{>0}$. At $\lambda = 1/6$, the G_{25} curve through G_{32} satisfies

$$5u + 29t = 28R/3 + \sigma, \quad 5R/6 + \sigma = 2t + 2 - 2g,$$

leaving $u = R = 2, \sigma = 1$. Its image is a smooth conic, but all basic degrees $12, 18, 24, 30$ exceed the axis degree six, so its differential vanishes, a contradiction. For $G(3,1,3)$ the equations

$$5u + 20t = 11R/3 + \sigma, \quad 2R/3 + \sigma = 2t + 2 - 2g, \quad -9t + R \geq u$$

leave $u = 1, R = 2 < 5u/2$. There are no order-three reflections left, and the integral bundle $10L = 2K + B$ excludes $10\lambda = 5/3$. □

The triality residue in (B.2). The curve is an actual line, with normal bundle $\mathcal{O}(1)^4$. Its two G_{33} points are p, q, and its other two inertia points have order two and trivial action on the root space. For D_4 put $e_j = e_j(x_1^2, \ldots, x_4^2)$ and

$$h_4 = e_2 - e_1^2/4, \quad P = x_1x_2x_3x_4, \quad h_6 = e_3 - e_1e_2/6 + e_1^3/48.$$

The outer S_3 preserves h_6 and $h_4^2 + 12P^2$. Its order-three eigenlines are $h_4 \pm 2\sqrt{-3}P$. The product relation in the fundamental group says that the two nontrivial local trialities at p, q are inverse. Thus these two eigenlines, though not preserved by arbitrary S_3, are globally defined here. For the source line bundles indexed by $2, 4a, 4b, 6$, one has $E_{4a}E_{4b} = E_2E_6$, including the scalar and automorphic characters. Their coarse degrees are all -2. The column zeros at p, q have respective orders $(1,0,2,1)$ and $(1,2,0,1)$. Consequently

$$S_2 = E_2(p+q), \quad S_6 = E_6(p+q), \quad S_{4a} = E_{4a}(2q), \quad S_{4b} = E_{4b}(2p)$$

are actual bundles of degree zero with $S_{4a}S_{4b} = S_2S_6$. After tangent removal the local two-by-two blocks have leading forms

$$\begin{pmatrix} a_2v & a_4 \\ a_2'v^2 & a_4'v \end{pmatrix}, \qquad \begin{pmatrix} b_4v^2 & b_6v \\ b_4'v^3 & b_6'v^2 \end{pmatrix}, \qquad \frac{a_4b_4}{a_2b_6} = \frac{1}{12}.$$

The ratio follows from the following polynomial identities. Use $\sum z_i = 0$, the permutation action of S_6, and the reflection in $(1, 1, \omega, \omega, \omega^2, \omega^2)$, where $\omega^2 + \omega + 1 = 0$. Put $p_j = \sum_i z_i^j$ and

$$\begin{aligned} F_4 &= p_4 - p_2^2/2, \qquad F_6 = 5p_2^3 - 30p_2p_4 - 20p_3^2 + 48p_6, \\ F_{12} &= 300p_2^3p_3^2 + 315p_2^3p_6 - 990p_2^2p_3p_5 - 450p_2^2p_4^2 - 300p_2p_3^2p_4 - 90p_2p_4p_6 \\ &\quad + 864p_2p_5^2 - 250p_3^4 + 840p_3^2p_6 - 540p_3p_4p_5 + 300p_4^3 + 72p_6^2. \end{aligned}$$

These three polynomials are invariant under the displayed generators. Set $\vartheta = \sqrt{-3}$, $\omega = (\vartheta - 1)/2$, $s = \sum_{i=1}^4 x_i$, and substitute

$$z_1 = u + \vartheta s/6, \quad z_2 = -u + \vartheta s/6, \quad z_{i+2} = x_i - (3 + \vartheta)s/12 \quad (1 \le i \le 4).$$

Write $H_A = h_4 + 2\vartheta P$, $H_B = h_4 - 2\vartheta P$, and $T_{12} = F_{12} + 3F_6^2/4$. Up to terms of degree at least seven in $x_1, \dots, x_4$, direct expansion gives

$$\begin{aligned} F_4 &= -2u^2e_1 + \omega H_A, \\ F_6 &= 16u^6 + 30(1 + \vartheta)u^2H_B - 36h_6 + 5e_1^3/4, \\ T_{12} &= 2700(1 + \vartheta)u^8H_B + 32400u^6h_6 - 1725u^6e_1^3. \end{aligned}$$

The terms in e_1^3 disappear in the indecomposable normal directions. The tangent invariant is $F_6 = 16u^6$, so use $v = u^6$ and the coarse frames $(e_1/u^4, H_A, H_B/u^4, h_6)$. Removing the tangent direction therefore gives the two displayed blocks and the required product

$$\frac{a_4b_4}{a_2b_6} = \Big(-\frac{\omega}{2}\Big)\,\frac{1 + \vartheta}{12} = \frac{1}{12}.$$

Choose global frames respecting the tensor identity. The four columns of the quotient differential, with entries in $\mathcal{O}(1)^4$, then satisfy

$$\begin{aligned} v_{4a}(p) &= A_p v_2(p), \qquad v_{4b}(p) = B_p v_6(p), \\ v_{4b}(q) &= A_q v_2(q), \qquad v_{4a}(q) = B_q v_6(q), \qquad A_pB_p = A_qB_q. \end{aligned}$$

For $p = (1 : 0), q = (0 : 1)$ their determinant is a constant multiple of $(A_pB_p - A_qB_q)x^2y^2$, hence zero. The full quotient map has generic normal rank 4, a contradiction. □

Proposition B.5. A nonuniform arithmetic ball quotient with compactification $\mathbb{P}^5$ cannot have $L = H/6$.

Proof. The actual integral bundle wL forces the Eisenstein field. The product comparison for parameter six in rank 4 gives $a \ge (17p/6 + 1)h_P + 30t + R + \sigma$. Using $a = 6u$, $h_P \ge u$, and $t \ge u/6$ gives a contradiction. For G_{32} the common equations and Euler leave only $(g, u, t, s, \sigma) = (0, 2, 1/3, 2, 2/3)$. The image is a smooth conic, but the generator degrees $(2, 3, 4, 5, 6)$ at cusps of width one make the differential zero. For F_4, the types with width one are $A = (1, 1, 2, 2, 3)$ and $B = (1, 2, 2, 3, 4)$; the unique residual row is $(g, t, s_A, s_B, \sigma, u) = (0, 1/2, 1, 2, 0, 3)$. The actual flags defined by the theta sections associated with the root lattice project onto $\langle dp_2\rangle$, $\langle dp_2, dp_6\rangle$, and $\langle dp_2, dp_6, dp_8\rangle$. These are nonzero since their moment determinants are $3\vartheta_2^4\vartheta_3^4\vartheta_4^4/8$, $-5760A\eta(\tau)^{12}\eta(2\tau)^{12}$, and $-696729600A^4\eta(\tau)^{16}\eta(2\tau)^{16}$, where $A = (\vartheta_3^4 + \vartheta_4^4)/2 \ne 0$ at an Eisenstein modulus. The final identity is an identity of weight 24 on $\Gamma_0(2)$; its valence bound is six, so the expansion of the lattice moments through q^6 is an exact finite verification. Successive divisions subtract $5, 6, 8, 10$ from the coefficient degrees $3 + d/2$. Thus $\dim \mathrm{Hom}(f^*T_{\mathbb{P}^5}, \mathcal{O}(3)) \le 1$, but Euler gives dimension at least two. For $G(3, 1, 4)$ only $u = 2, t = 1/3, R = 0, s = 2, \sigma = 2/3$ remains, with both ends $(1^2, 3^3)$. Differentiating the reduced order-three branch divisor on the product line counts one interior zero and three at each

end. Its degree equation is $2d_3 + 4 = 7$, impossible. The B_4 equations give $10t \le 4(1-g) - 2\sigma$ with $2t \in \mathbb{Z}_{>0}$, impossible. For D_4 retain its G_{33}, D_5 enlargements: $g = 0, u, s \le 2$ contradict $2s \equiv 3u + 2 \pmod 7$. Only after excluding them use $G(3,3,4)$: $u = 2, t = 1/3$ requires $12s + 5R = 26$, $s \le 2$. Persistent factors of lower rank contradict Euler. Finally A_4 gives $5h = 27t - \sigma < 5u$. □

Proposition B.6. A nonuniform arithmetic ball quotient with compactification $\mathbb{P}^5$ cannot have $L = H/4$.

Proof. The field is Gaussian. For G_{31} with a common fixed point the only numerical row is $(g, u, t, s, \sigma) = (0, 2, 1/2, 3, 0)$. The matrix of ratios of the primitive theta sections, in degrees $8, 12, 20, 24$, is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 6 & 3 & 1/2 & -1/4 \end{pmatrix}.$$

It follows from the linearization of $i_2^3 - 4i_2i_4 + 8i_6 - 16e_6$ at $x_i = 1 + \xi_i$, $\sum \xi_i = 0$: $dq_4 = dq_5/2 - dq_6/4$. The coefficient degrees $2 + d/2 - 3k$, successively killed at their two constant directions, bound $\dim \mathrm{Hom}(f^*T_{\mathbb{P}^5}, \mathcal{O}(2))$ by two; Euler gives at least three. The G_{31} torsor has sixteen fixed points for $J = 2^{2+4}$, in residual orbits of sizes six and ten. Its stabilizer orders $120, 72$ are not divisible by $|G(4,2,4)/J| = 48$. It therefore contains no full $G(4,2,4)$ fixed point. For $G(4,2,4)$ with a fixed point, the four types with width one are $A = (1^3, 2^2)$, $B = (1^2, 2^3)$, $C = (1, 2^2, 4^2)$, $D = (1, 2, 4^3)$. Their product degrees are half their respective highest degrees. The seven residual end lists are C, AA, CD, AAD, BBC, CDD, $CDDD$, with $u = 1, 1, 2, 2, 2, 3, 4$ and $\nu = 4u$. Every row has $R = 0$ and a nonzero product target of degree $-8t + \nu/2 = 0 < u$. The remaining factors are parabolics of $A_5, B_5, D_5, G(4,4,5), F_4, G_{29}$, but smaller persistent factors contradict Euler. Their rank 2 orders divide 24, so $24c_2^{\mathrm{orb}} = 360L^2$ extends across the finite cusp set. Its value on a plane is $45/2$, contradicting integrality. □

Lemma B.7. For a fixed curve in the quotient of ten points, with full generic group E_6, product enlargements, cusps with full group E_6 and width one, and possibly R points of type E_7, the bound $a \le 32t/5$ excludes the curve.

Proof. For $R > 0$ the odd coarse source coordinates are $p_5/u, p_9/u$, with $v = u^2$. Their residues give

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = 12s + 27R/2 + \sigma - 44t, \quad h = -12t + 3R + 3s.$$

Here $2t \in \mathbb{Z}_{>0}$, $2t + R \in 2\mathbb{Z}$, and a is even. The case $h \ge 0$ reduces to $2t = 1, 2, 3$ and has no row satisfying the parity conditions. For $h < 0$, the first five residual coefficient degrees are

$$-10t + R + 2s,\ -7t + 3R/2 + 2s,\ -6t + R + s,\ -4t + R + s,\ -3t + R/2 + s.$$

They are negative for $g > 0$. For $g = 0$, a residual coefficient can be nonnegative only if $t \le 2$: the first, third, and fourth degrees are negative, the second is less than $2 - t$, and the fifth is at most $2 - t$. All vanish except at $(t, R, s, \sigma, a, h) = (3/2, 1, 4, 1/2, -4, -3)$, whose evaluation rank violates (5.1). Thus the Hom space for the target associated with the invariant of greatest degree is one-dimensional and $e \ge -h$. On its kernel, the row associated with the invariant of degree nine extends to a line of degree $h_9 = -9t + 5R/2 + 2s$. Since $h + h_9 \le -R/8 - 5\sigma/12 < 0$, one has $h_9 < e$, contradicting Proposition 2.3. For $R = 0$ the degree equations are $a = 24(1-g) - 20t - 11\sigma$, $h = 6(1-g) - 6t - 3\sigma$, and $t \in \mathbb{Z}_{>0}$. Two end divisions leave degrees $(d - 12)t + s$ and exclude every case except $(g, t, s, \sigma, a, h) = (0, 1, 4, 0, 4, 0)$. Here the normal bundle splits under the odd involution as $N^+ \oplus N^-$, with $\mathrm{rank}\, N^- = 2$ and $\deg N^- = -14t + 3s = -2$. A negative line quotient exists, but the flags defined by the

theta sections give $\mathrm{Hom}(N^+, \mathcal{O}(-1)) = 0$. The splitting in Proposition 2.3 then forces every summand of N^+ to have positive degree. Its nonzero map associated with the invariant of greatest degree to $E_{12}(3S) = \mathcal{O}$ is impossible. □

C Finite ranges and the remaining monomial comparisons

This appendix specifies the numerical domains and every rejection test used in Sections 5 and 6. All operations are on integers and rational numbers. The bounds below define finite sets of end multisets, and (C.8) treats every marked degree, including unbounded intervals. Repeated affine actions remain distinct geometric possibilities; a numerical row is counted once when its group, section degrees, width, and fixed-point requirement agree. Every such action is excluded by the same displayed test.

C.1 Cusps and their widths

For horizontal rank $r \geq 6$, append m to each prefix until its length is $r + 1$. Rains's strongly crystallographic classification gives the following raw section degrees [25, Section 4.4].

(m, q)	prefixes	(m, q)	prefixes
$(2, 1)$	$(1, 1, 1), (1, 1), (1), ()$	$(6, 1)$	$(1), (2, 3), (2), (3), ()$
$(2, 2)$	$(1, 1, 1, 1)$	$(6, 2)$	$(1, 3), (2, 3, 3), (3, 3)$
$(3, 1)$	$(1, 1), (1), ()$	$(6, 3)$	$(1, 2), (2, 2, 3), (2, 2)$
$(3, 3)$	$(1, 1, 1)$	$(6, 6)$	$(1, 2, 3)$
$(4, 1)$	$(1, 2), (2, 2), (1), (2), ()$	$(4, 4)$	$(1, 1, 2)$
$(4, 2)$	$(1, 2, 2), (2, 2, 2), (1, 1)^*, (1, 2), (2, 2)$		

The $(6, 1)$ prefixes $(2, 3), (2), (3)$ occur three, two, and two times, respectively; there are 36 entries before normalization. The asterisk denotes the nontrivial torsor, which has no full group fixed point. It does not exclude smaller stabilizers or nontrivial fiber characters.

Here is the complete normalization rule. First divide the raw tuple by its greatest common divisor, obtaining $a = (a_0, \ldots, a_r)$. Put

$$h_i = \gcd(a_j : j \neq i), \qquad H_0 = \operatorname{lcm}_i h_i.$$

For every $v \mid H_0$, the smooth cyclic Veronese has generator tuple

$$\alpha_i = \gcd(v, h_i) a_i / v. \tag{C.1}$$

These are precisely the smooth cyclic Veroneses. The coordinate reflection subgroups have pairwise coprime orders $\gcd(v, h_i)$, and Chevalley–Shephard–Todd gives the criterion. Removing an initial scalar kernel of order g replaces q_{old} by q_{old}^g and divides the central period by g. This gives 64 entries after normalization and 32 distinct tuples with their group parameters retained. Every record satisfies

$$(r - 1)b \leq A = \sum_i \alpha_i \leq rm + 1, \qquad b = \max_i \alpha_i \leq m. \tag{C.2}$$

The raw maximum occurs at least twice and $\sum_i (b - \alpha_i) \leq 2b$; applying (C.1) proves the same bounds after normalization.

The lifts of the entire groups at the cusps follow directly from the reflection roots. For $G(m, q, r)$, adjacent and primitive phase transpositions lift with their positive horizontal Gram matrix. Their rational span fixes a Hermitian plane through the cusp line, and the generic disk group is $G(m, p, r)$, $q \mid p \mid m$. For $q = 1, m = 2, 3, 4$, a diagonal image of order m supplies a lift

of the entire group. For $m = 6$, if the fiber character kernel is $G(6, p, r) \lhd G(6, q, r)$, its order is exactly

$$\kappa = p/q. \tag{C.3}$$

To see this, $2I - D_{ij} - D_{ij}^{-1} = E_i + E_j$ and $(E_i + E_j)(E_i + E_k) = E_i$ split the horizontal lattice by coordinates. Since $1 - \zeta_6$ is a unit, the kernel has a unique fixed point; the full group therefore fixes it, and the character image is the full quotient.

For later exceptional comparisons, the Weyl groups E_6, D_7, E_7, E_8 lift in their entirety, have no torsor, and have width one and section tuples

$$(1^3, 2^3, 3), \quad (1^4, 2^4), \quad (1, 1, 2, 2, 2, 3, 3, 4), \quad (1, 2, 2, 3, 3, 4, 4, 5, 6). \tag{C.4}$$

For these groups $(s_\alpha - 1)\beta = \alpha$ gives zero coinvariants and no invariant Pic^0. The polarization is an integral power of the minimal one even at CM moduli. Connected reflection divisors through the origin force trivial linearization, and the criterion using greatest common divisors after deleting one entry in (C.1) forces that power to be one. The positive rational root span supplies the Hermitian plane, and the trivial full fiber character gives width one.

C.2 The finite domain and the unbounded marked degree

For fixed r, m, p, j, after $g = 0$ has been proved, enumerate

$$1 \le s \le \lfloor 2 + 2j/m \rfloor, \quad 0 \le R \le \left\lfloor \frac{2 + 2j/m - s}{1 - 1/m} \right\rfloor, \quad \sigma = 2 + 2j/m - s - (1 - 1/m)R. \tag{C.5}$$

The allowed σ are sums of $1 - 1/e$, $1 < e \mid w$, with at most $\lfloor 2\sigma \rfloor$ summands. Use $w = 4$ for $m = 4$, $w = 6$ for $m = 3, 6$, and the union of the three possibilities $w = 2, 4, 6$ for $m = 2$. At each end independently use every record above with $q_c \mid p$ and every $\kappa_c \mid p/q_c$, before applying the stricter exclusions using the actual widths. The quantities B_c, ν_c, a, h are those in (6.1)–(6.3). When all widths are one, the product line imposes

$$rj/p + (1 - 1/p)R \in \mathbb{Z}.$$

Take only integral a, impose $a \le rwt$, and impose even a for twelve labels.

For completeness, the coefficient bound used in the enumeration is as follows. Let v_c be the zero forced at the c-th end after the map kills df, and set $V = \sum_c v_c$, $\nu = \sum_c \nu_c$. If $\kappa_c \neq 1$, use $v_c = 1$; if $\kappa_c = 1$, use $v_c = \min_i \alpha_{c,i}$. In the latter case, if the tuple has a unique generator of degree one and the other degrees have a common divisor greater than one, the radial equation instead forces the least of those other degrees. Nonvanishing follows from the greatest common divisor test using the actual width in Lemma 3.8. The symmetric coefficients have degrees

$$R + \nu - V - (r - 1 - i)j \quad (p > 1), \qquad R + \nu - V - (r - i)j \quad (p = 1).$$

The triangular enlargement matrix imposes R zeros. There is at most one extra product coefficient. Thus the bounds

$$\ell \le \left\lfloor \frac{\nu - s}{j} \right\rfloor + 3, \qquad \ell \le \left\lfloor \frac{\nu - V}{j} \right\rfloor + 2 \quad (p = 1) \tag{C.6}$$

are valid. The sharper count used in Appendices C.3–C.5 is

$$\ell \le 1 + \max\left(0, \min\left(q_{\mathrm{sym}}, \left\lfloor \frac{\nu - V}{j} \right\rfloor + 1\right)\right) + \epsilon_P, \tag{C.7}$$

where $q_{\mathrm{sym}} = r$ for $p = 1$, and $r - 1$ otherwise. Here $\epsilon_P = 0$ for $p = 1$; for $p > 1$, take $\epsilon_P = 1$ unless the following test forces $\epsilon_P = 0$. When all widths are one and $p > 1$, the product line has degree $E_P = -rj/p - (1 - 1/p)R$, and $h - E_P - V < R$ forces $\epsilon_P = 0$.

There is no search cutoff on the marked integer e. For each permitted ℓ, solve exactly

$$e \geq \max(1, -h), \qquad a - (n - 2\ell)e \in 2\mathbb{Z}_{\geq 0}. \tag{C.8}$$

If $n - 2\ell > 0$, this is a finite interval with its actual parity condition. If $n - 2\ell = 0$, it is either empty or the entire lower half-line, depending on $a \in 2\mathbb{Z}_{\geq 0}$. If $n - 2\ell < 0$, it is an unbounded half-line with the indicated congruence. When the coefficient is even and a is even, both parities of e must be retained. All bundle tests are monotone in e, so failure at the smallest permitted integer excludes its whole interval or half-line. In particular, no positivity condition on a is added to the enumeration.

C.3 Ranks at least 9: the 170 rows

The analytic range proved in Proposition 6.2 is

$$9 \leq r \leq 28, \qquad 1 \leq j \leq \begin{cases} 5 & r = 9, \\ 4 & r \geq 10. \end{cases}.$$

With (C.7), the domain (C.5)–(C.8) gives

r	9	10	11	14
rows	127	41	1	1

and zero rows in every other rank. At rank 9 the nonzero counts by (m, p) are $3, 4, 4, 30, 3, 4, 79$ at $(3,1), (3,3), (4,2), (4,4), (6,1), (6,2), (6,6)$, respectively. The remaining tangent tests have the following inputs. Here ρ is the cover degree and β its total allowed pole order.

r	(m,p)	j	s	ρ	β	a	ℓ	$e_{\min}$
9	$(4,2)$	2	3	1	12	17	5	5
9	$(4,4)$	2	3	2	20	$6,7,8,9$	6	6
9	$(6,1)$	$1,2,3$	j	1	$6j$	$7j+2$	≤ 6	$3j$
9	$(6,6)$	3	3	2	30	$8,7$	7	$10,9$
9	$(3,1)$	1	2	1	6	7	6	3
9	$(3,3)$	1	2	1	6	9	$5,6$	3
10	$(6,2)$	2	2	1	12	12	$6,7$	6
10	$(4,4)$	1	2	2	12	$8,10$	$6,7$	3
10	$(4,4)$	2	3	1	12	$12,14,16,18$	6	6

For the tangent test, sort

$$2 + \beta, \quad \beta - \rho i j \ (1 \leq i < r), \quad \beta - \rho r j / p,$$

with the last entry interpreted as the $i = r$ symmetric entry when $p = 1$. The required positive rank and degree are $n - \ell - 2$ and $\rho a + (\ell - 1)\rho e$. Rows 1,2,3,5,7 have available/required ranks $(4,6), (3,5), (4,5), (4,5), (4,5)$. Row 6 gives $5 < 6$, or $9 + 5e \geq 24 > 23$. Row 4 requires degree $122 > 95$ in rank 4. The last two rows give $5 < 6$, or $2a + 6\rho e \geq 52 > 45$. Every comparison improves as e increases. The other rank 10 rows with parameter six violate (C.3). The unique rank 11 and rank 14 rows are

$$(m, p, j, s, R, \sigma, a, h) = (4, 2, 1, 2, 0, 1/2, 4, -4), \quad (4, 2, 1, 2, 0, 1/2, 10, -5).$$

They require respectively one and two nonexistent full fixed points on the Gaussian torsor. Only the two rank 9 rows

$$(m, p, R, a, t, h) = (3, 1, 1, 16 \text{ or } 18, 1/3, -2)$$

remain. They have two cusps of width one, with section degrees one and three. The comparison using the normal bundle has normalized target degrees $7 - i$, $1 \leq i \leq 9$. For $\ell = 5, \ldots, 10$, required degrees $22, 24, 26, 28, 30, 32$ exceed available sums $20, 18, 15, 11, 6, 0$. For $\ell < 5$, too few target summands have degree at least $e \geq 2$. This completes all 170 exclusions.

C.4 Rank 8: the 117 rows

For twelve labels a is even and $a \le 8wt$. The bounds

$$h \le -5j + 2m(1-g) + (2-m)R - m\sigma, \qquad a \le (16m+2)(1-g) - (D-16m)t$$

show that $g \ge 1$ is impossible: the coefficient count gives $\ell \le 4$ and $a > 0$, but the second bound gives $a < 0$. If $h \ge 0$, then $j \le 2$; all records and widths in that finite range contradict integral $a \le 8wt$. If $j \ge 8$, then $\ell \le 6$ and $a \ge 0$, contrary to $a \le 16m + 2 - j(12 + 8/p) < 0$. Thus $g = 0, h < 0, 1 \le j \le 7$. For this rank, enlarge the correction condition to $\sigma = 0$ or $\sigma \ge 1/2$, as a necessary condition containing the exact semigroup. Using (C.7) gives

(m,p)	$(2,1)$	$(2,2)$	$(3,1)$	$(3,3)$	$(4,2)$	$(4,4)$	$(6,2)$	$(6,3)$	$(6,6)$
rows	9	20	2	2	12	40	1	8	23.

The pairs $(4,1), (6,1)$ leave no row. Two rows have $a = 0$, and 27 domains for the marked degree are unbounded; these are retained.

Apply Proposition 3.7 with the actual widths and the sharper comparison using the normal bundle whenever the end is immersed. Lifts of the entire groups exclude $G(4,1,8)$ and $G(3,1,8)$ cusps after the comparisons on their fixed curves. The tuple $(1,1,4^7)$ is excluded only when its width requires an absent full $G(4,2,8)$ fixed point. Equation (C.3) removes 29 rows with parameter six. For the three remaining rows with parameter six,

$$T = G(6,3,8), \quad t = 1/2, \quad s = 3, \quad R = \sigma = 0, \quad h = -6, \quad a = 20, 18, 16,$$

the end types are $A = (1,1,3^7)$, $B = (1,2,6^7)$, and $C = (2,2,3,6^6)$, with combinations ABB, ABC, ACC. Width one forces a nonzero generator of degree one at A, B, and nonzero generators of degrees two and three at C. Thus the actual critical degrees are $c_f = 0, 1, 2$, respectively. The normal targets in Proposition 3.7 have normalized degrees

$$(12, 9, 7, 6, 3, 0, -3, -6).$$

Each row has $\ell = 5, e \ge 6$, so its positive normal part has rank 4 and degree at least $42, 39, 36$, respectively. All exceed the sum $12 + 9 + 7 + 6 = 34$. No assertion of immersion at those ends is required.

The two residual Gaussian comparisons using the normal bundles have

T	a	j	s	widths
$G(4,4,8)$	6	1	2	$(2,1)$
$G(4,4,8)$	10	2	3	$(1,1,1)$.

On the double and identity covers the upper bounds on the degrees of the normal targets are $(8,6,6,4,2,0,-2,-4)$. Only $\ell = 5, e' = 4$ remains, and its positive part has rank 4 and degree $24 = 8 + 6 + 6 + 4$. It would have to equal the four largest target lines. The product direction instead has pole at most one at every full $G(4,4,8)$ fixed point with trivial fiber character. Write

$$\Omega = \{z \in \mathbb{Z}[i]^8 : \textstyle\sum z_i \in (1-i)\mathbb{Z}[i]\}, \quad A = \mathbb{C}^8/\Omega, \quad M = p^*\mathcal{O}([0])^{\boxtimes 8},$$

where $p : A \to E_i^8$ has degree two. The character summands of $p_*\mathcal{O}_A$ give odd and even theta products. The even product centered at $(1+i)/2$ is nonzero at the origin; their ratio has initial indecomposable term $c\prod z_i$, $c \ne 0$. Changing one coordinate sign separates this term from symmetric invariants. The four fixed points are $\Omega^\#/\Omega = K(M)$, with $\Omega^\# = \mathbb{Z}[i]^8 + \mathbb{Z}(1+i)(1,\dots,1)/2$. Normalized translations with trivial fiber character preserve M and transport the calculation. Thus the contraction contains $q\,d(\prod z_i)$ at each required fixed point. This

lowers the product line from degree six to zero on the double cover, or to at most three on the identity cover, contradicting its required minimum four.

The remaining B_8 row is

$$a = 8, \quad t = 1/2, \quad s = 3, \quad R = \sigma = 0, \quad h = -2, \qquad \alpha_c = (1^3, 2^6).$$

For the minimal D_8 polarization, three even and one odd theta sections of degree one give the first two independent derivative projections. For a map to $\mathcal{O}(-2)$, the first cusp division leaves coefficient degrees $i - 5$, so those first two projections vanish. The radial equation forces a second division in all coefficients of degree one. The resulting degrees $i - 8$ leave only the projection of the inverse differential to the coordinate of greatest invariant degree. Hom dimension one now gives $a \geq 8(-h) = 16$, a contradiction. All twenty D_8 rows require a B_8 end. The complete rejection counts are

bundle tests	16	B_8 theta	1
B_8 ends	20	$G(3,1,8)$ tests/ends	$2+2$
$G(4,1,8)$ ends	33	Gaussian torsors/product tests	$9+2$
Eisenstein widths	29	normal bundles over the Eisenstein field	3.

These sum to 117.

C.5 Rank 7: the 413 rows

Use the preliminary bound $a \leq 7wt$, deferring D_7 when an E_8 enlargement is possible. The inequalities

$$h \leq -4j + 2m(1-g) + (2-m)R - m\sigma, \qquad a \leq (14m+2)(1-g) - (D-14m)t$$

exclude $g \geq 1$ by $\ell \leq 4$, and leave no $h \geq 0$ record with $j \leq 3$. For $j \geq 8$, the inequalities $a \geq -h \geq 4j - 12$, $a \leq 86 - 49j/6$, and $\ell \leq 5$ leave only $m = p = 6, j = 8$, where $-h \geq 24 > a$. Hence $g = 0, h < 0, 1 \leq j \leq 7$. Use the relaxed necessary condition $\sigma = 0$ or $\sigma \geq 1/2$, which contains every actual correction. With (C.7), the full domain gives

(m,p)	$(2,1)$	$(2,2)$	$(3,1)$	$(3,3)$	$(4,1)$	$(4,2)$	$(4,4)$	$(6,2)$	$(6,3)$	$(6,6)$
rows	21	43	2	5	1	51	132	18	16	124.

All 25 unbounded domains for the marked degree are retained. The exclusions have two stages, with different available nef bounds. Before (6.5), use only $a \leq 7wt$; the following tests exclude 166 rows.

groups	number	test
$G(6,p,7)$	148	$\kappa_c = p/q_c$
$G(6,p,7)$	10	cyclic covers
$G(3,1,7)$	2	normal bundle
$G(3,3,7)$	5	excluded $G(3,1,7)$ ends
$G(4,1,7)$	1	normal bundle.

These preliminary tests suffice for the list of possible factors used to prove (6.5); no row is excluded here using $a \leq 10t$. Keep the other 247 rows, including D_7, until (6.5) and the exclusion of E_8 have been proved. The new bound excludes 205 of these rows by $a > 10t$. The remaining 42 rows are excluded by a B_7 end in one case, a $G(4,1,7)$ end in 25 cases, absence of a full fixed point on the Gaussian torsor in eight cases, and the comparison using the normal bundle in eight cases. The total is $166 + 205 + 42 = 413$. The ten rows tested using cyclic covers have $R = 0$ and at most two nontrivial widths. At the least permitted e, one of the two monotone inequalities in Proposition 3.7 fails. The same comparison using the

normal bundle excludes the two $G(3,1,7)$ rows and the $G(4,1,7)$ row, and lifts of the entire groups exclude the corresponding cusps. The fixed-point rejection again concerns only $(1,1,4^6)$ when a full $G(4,2,7)$ fixed point is required on its torsor. In the eight Gaussian normal cases, $R = \sigma = 0$ and the section degrees are one, two, and four. The actual width is the least nonzero degree, so the end is immersed in $Q = q^\kappa$. Seven cases have $j = 2, s = 3$ and one cusp of width two. After the double cover, the upper bounds on the degrees of the normal targets $(12, 9, 8, 4, 0, -4, -8)$ have sum at most 29 in rank 3, but the marked positive part requires 36. The last case $j = 4, s = 4, a = 9, \ell = 5, e \geq 9$ has width one and requires 43 against the same bound. The arguments for D_7 and for E_7 without enlargements are given explicitly in Proposition 6.5, after the bound $a \leq 10t$ has been established.

C.6 Projective space

Here we prove the auxiliary assertion of Proposition 6.1 independently of the theorem for weighted quotients. On $\mathbb{P}^k$, put $a = (k+1)u$, $u > 0$. The Euler sequence gives $h \geq u$ for every nonzero inverse line quotient. A persistent factor of rank at most $k-2$ contradicts this on a complete curve avoiding the finite boundary. Parabolic classification thus leaves the corank 1 curves listed below. At a smooth cusp its spherical link makes the horizontal group irreducible.

For $k = 7$, first remove the primitive group G_{34}, before applying ordinary monomial formulas at its possible horizontal subgroups. Its polynomial degrees are $(6, 12, 18, 24, 30, 42)$, its section tuple is $(1, 3, 4, 6, 7, 9, 12)$, and

$$a = 41s + 2 - 2g - 132t, \quad h = -42t + 12s, \quad 6t \in \mathbb{Z}_{>0}.$$

There is no torsor, and the reflection divisors give a lift of the entire group with trivial character at the origin. Without ends $h < 0$, excluding compact disks immediately. With ends the logarithmic identity is $2g - 2 + s + \sigma = 2t$. Its combination with $h \geq a/8 > 0$ gives $94t + 55\sigma \leq 108(1-g)$, hence $g = 0$ and in particular $6t \leq 9$. In that complete finite range, $a \in 8\mathbb{Z}_{>0}$, $a \leq 48t$, and $h \geq a/8$ leave no possibility. The lift of the entire group then excludes G_{34} cusps as well.

Now consider the monomial groups. In both the compact and nonuniform comparisons, defer D_6 and D_7 until their possible E_7 and E_8 enlargements have been removed below. For disks without ends and every $m \geq 2$, the logarithmic identity gives

$$h \leq -(r-1)j + \frac{2j + 2m}{m-1} \leq 7 - r, \qquad r = k - 1 \geq 7.$$

For $r = 6$, use $h \leq 4 - 4j$ for B_6 and $h \leq -5j + (2j + 2m)/(m-1)$ for $m \geq 3$; these again contradict $h > 0$. For nonuniform disks, use the complete cusp records above in descending parameter order. For $r = 6, 7$, omit D_r for the later ADE comparison. The cusp sum bound and the logarithmic equation give

$$a \leq (2rm + 2)(1 - g) - (D - 2rm)t,$$

since the omitted coefficient of R is $\eta - r(m-1) = 2r - 1 + 1/p - rm \leq 0$. As $a = (r+2)u \geq r+2$ and $D > 2rm$ for $r = 6, 7$, this proves $g = 0$ and

$$1 \leq j \leq \left\lfloor \frac{mr(2m-1)}{D - 2rm} \right\rfloor.$$

These are the following complete j-ranges:

(m,p)	$(2,1)$	$(3,1)$	$(3,3)$	$(4,1)$	$(4,2)$	$(4,4)$	$(6,1)$	$(6,2)$	$(6,3)$	$(6,6)$
$r = 6$	2	3	6	4	7	9	7	11	13	16
$r = 7$	1	2	3	3	4	5	5	7	8	9.

In (C.5), impose $a \in (k+1)\mathbb{Z}_{>0}$, $a \le (k+1)wt$, and $h \ge a/(k+1)$. For every listed pair there are zero survivors. The comparisons of the orders of the root groups exclude exceptional ends for these non-ADE monomial groups. In rank 7, for example, $m^6\,7! > 576\,7!$ for $m \ge 3$, but $|W(E_7)|/|G(2,1,7)| = 9/2$ is not integral.

For $r = 8$, there is no primitive rank 9 enlargement and

$$h \le 2m(1-g) - (r-3)j + (2-m)R - m\sigma.$$

The estimate is nonpositive unless $g = 0$ and $(m,j) = (3,1),(4,1),(6,1),(6,2)$. For $(m,j) = (3,1)$, $s = 2$ gives $R \le 1, h \le 0$, and $s = 1$ gives $R \le 2, h \le -2$. For $(m,j) = (4,1), s = 1$, one has $h \le -1$. For $(6,1), s = 1$, one has $h \le 0$, but $s = 2$ is impossible because it requires $5R/6 + \sigma = 1/3$. For $(6,2)$, the two cases have $h \le -6, -2$. The sole remaining case $m = 4, j = 1, s = 2$ has $h > 0$ only when $p > 1, R = 0, \nu_1 = \nu_2 = 4$, both widths one, and $B_c \ge 28$. Thus $a \ge 24$, contrary to $a \le 40t = 10$. This excludes all these monomial disks and then their full cusps.

For $k = 7$, after these non-ADE groups have been removed, the orders of stabilizers of rank 4 in the remaining ADE factors divide 2880. For $16L = bH$, Chern integrality gives $1575b^4/512 \in \mathbb{Z}$, hence $8 \mid b$ and $a \le 16t$. An E_7 point or E_6 cusp gives an E_6 disk with

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = 11s + 13R + 2 - 2g - 42t, \quad h = -12t + 3R + 3s, \quad 2t \in \mathbb{Z}_{>0}.$$

The coefficient $13R$ includes the sign character of the odd invariant degrees. Eliminating s in the Euler inequality gives $10t + 22\sigma \le 42(1-g)$, hence $g = 0, 2t \le 8$. The conditions $a \in 8\mathbb{Z}_{>0}$, $a \le 16t$ leave only $(t,s,R,\sigma,a,h) = (3/2,3,4,0,24,3)$, whose odd line would have degree $-19/2$. For D_6, the symmetric inverse gives $5h = a - 12t - R - \sigma$. Together with $h \ge a/8$, this requires $a \ge 32t$, a contradiction. The last A_6 disk has $7h = a - 20t - \sigma < 0$.

For $k = 8$, the orders of stabilizers of rank 3 in the ADE groups divide 24. The integral $18L = bH$ and $24c_3^{\mathrm{orb}} = -2016L^3$ give $28b^3/81 \in \mathbb{Z}$, hence $9 \mid b$ and $a \le 18t$. An E_8 point or E_7 cusp gives an E_7 disk. At E_8, the coarse parameter is $v = u^2$. Homogeneity gives pole order six for the row of the inverse differential associated with the invariant of degree eighteen, and determinant order 28; the invariant directions have trivial fiber character on the coarse quotient. Consequently

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = 17s + 28R - 70t + 2 - 2g, \quad h = -18t + 6R + 4s, \quad 2t \in \mathbb{Z}_{>0}.$$

Elimination gives $9h = 2a - 18t - 3R - 2\sigma$. The Euler inequality and $a \le 18t$ force $R = \sigma = 0, a = 18t, s = 5t, 3t = 2 - 2g$. Thus $g = 0, t = 2/3$, contradicting $2t \in \mathbb{Z}$. For the remaining D_7 disk, $6h = a - 21t - R - \sigma$ gives $a \ge 63t + 3R + 3\sigma$, again impossible. The last A_7 disk has $8h = a - 27t - \sigma < 0$.

For $k = 9$, the only remaining primitive group is E_8. Its tuple in (C.4), with width one and the full group, gives

$$2g - 2 + s + \sigma = 2t, \quad a = 29s + 2 - 2g - 128t, \quad 2t \in \mathbb{Z}_{>0}, \quad a \in 10\mathbb{Z}_{>0}.$$

The upper bound $a \le 60(1-g) - 70t$ forces $g = 0, t = 1/2, s = 3$, and then $a = 25$, a contradiction. Only type A groups remain in $\mathbb{P}^9$. Write $20L = bH$. Chern integrality gives $27b^2/20, 9b^3/25 \in \mathbb{Z}$, hence $10 \mid b$ and $a/t \le 20$. The last A_8 disk satisfies $9h = a - 35t - \sigma < 0$, contrary to the Euler lower bound. This completes the independent exclusions of $\mathbb{P}^7, \mathbb{P}^8, \mathbb{P}^9$.

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